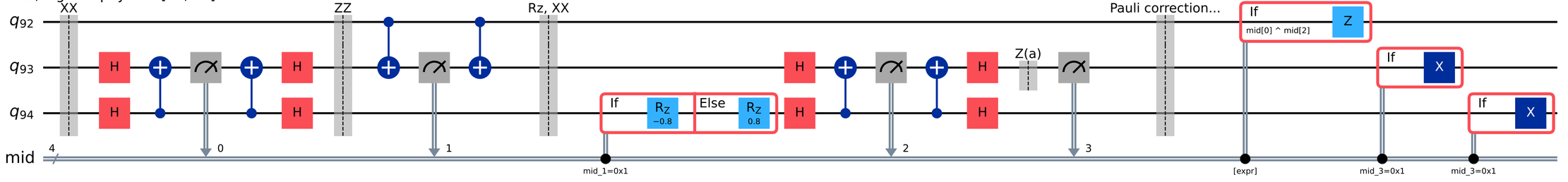
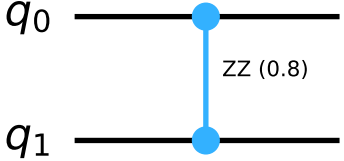
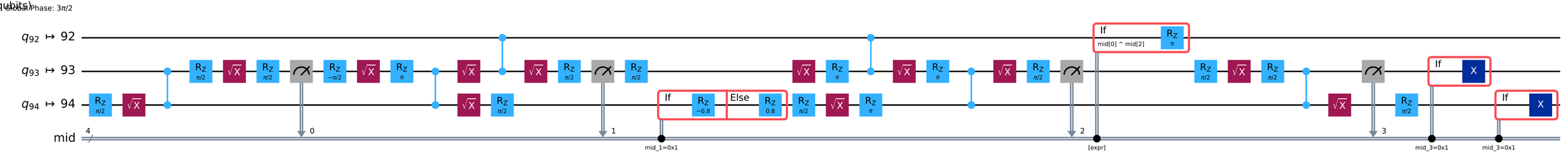
[Z2] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 6, 4 mid-circuit measurements, logical→physical [92, 94]



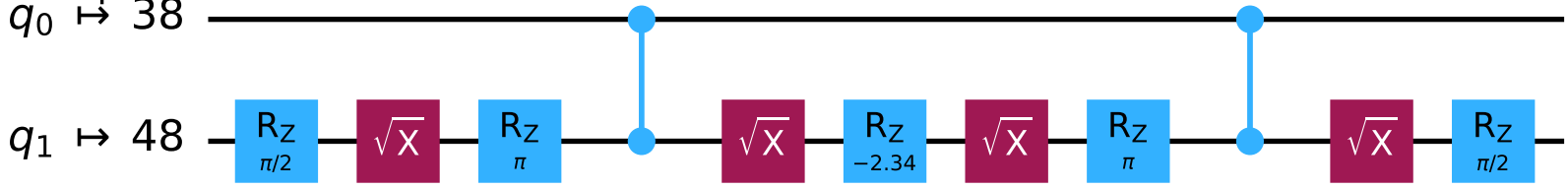
[Z2] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 1

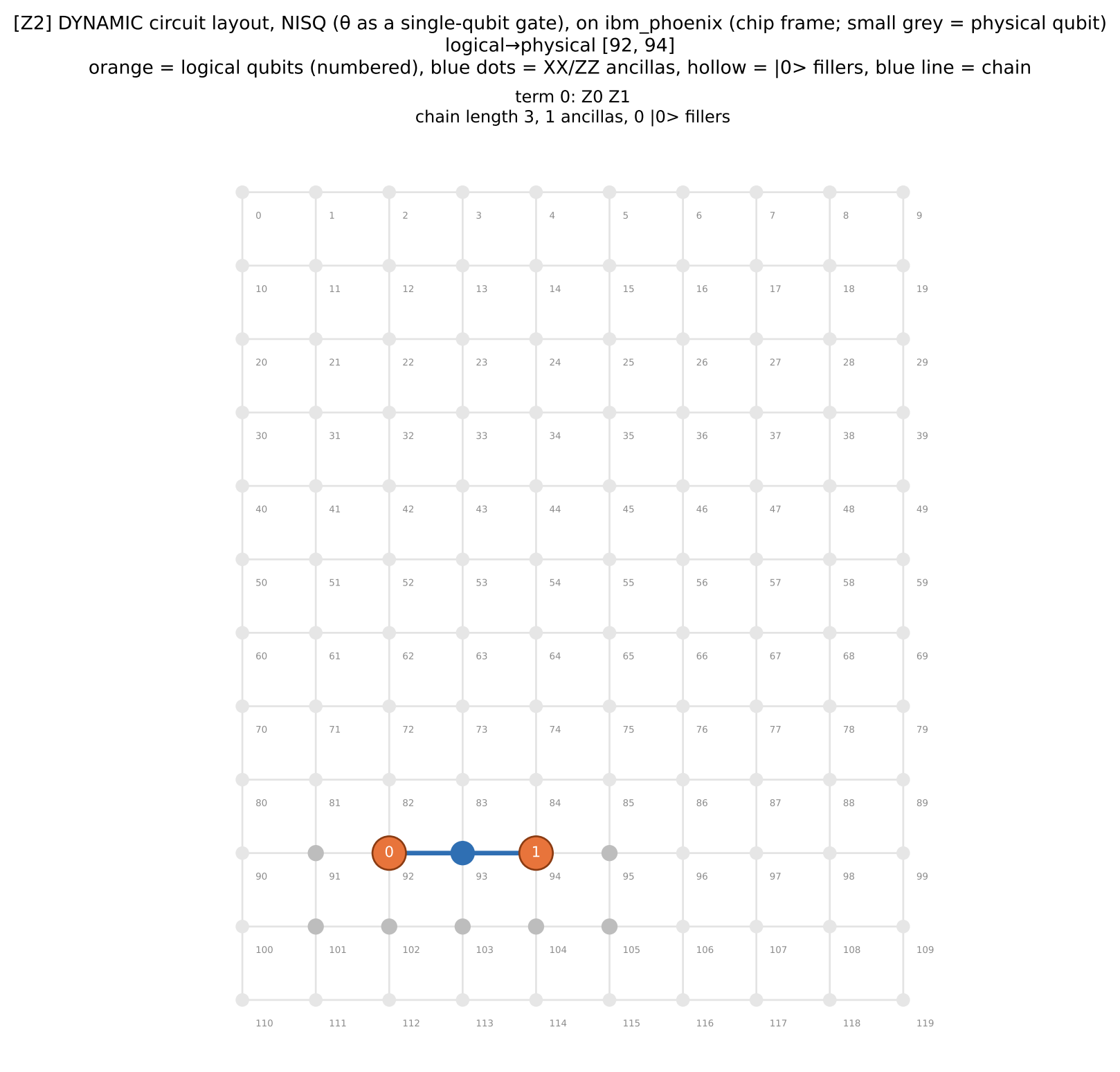


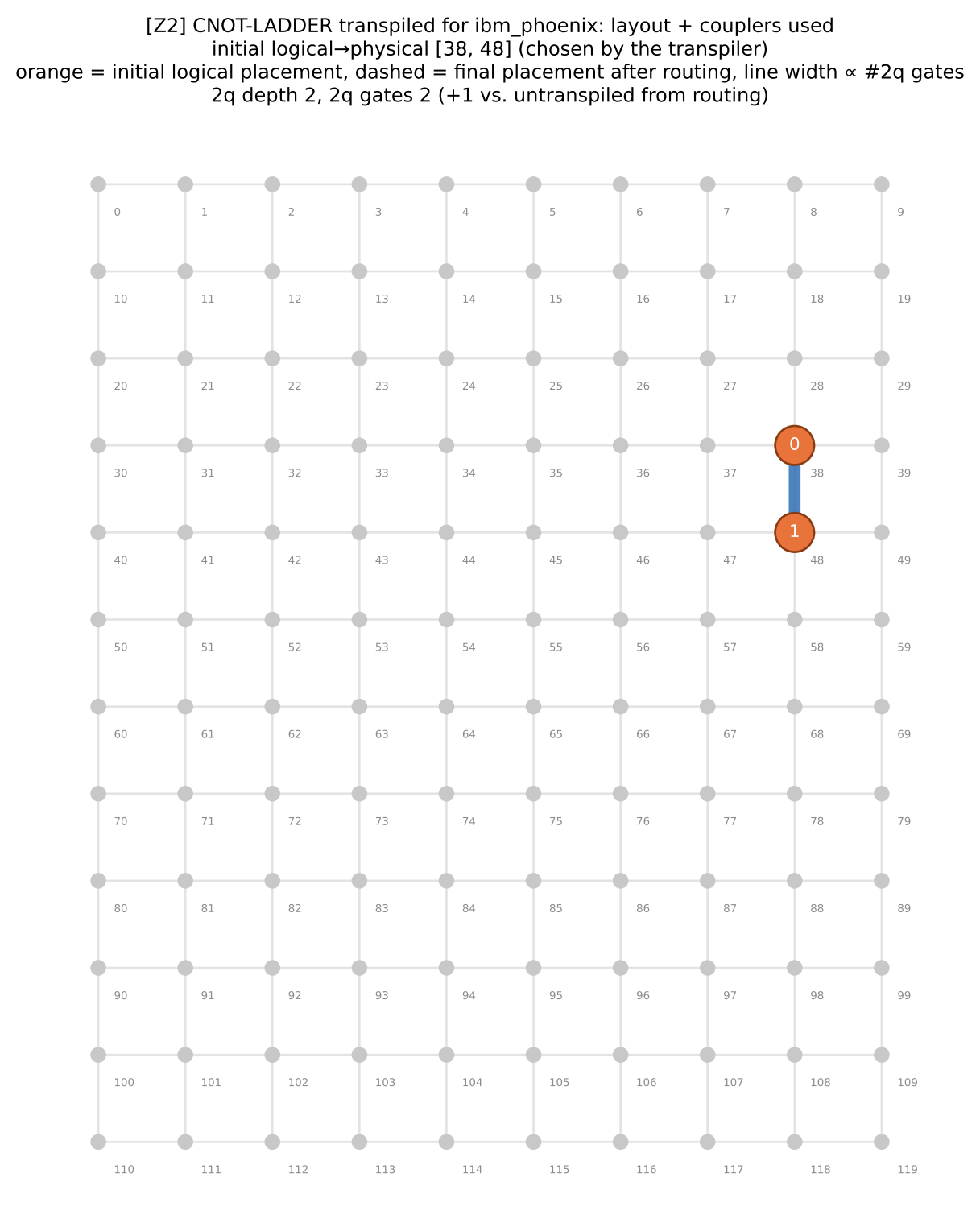
[Z2] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:16 sx:12 cz:6 measure:4 if_else:4)
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits).
2q depth 6, total depth 33



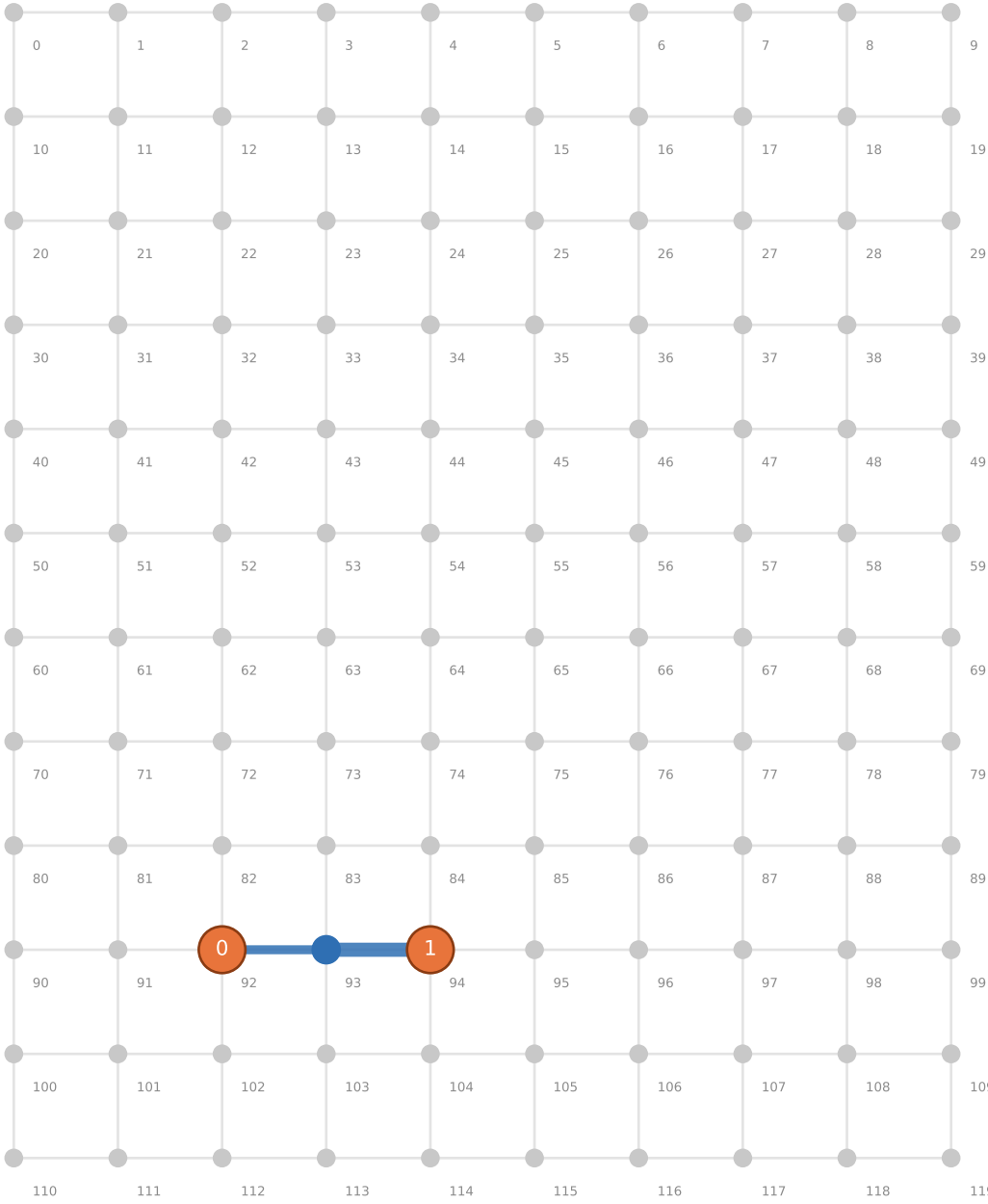
[Z2] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:5 sx:4 cz:2)
Global Phase: 0
exp(-i θ P), θ=0.4: P = Z0 Z1   (2 logical qubits)
2q depth 2, total depth 11



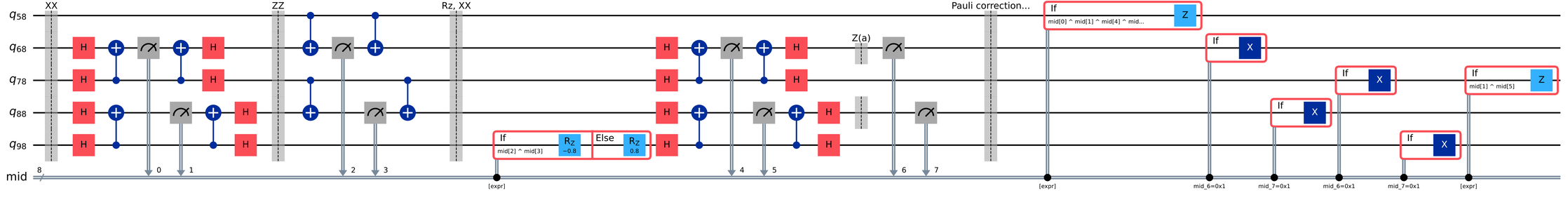

[Z2] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [92, 94]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1
chain length 3, 1 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1


[Z2] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 2, 2q gates 2 (+1 vs. untranspiled from routing)
0
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113
114
115
116
117
118
119
0
1

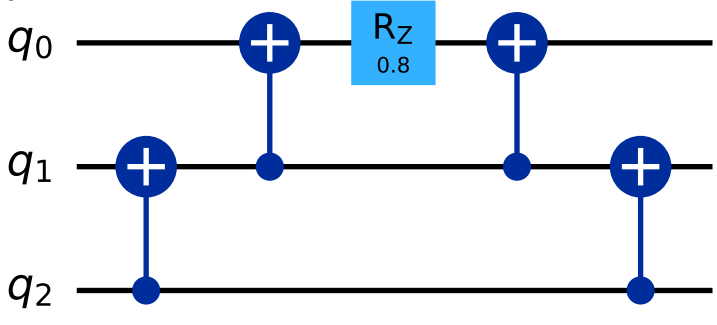
[Z2] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [92, 94]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 6



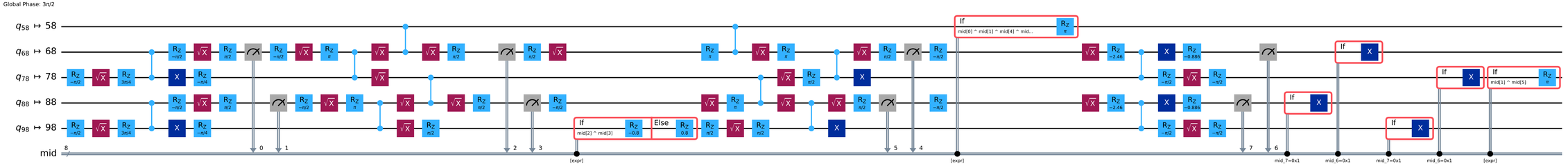
[Z3] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, 8 mid-circuit measurements, logical→physical [58, 78, 98]



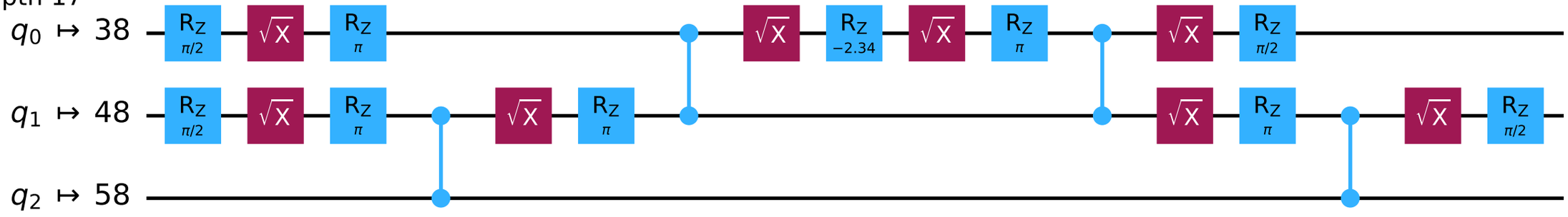
[Z3] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4



[Z3] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:37 sx:24 cz:12 measure:8 if_else:7 x:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 6, total depth 37

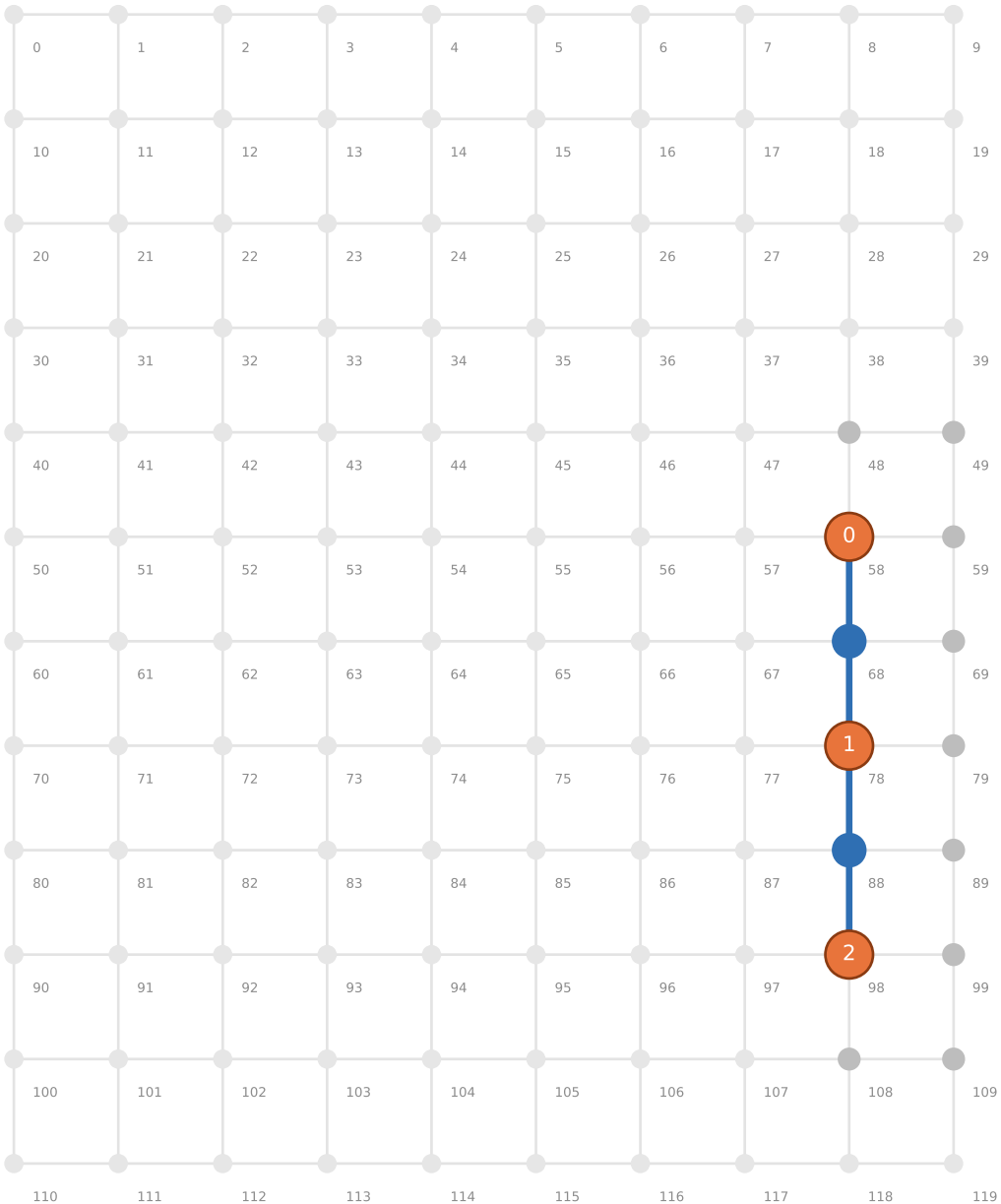


[Z3] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:10 sx:8 cz:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2   (3 logical qubits)
2q depth 4, total depth 17

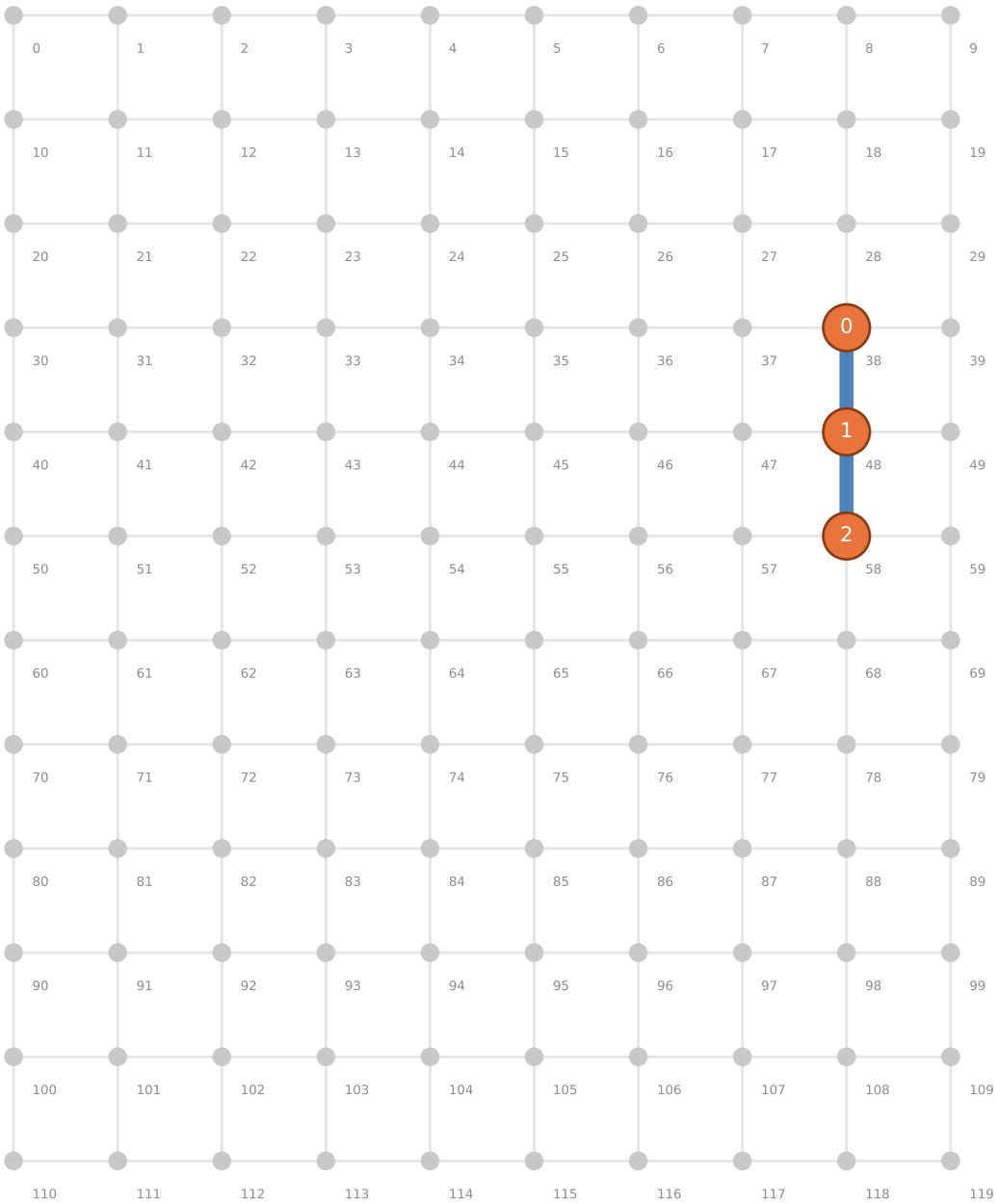


[Z3] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

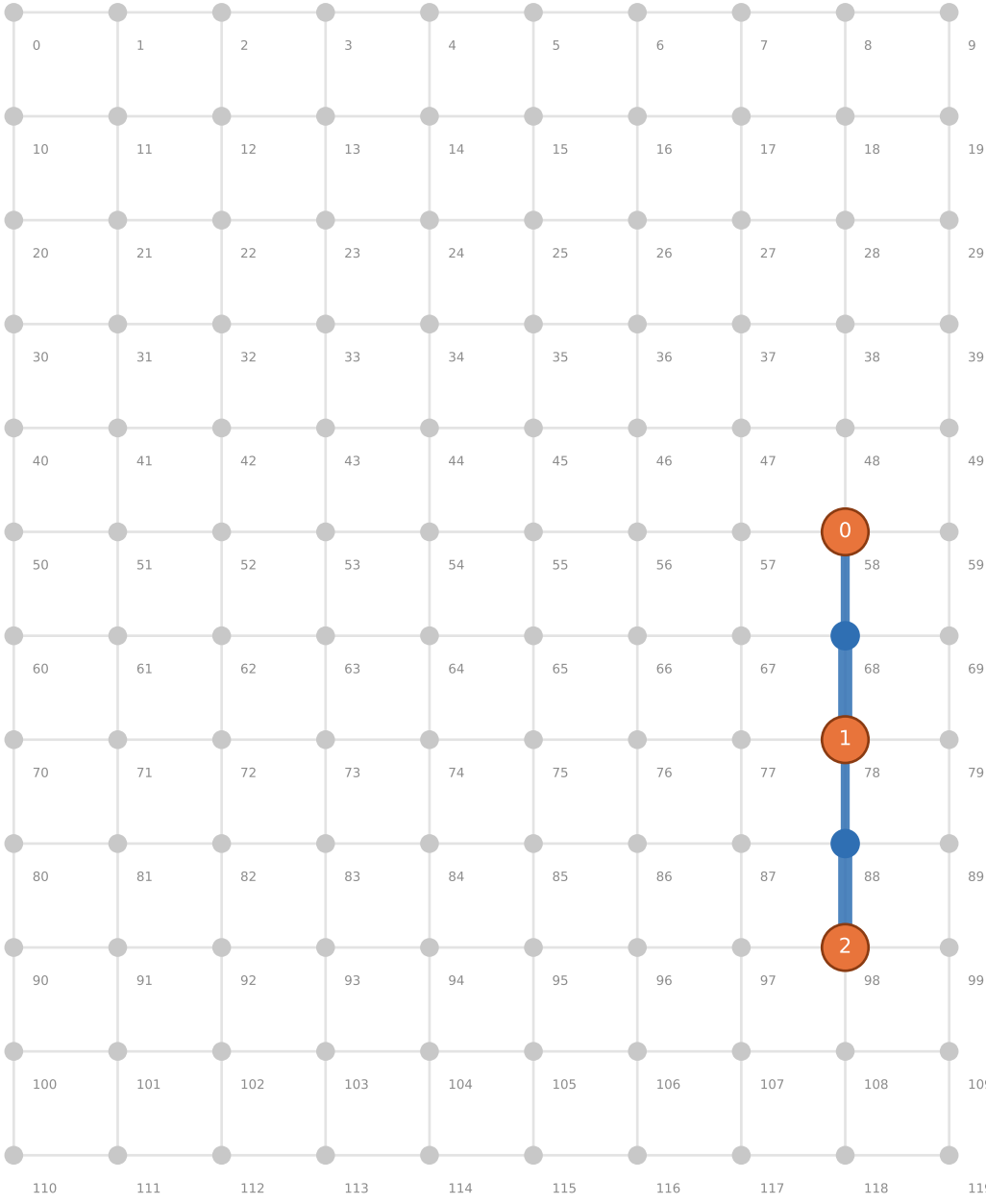
term 0: Z0 Z1 Z2
chain length 5, 2 ancillas, 0 |0> fillers



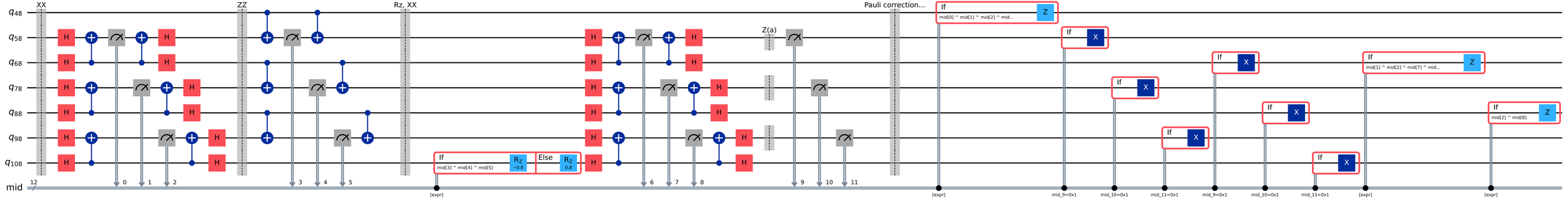
[Z3] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 4, 2q gates 4 (+0 vs. untranspiled from routing)



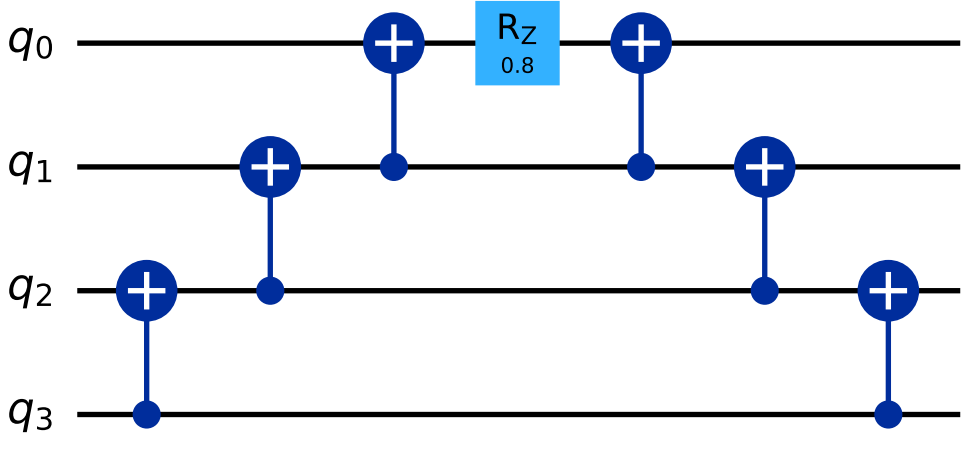
[Z3] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 12



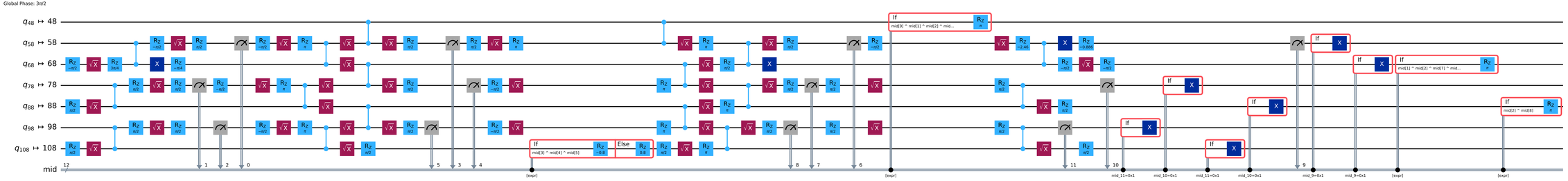
[Z4] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, 12 mid-circuit measurements, logical→physical [48, 68, 88, 108]



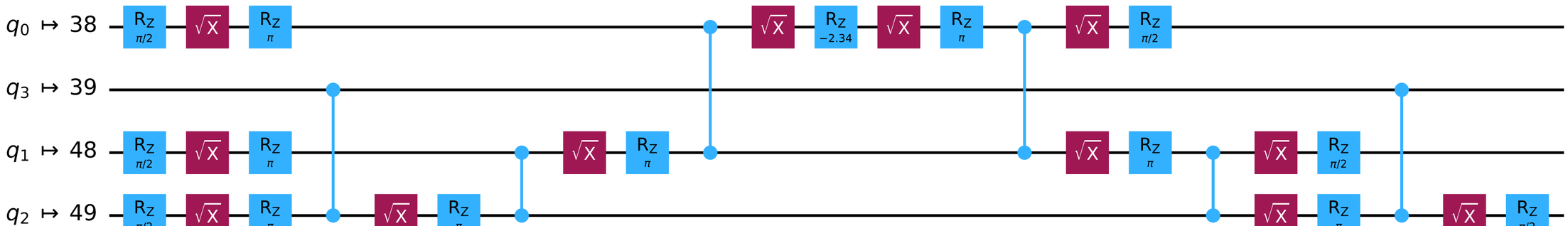
[Z4] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6

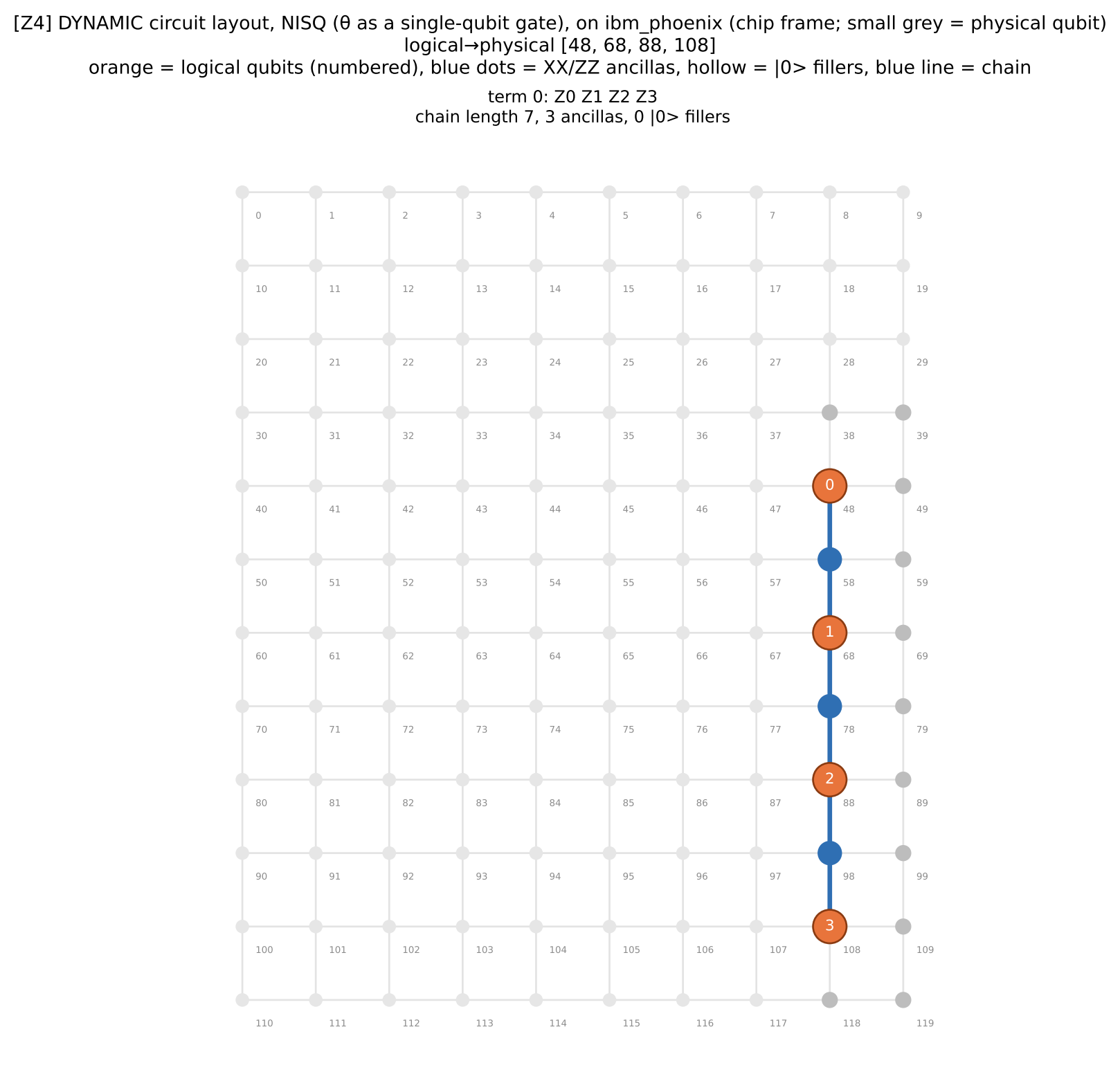


[Z4] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:46 sx:36 cz:18 measure:12 if_else:10 x:3)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
2q depth 6, total depth 38

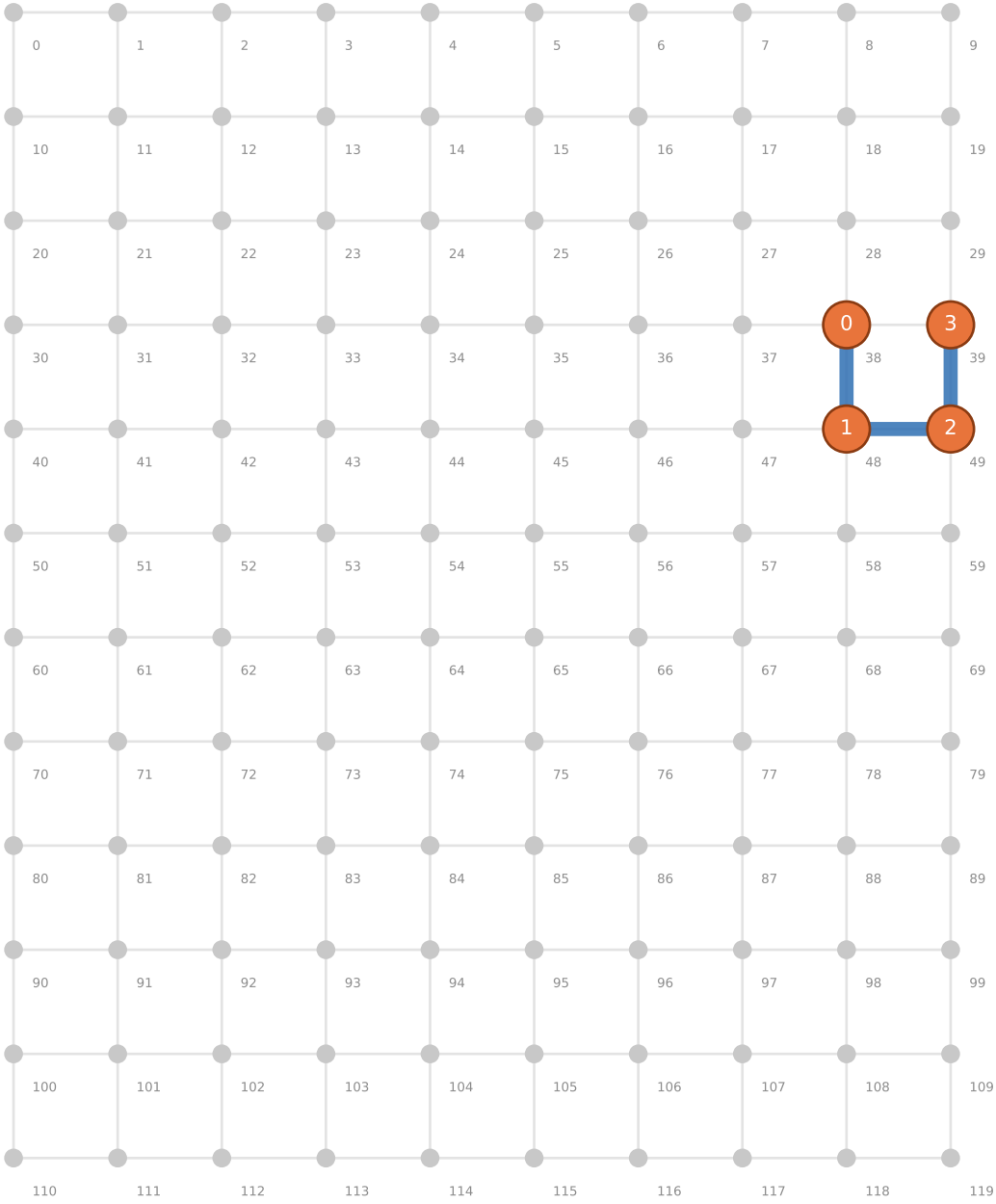


[Z4] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:15 sx:12 cz:6)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3   (4 logical qubits)
Global Phase: 0
2q depth 6, total depth 23

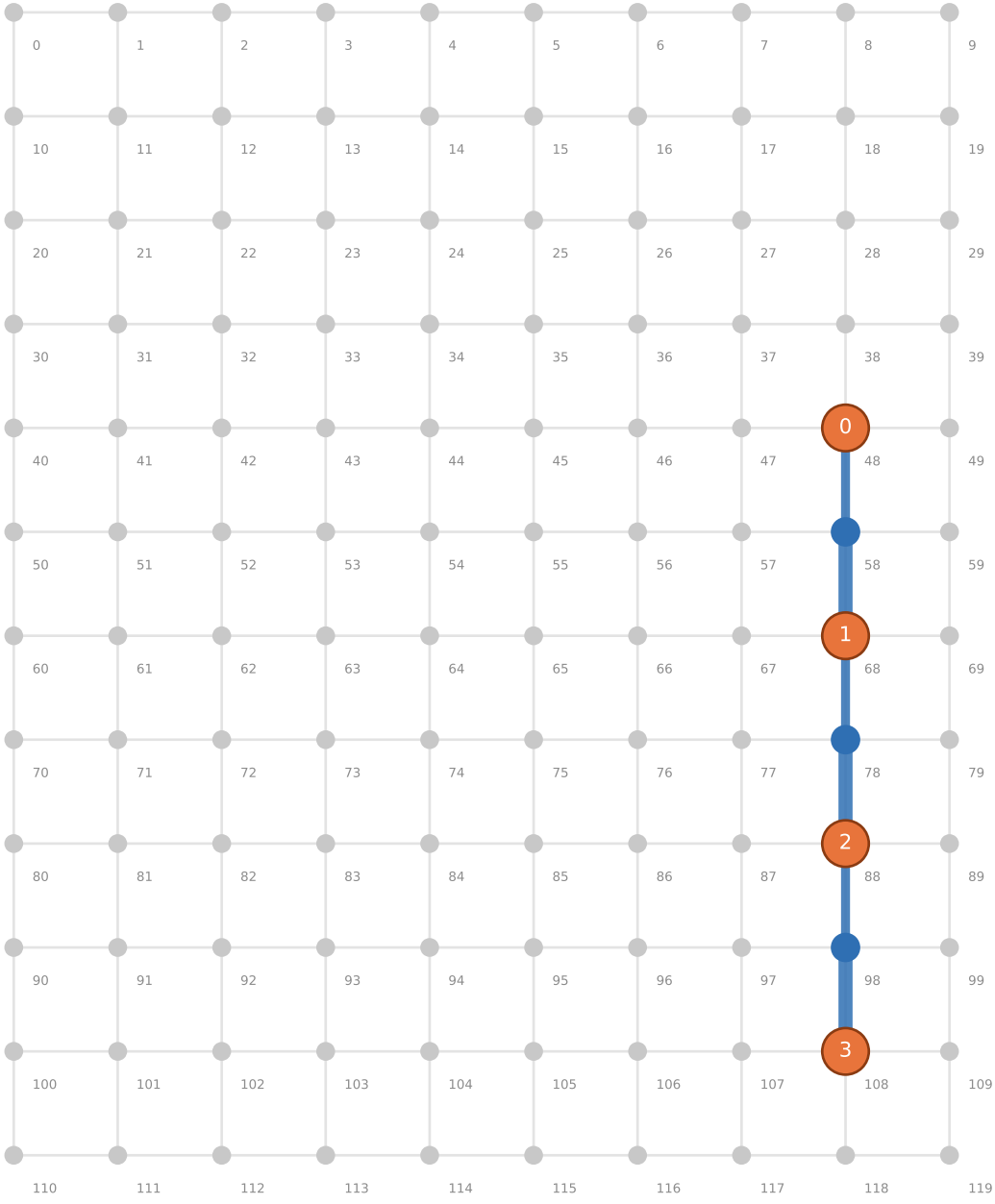



[Z4] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3
chain length 7, 3 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
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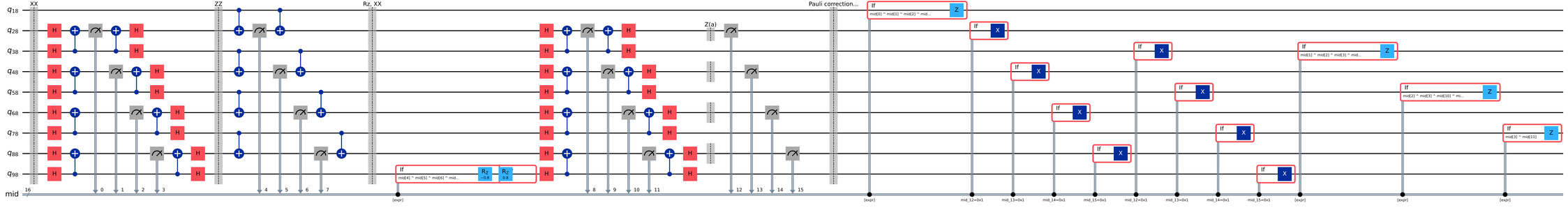
[Z4] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)



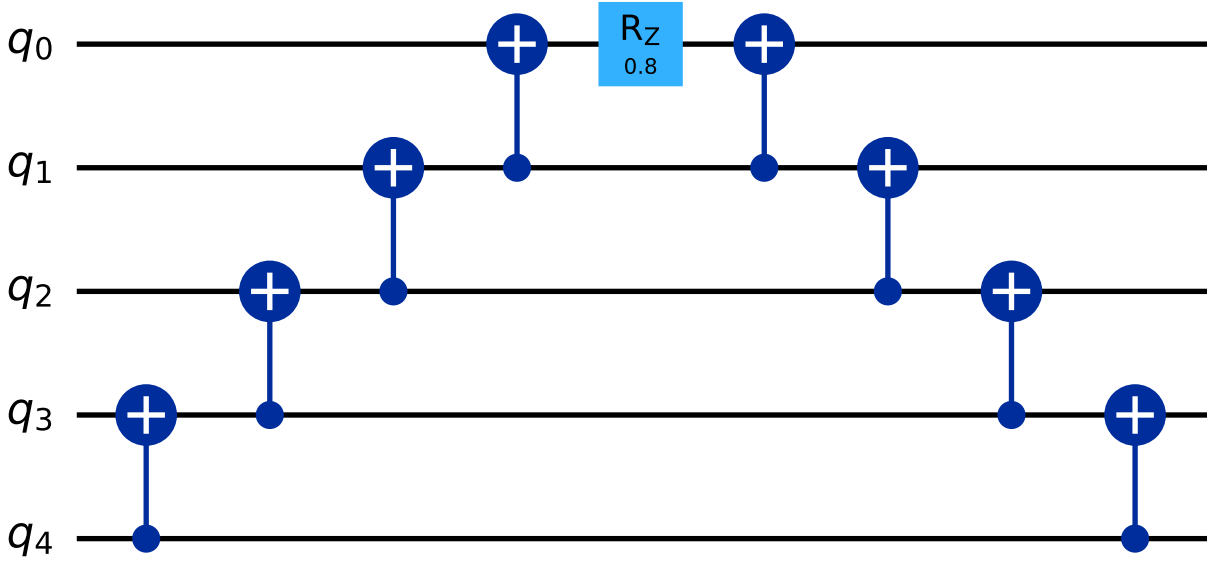
[Z4] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 18



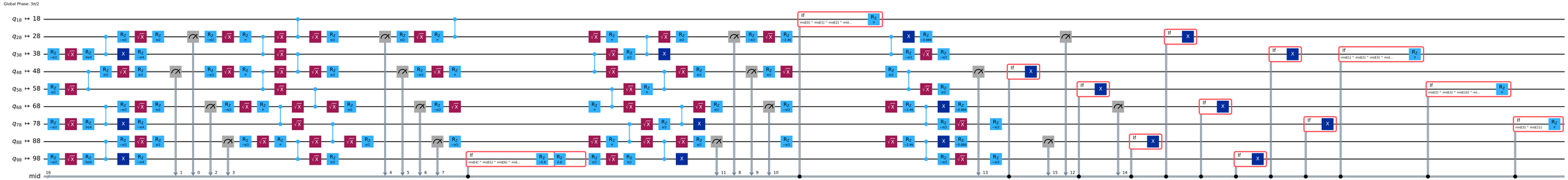
[Z5] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 (5 logical qubits)
2q depth 6, 16 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



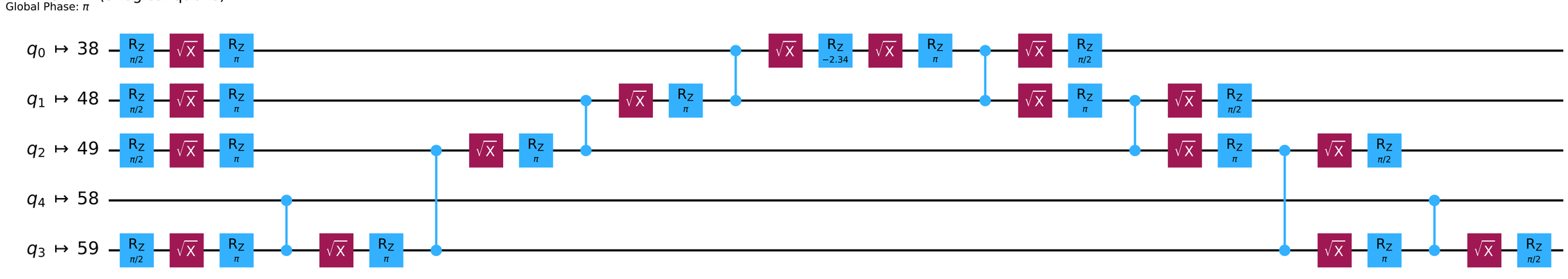
[Z5] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8

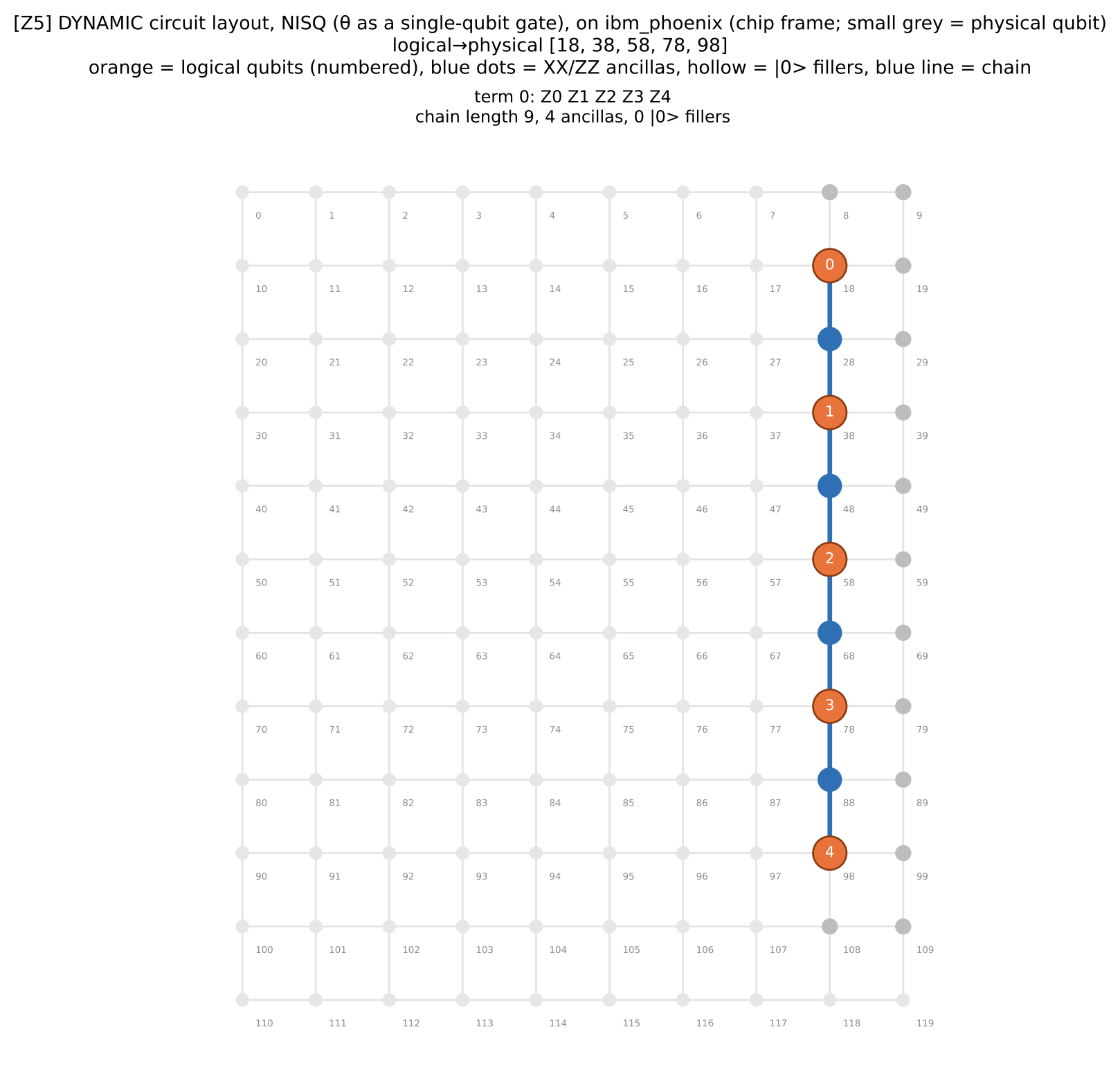


[Z5] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:67 sx:48 cz:24 measure:16 if_else:13 x:9)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 6, total depth 39

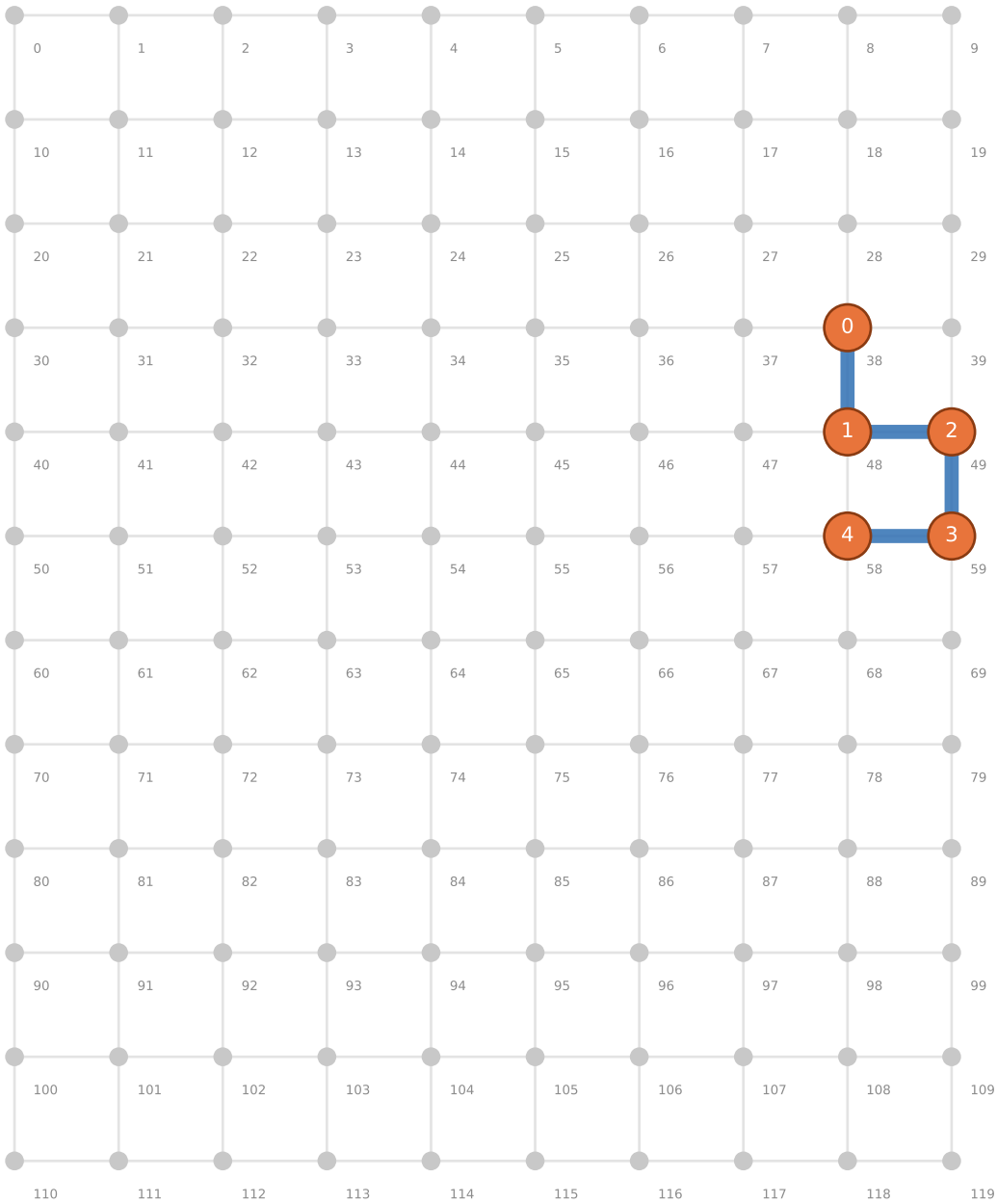


[Z5] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:20 sx:16 cz:8)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4   (5 logical qubits)
2q depth 8, total depth 29

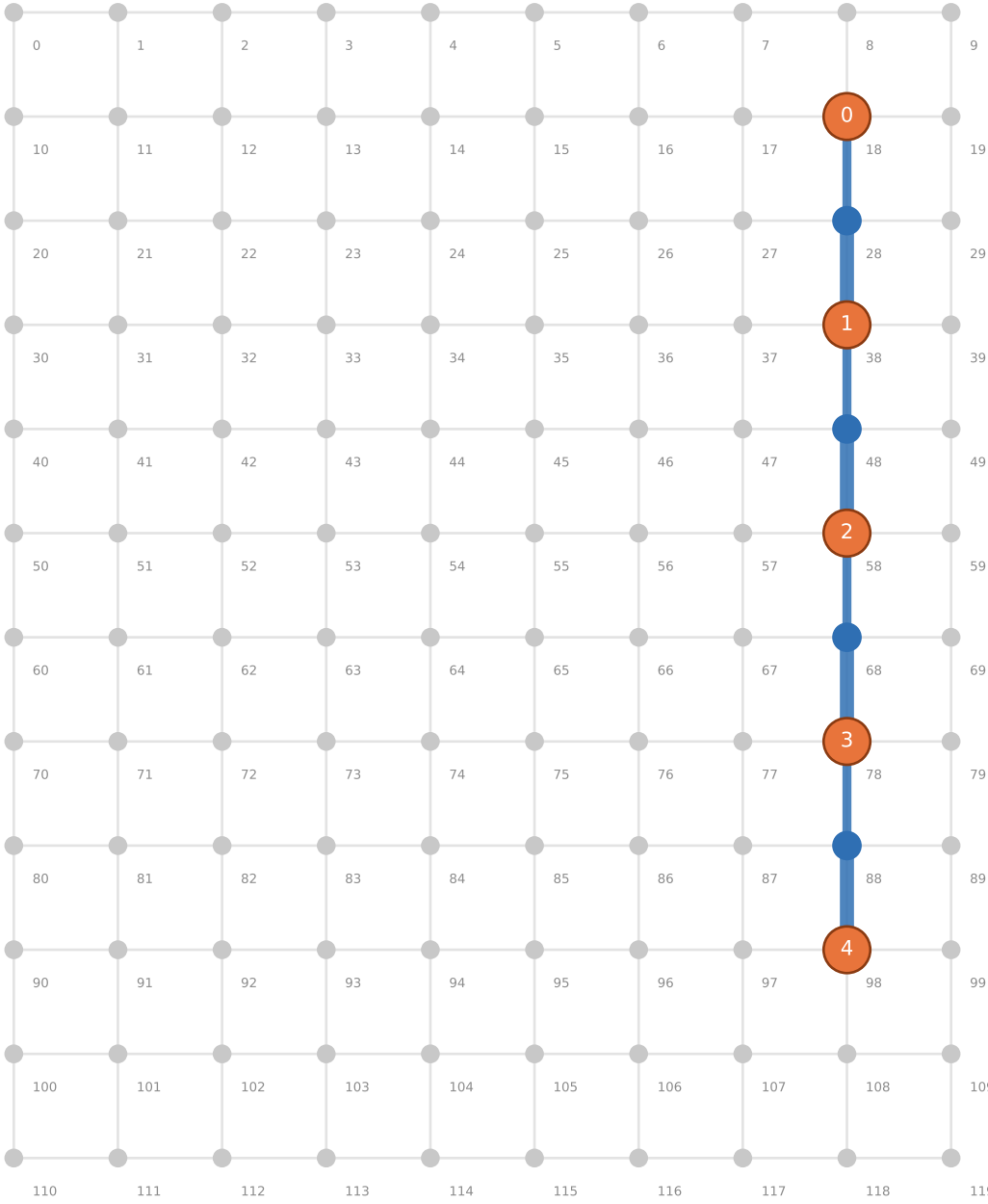



[Z5] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4
chain length 9, 4 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
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2
3
4

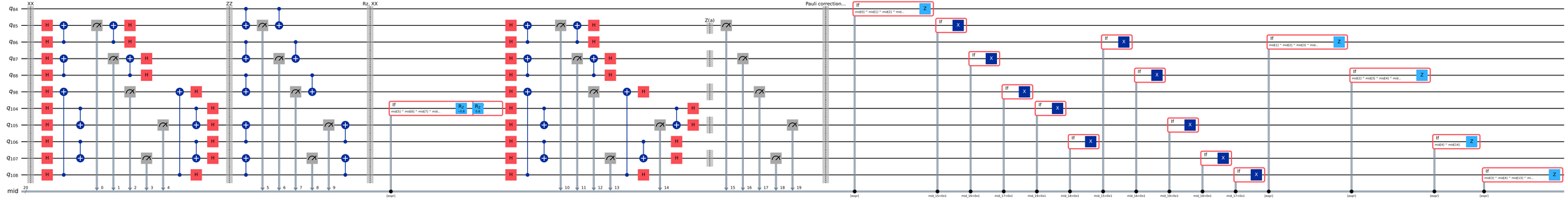
[Z5] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 8, 2q gates 8 (+0 vs. untranspiled from routing)



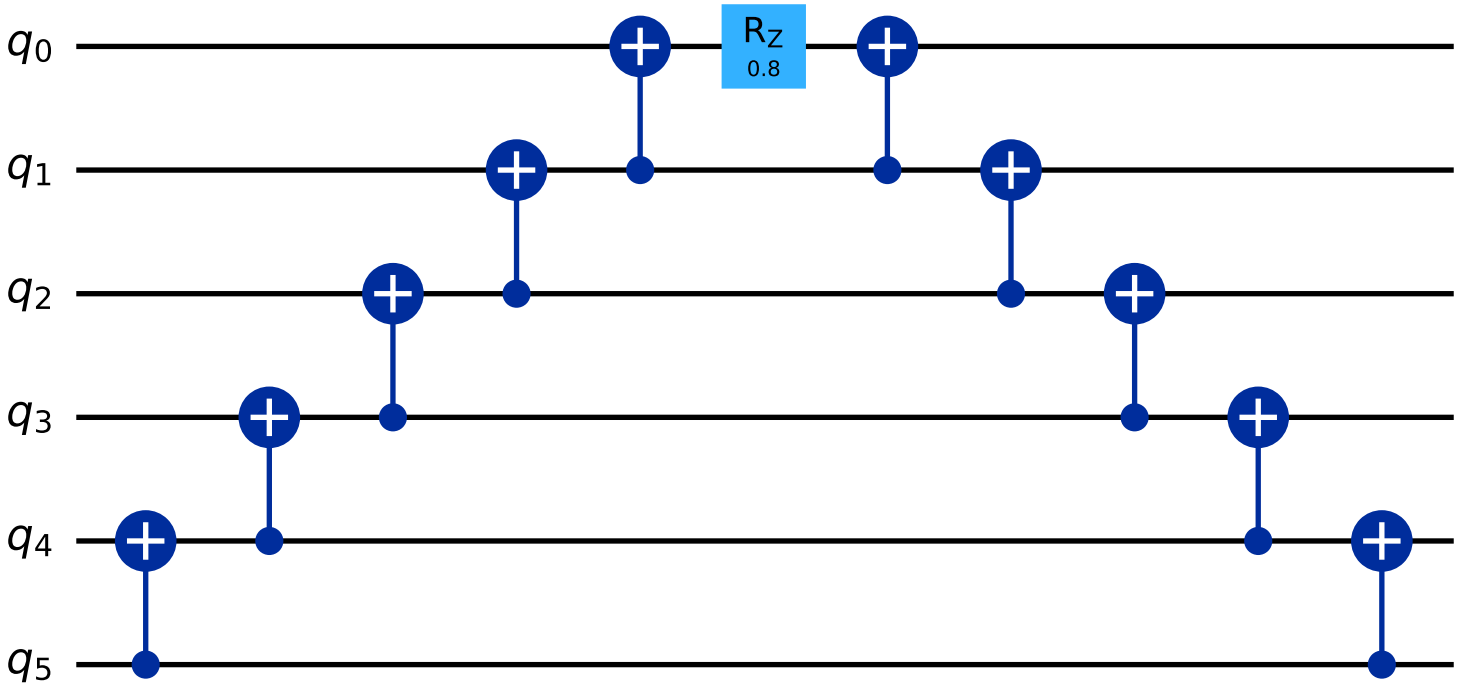
[Z5] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 24



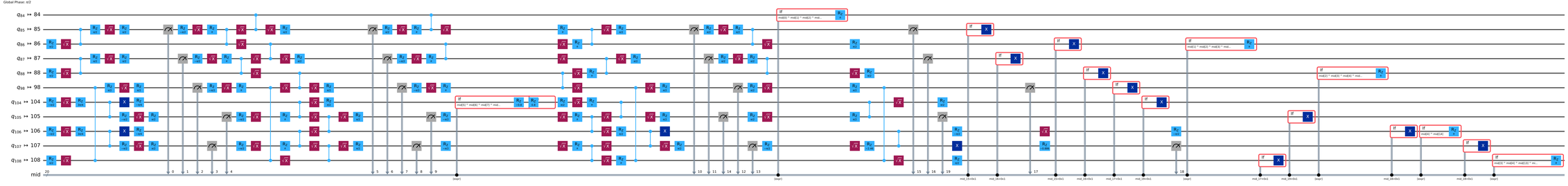
[Z6] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 6, 20 mid-circuit measurements, logical=physical [84, 86, 88, 108, 106, 104]



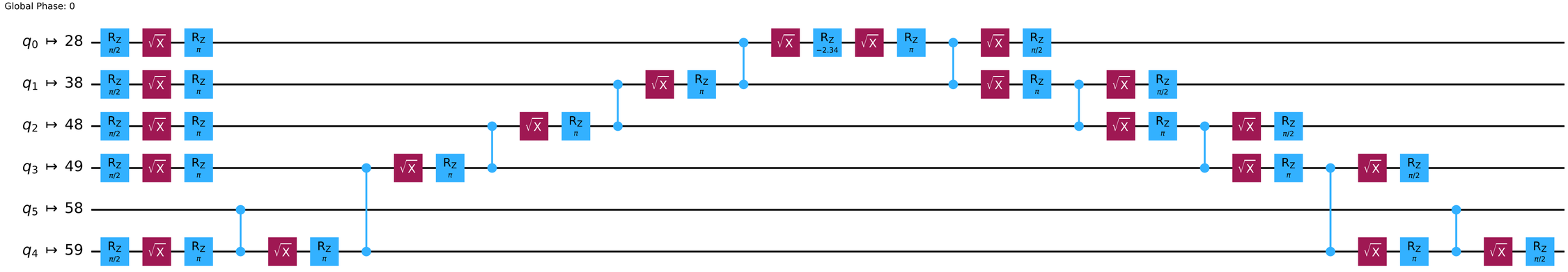
[Z6] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10

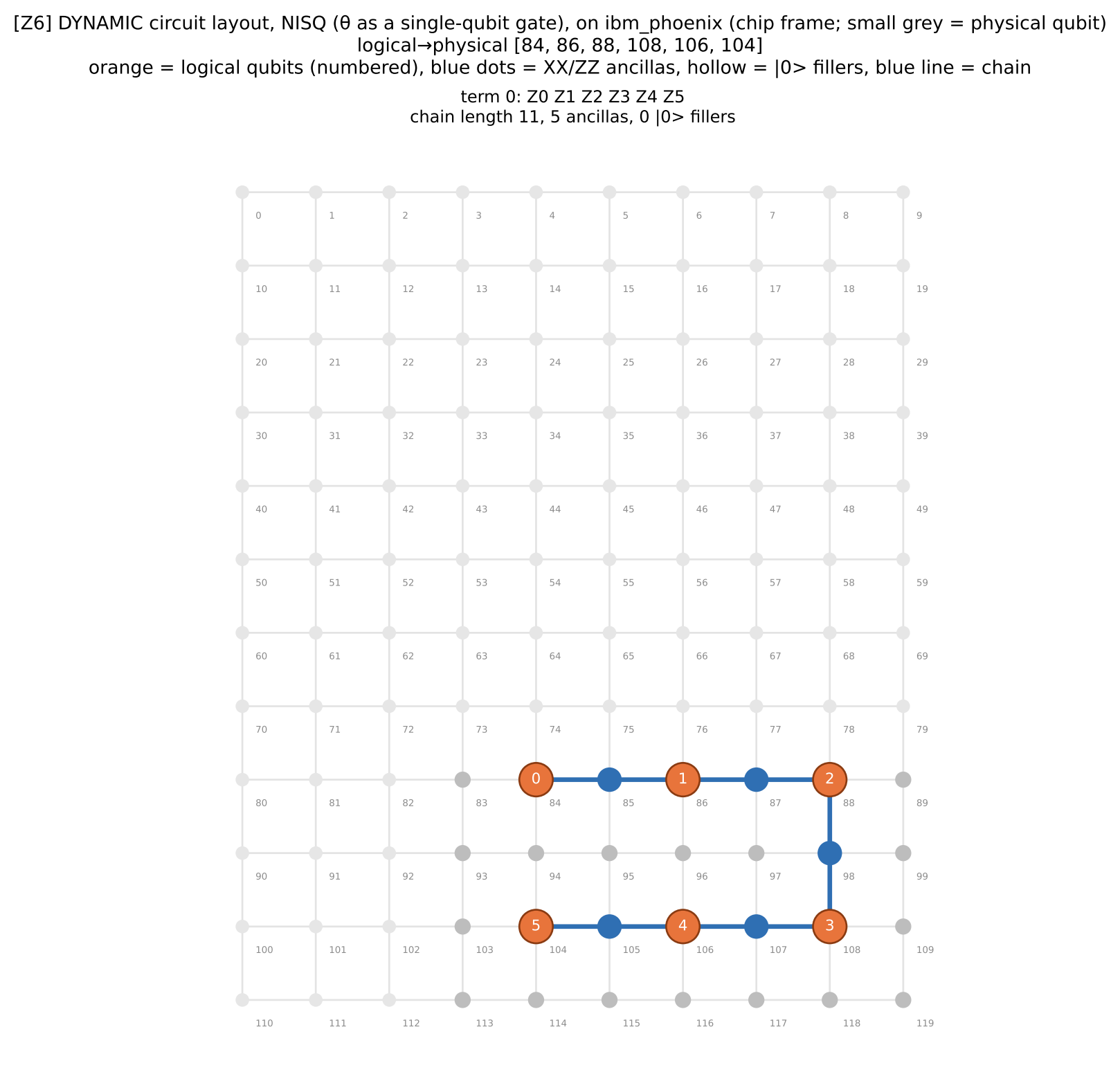


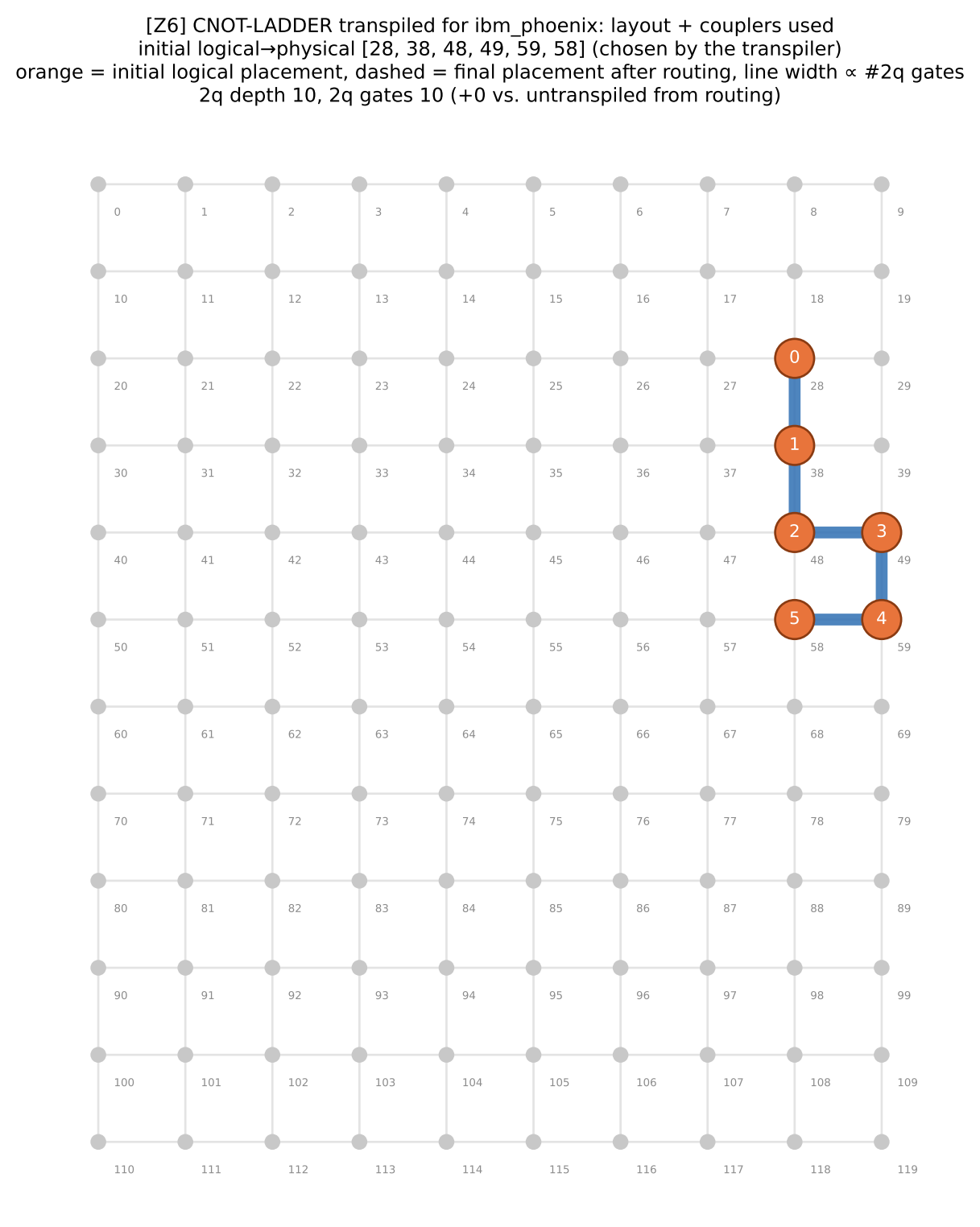
[Z6] DYNAMIC circuit, NISQ (θ as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:74 sx:60 cz:30 measure:20 if_else:16 x:4)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 (6 logical qubits)
2q depth 6, total depth 37



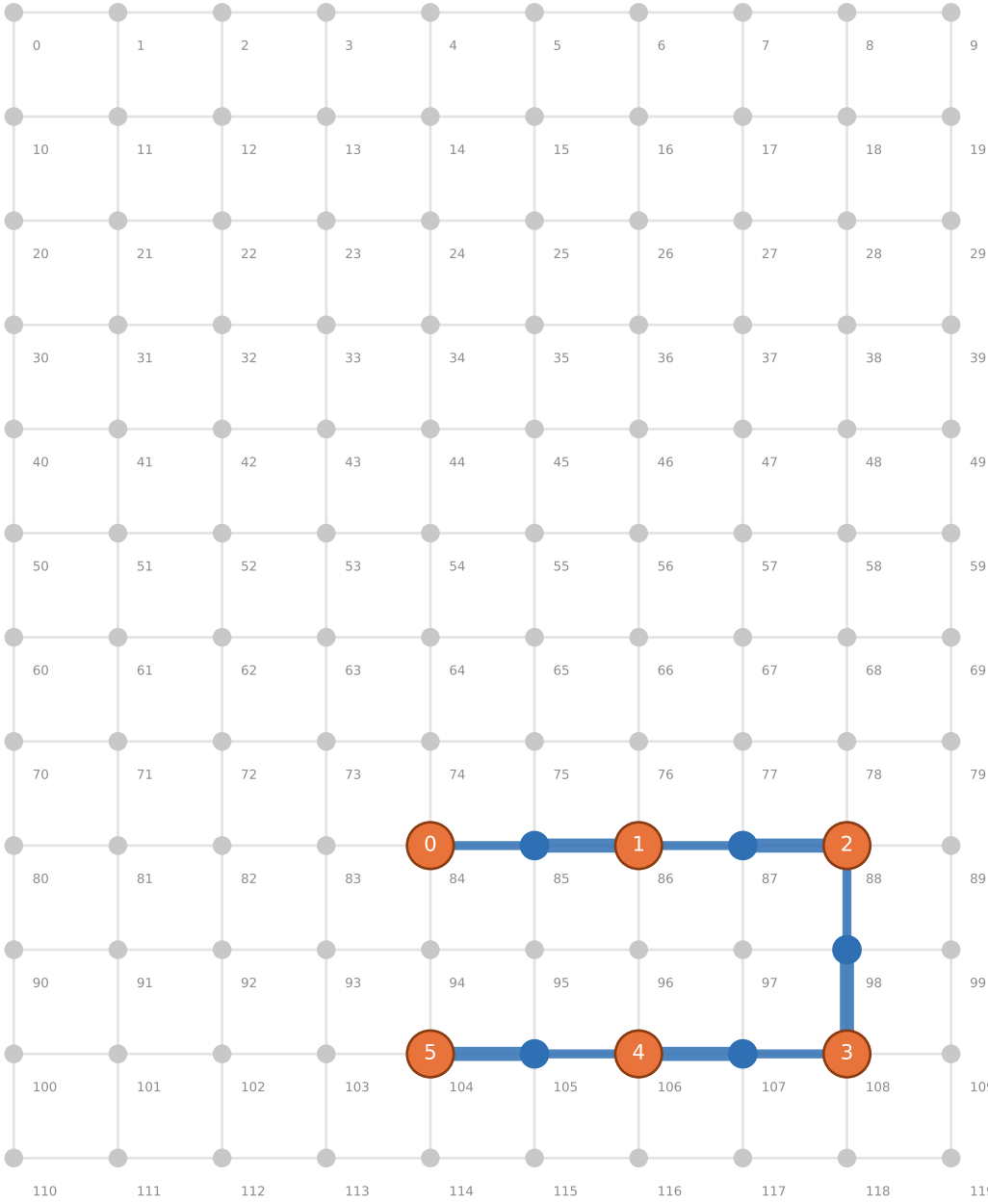
[Z6] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:25 sx:20 cz:10)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5   (6 logical qubits)
2q depth 10, total depth 35



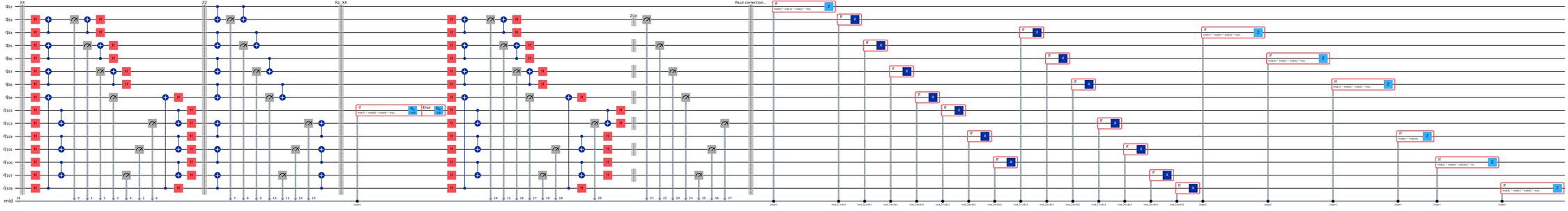

[Z6] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4 Z5
chain length 11, 5 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5


[Z6] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [28, 38, 48, 49, 59, 58] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 10, 2q gates 10 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5

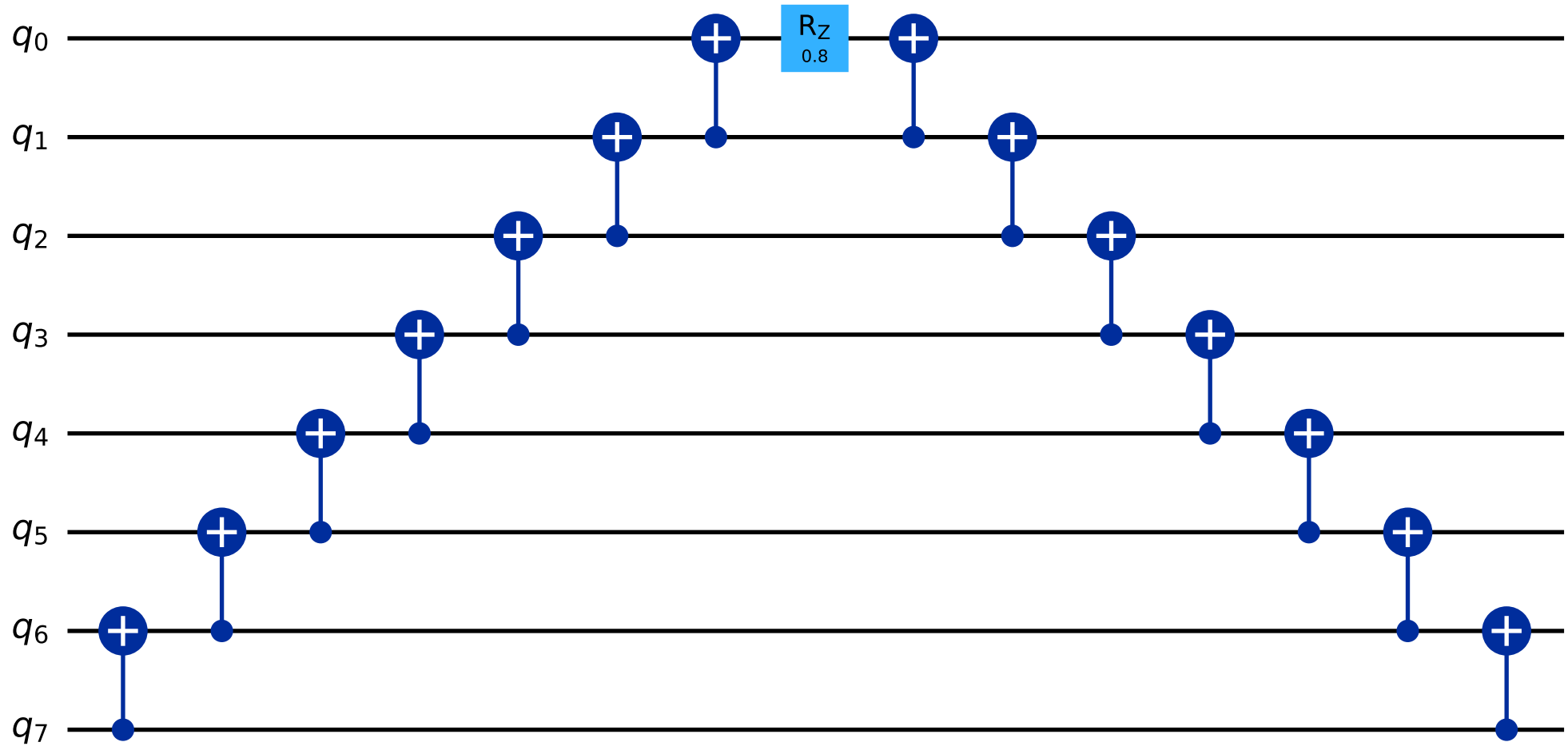
[Z6] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [84, 86, 88, 108, 106, 104]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 30



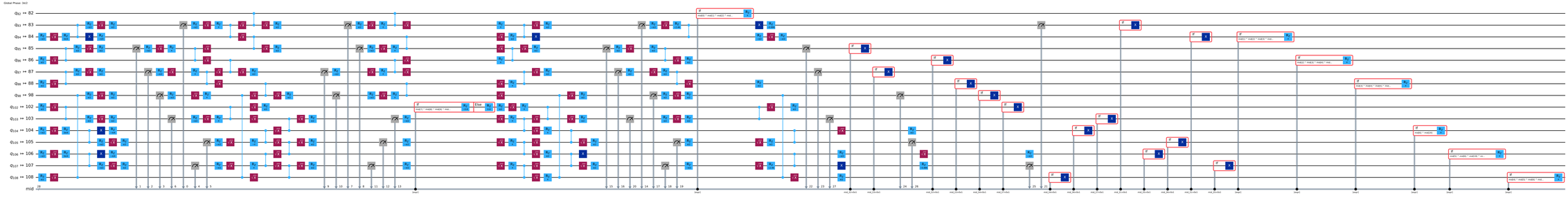
[Z8] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 (8 logical qubits)
2q depth 6, 28 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



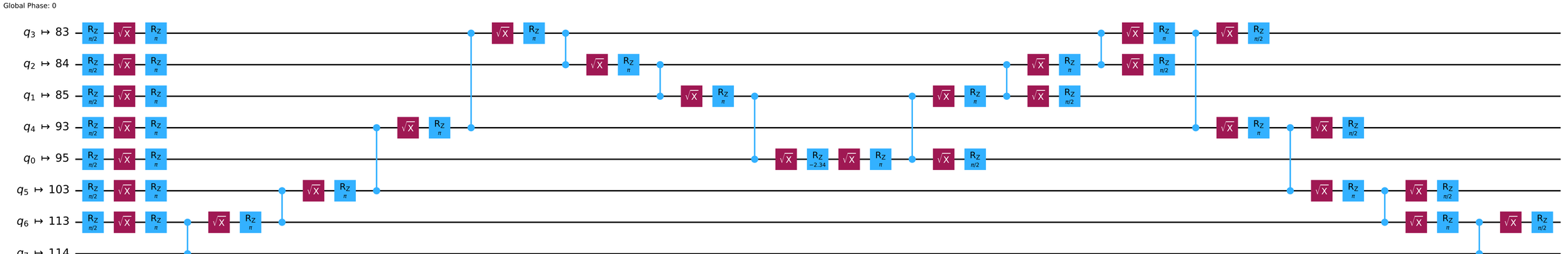
[Z8] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14



[28] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:104 sx:84 cz:42 measure:28 if_else:22 x:7)
exp(-i 9 P1 θ=0.4 · P = ZI Z1 Z2 Z3 Z4 Z5 Z6 Z7 (8 logical qubits)
2q depth 6, total depth 42

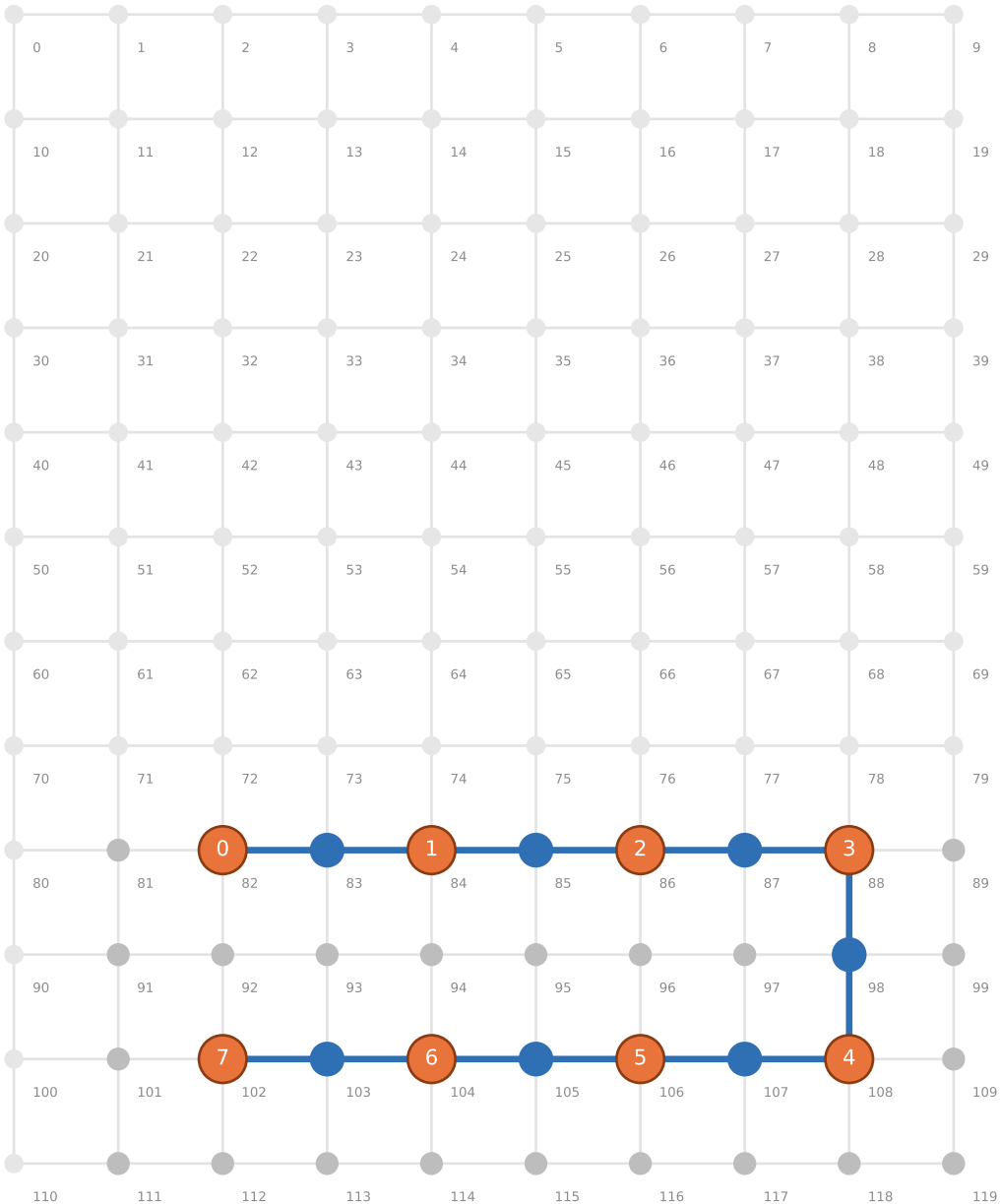

Global Phase: 3π/2
q82 → 82
q83 → 83
q84 → 84
q85 → 85
q86 → 86
q87 → 87
q88 → 88
q98 → 98
q102 → 102
q103 → 103
q104 → 104
q105 → 105
q106 → 106
q107 → 107
q108 → 108
mid
if
Else
meas[1] ^ meas[2] ^ meas[3] ^ mea...
meas[1] ^ meas[2] ^ meas[3] ^ mea...
meas[1] ^ meas[2] ^ meas[3] ^ mea...

[Z8] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:35 sx:28 cz:14)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7   (8 logical qubits)
2q depth 14, total depth 47

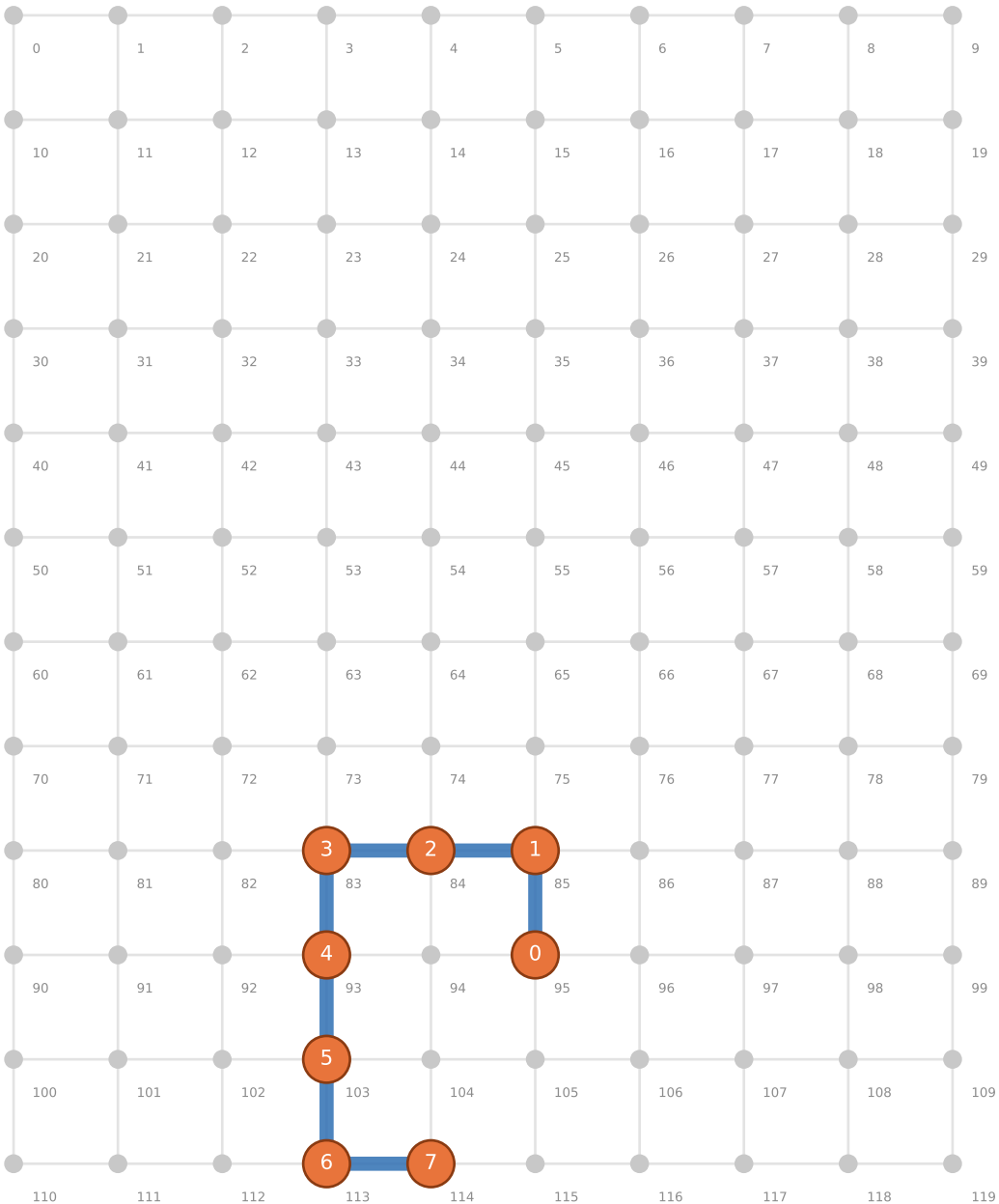


[Z8] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

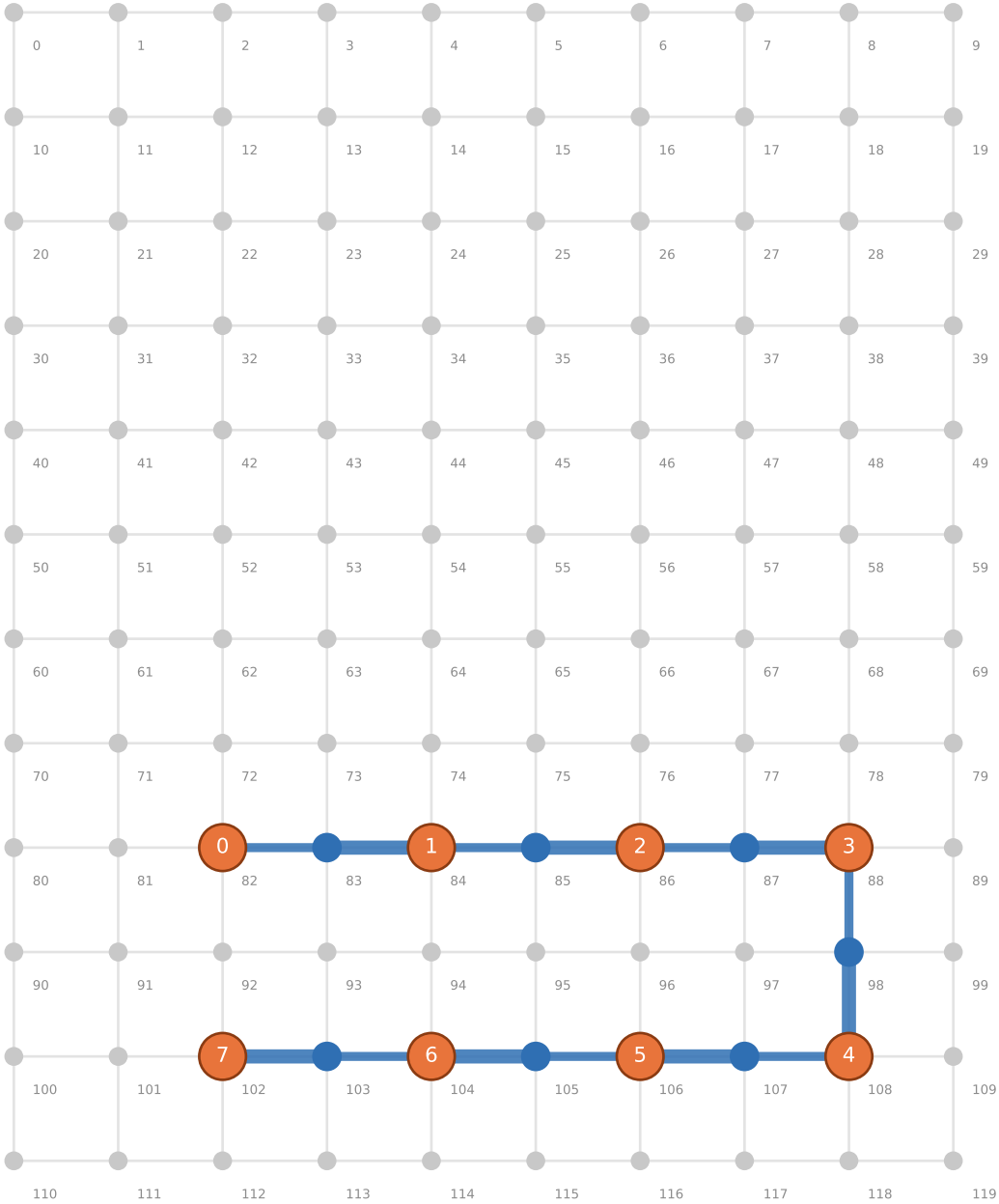
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7
chain length 15, 7 ancillas, 0 |0> fillers



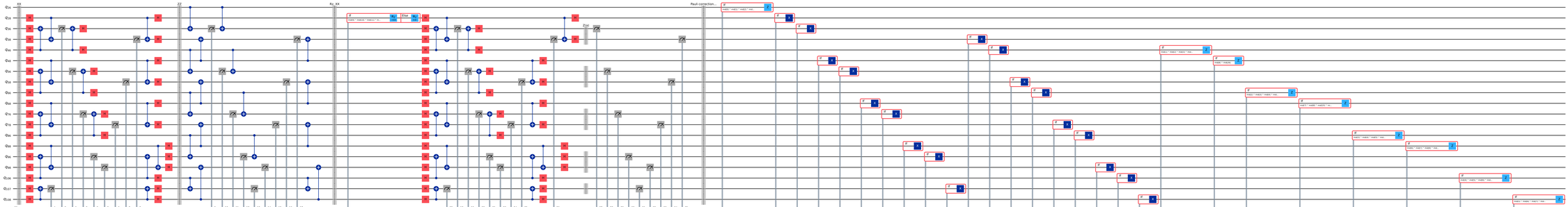
[Z8] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 83, 93, 103, 113, 114] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)



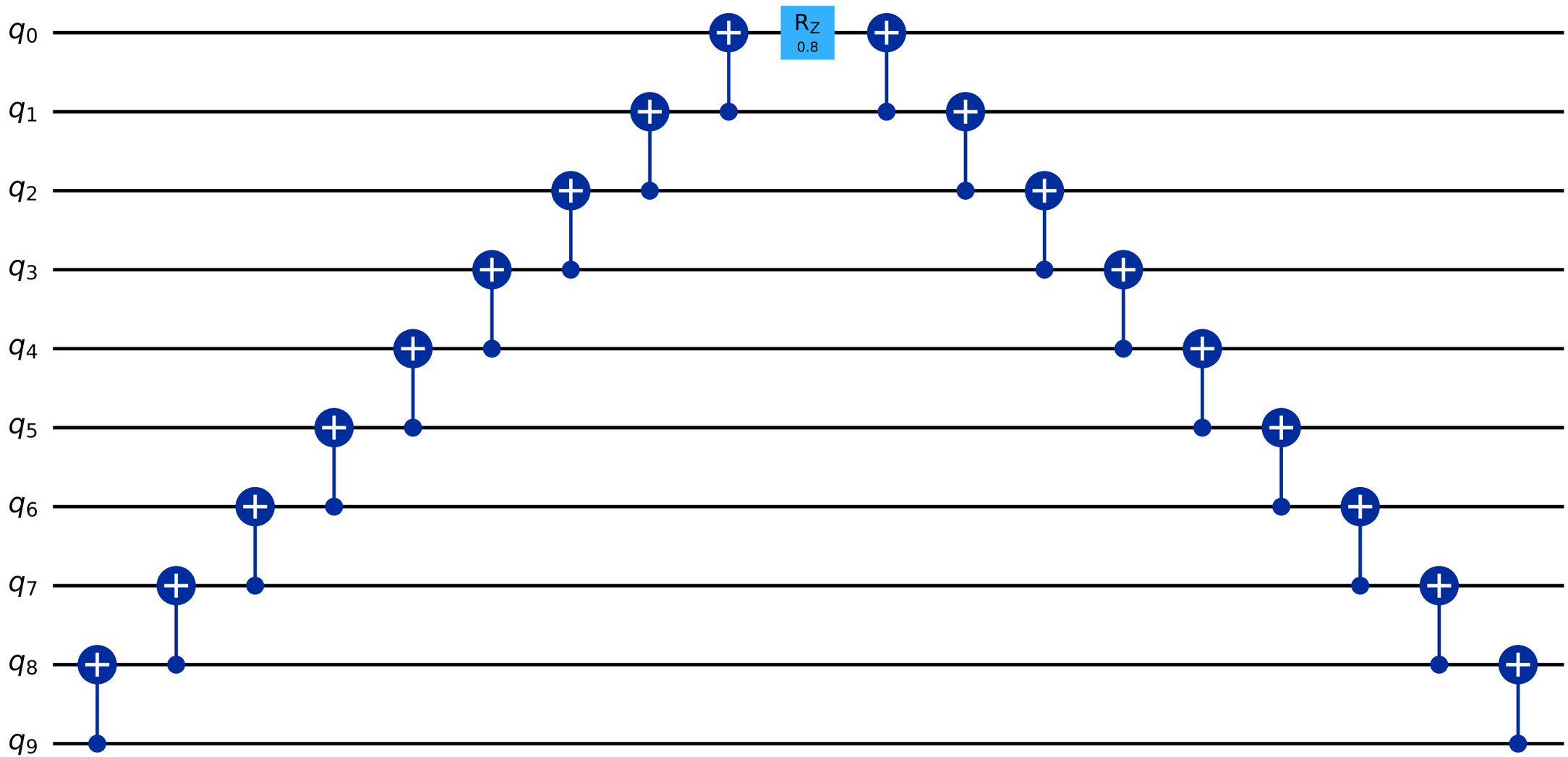
[Z8] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 42



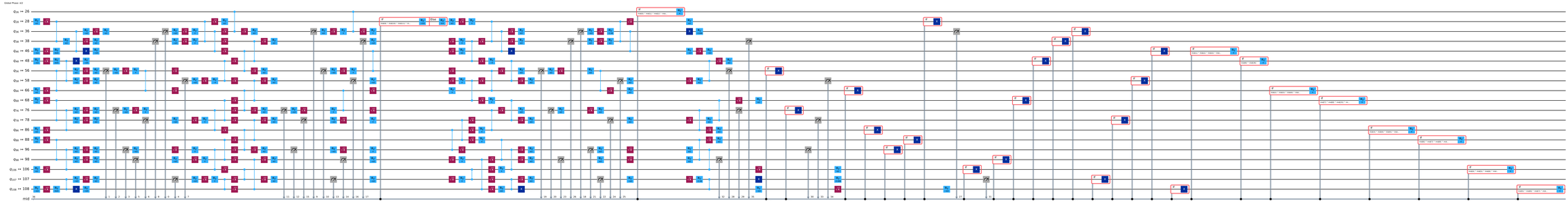
[[210] DYNAMIC constant-depth circuit, N8SQ I9 as a single-qubit gate] (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = ZI Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 6, 36 mid-circuit measurements, logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]



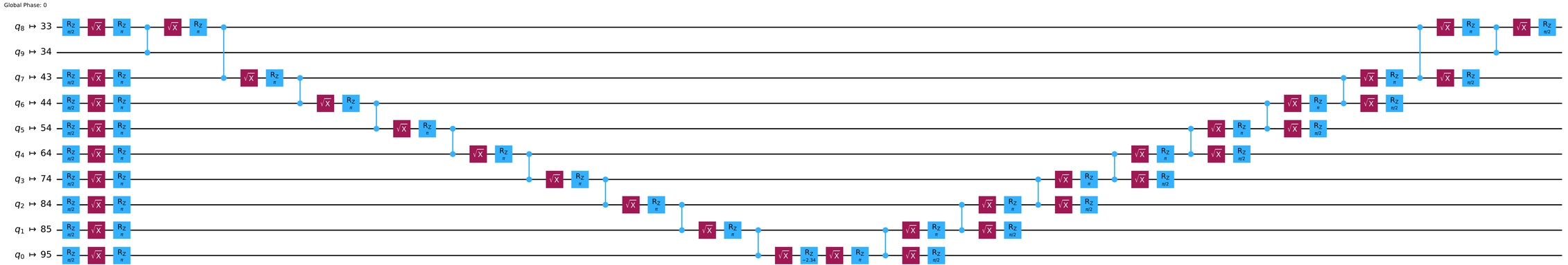
[Z10] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9   (10 logical qubits)
2q depth 18

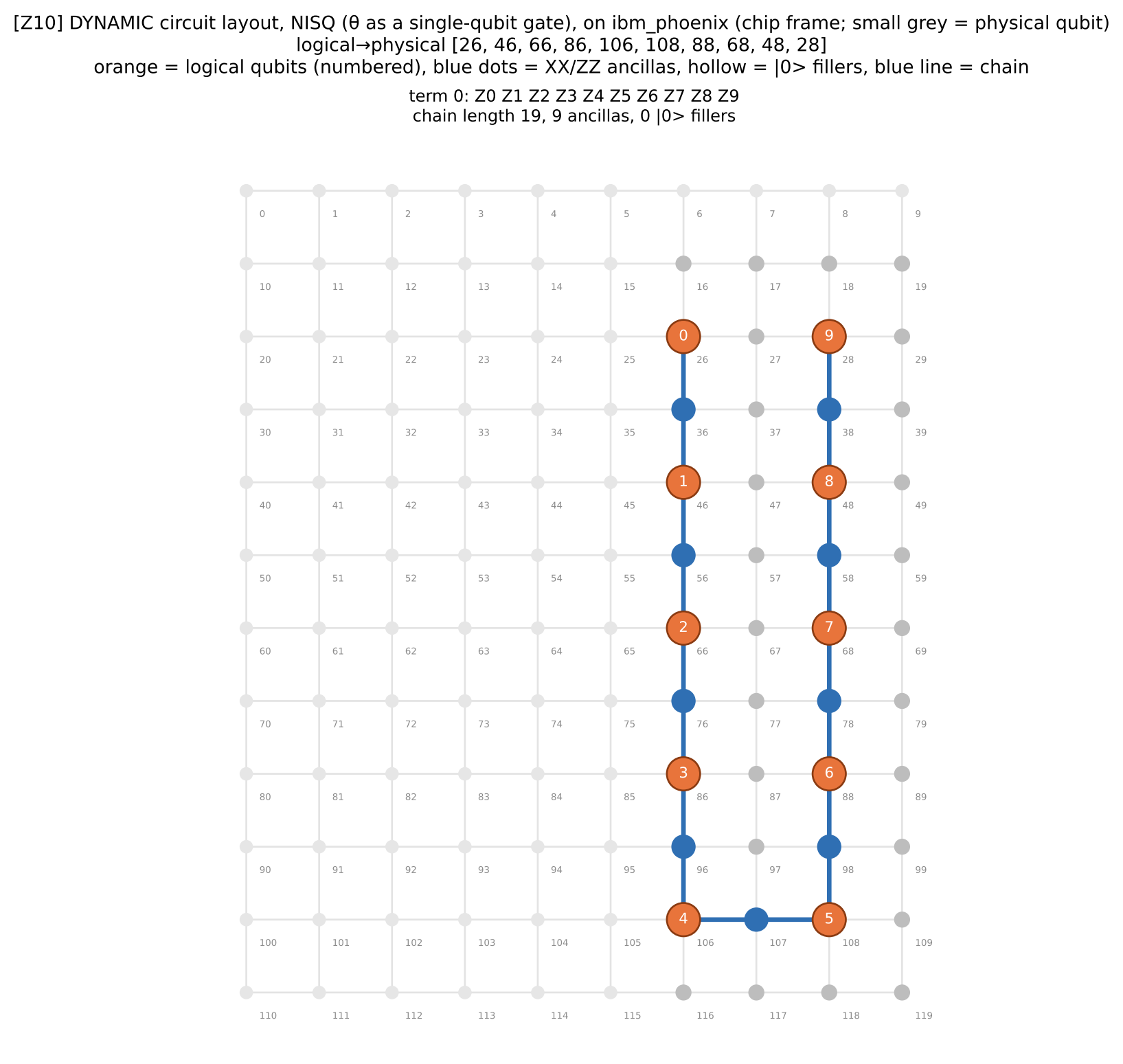


[210] DYNAMIC circuit, NISQ IR as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:130 sx:108 cz:54 measure:36 if_else:28 x:7)
expl:4 ff:1, fn:0, K: P = 20,21,22,23,24,25,26,27,28,29 · (10 logical qubits)
2q depth:6, total depth:44

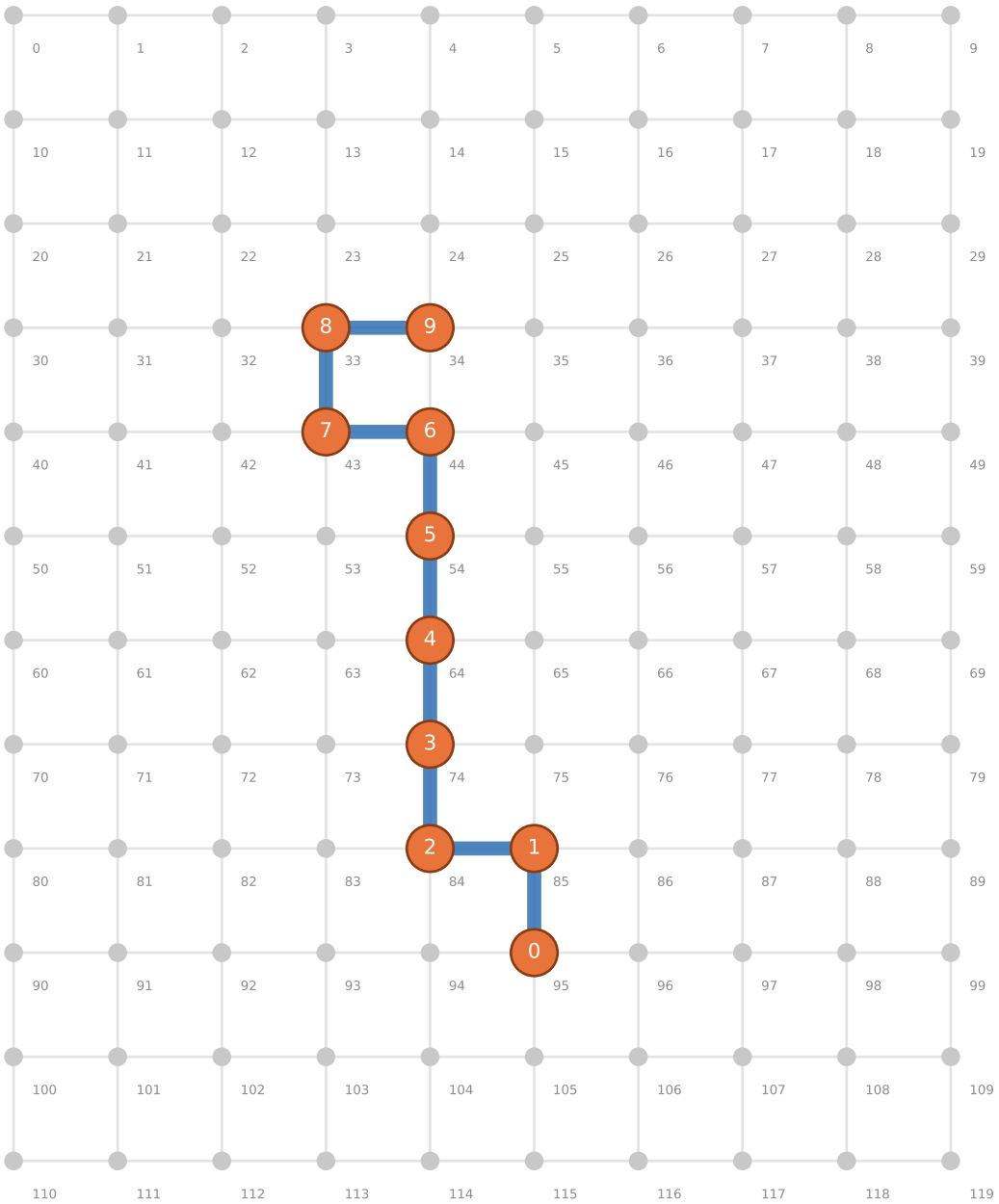

Global Phase: π/2
q_{26} ↦ 26
q_{28} ↦ 28
q_{36} ↦ 36
q_{38} ↦ 38
q_{46} ↦ 46
q_{48} ↦ 48
q_{56} ↦ 56
q_{58} ↦ 58
q_{66} ↦ 66
q_{68} ↦ 68
q_{76} ↦ 76
q_{78} ↦ 78
q_{86} ↦ 86
q_{88} ↦ 88
q_{96} ↦ 96
q_{98} ↦ 98
q_{106} ↦ 106
q_{107} ↦ 107
q_{108} ↦ 108
mid
meas: 36
if (meas == mask) state = ...
Reset
Reset
if (meas == mask) state = ...
if (meas == mask) state = ...
if (meas == mask) state = ...

[Z10] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:45 sx:36 cz:18)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 (10 logical qubits)
2q depth 18, total depth 59

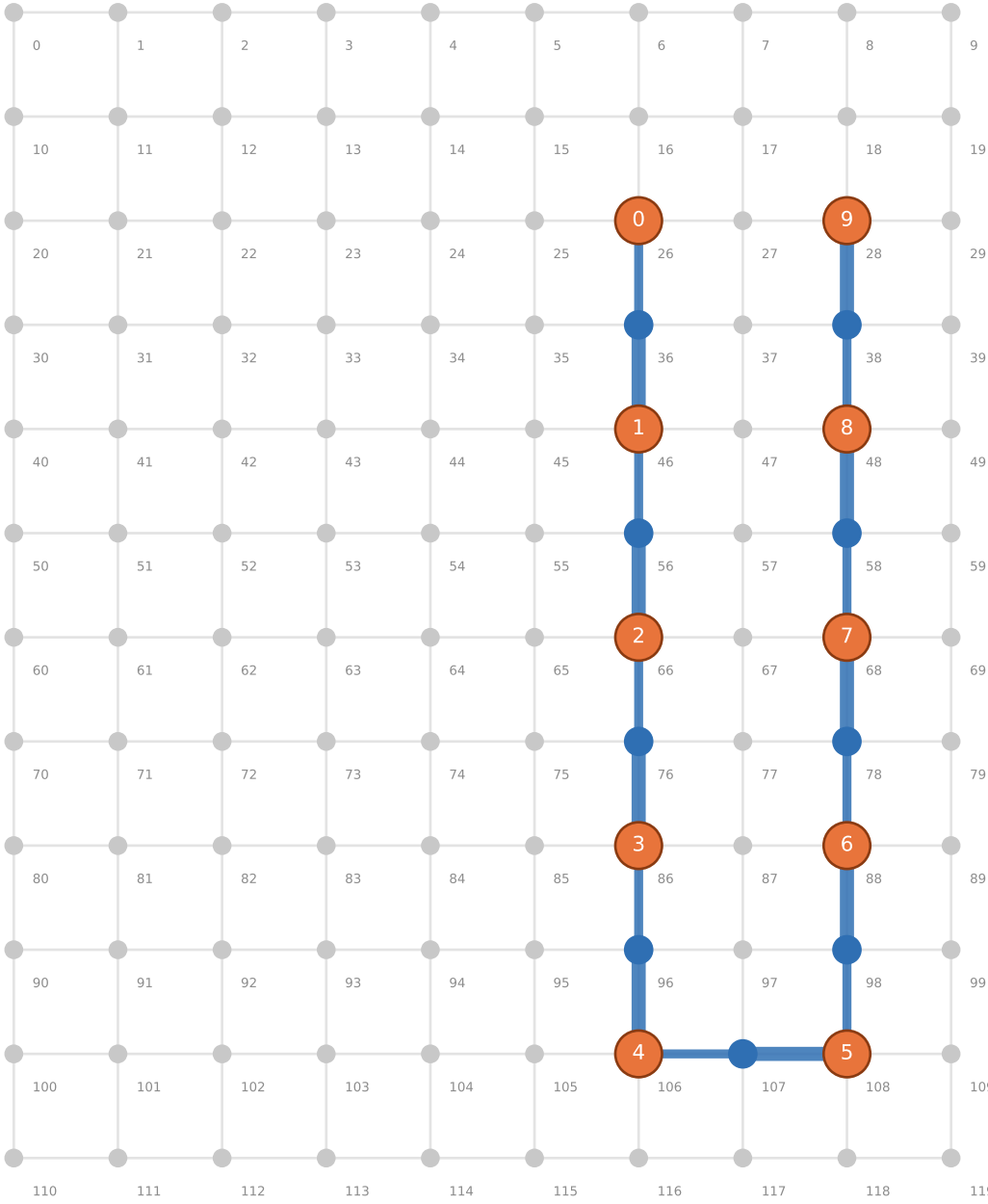



[Z10] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
chain length 19, 9 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
9
1
8
2
7
3
6
4
5

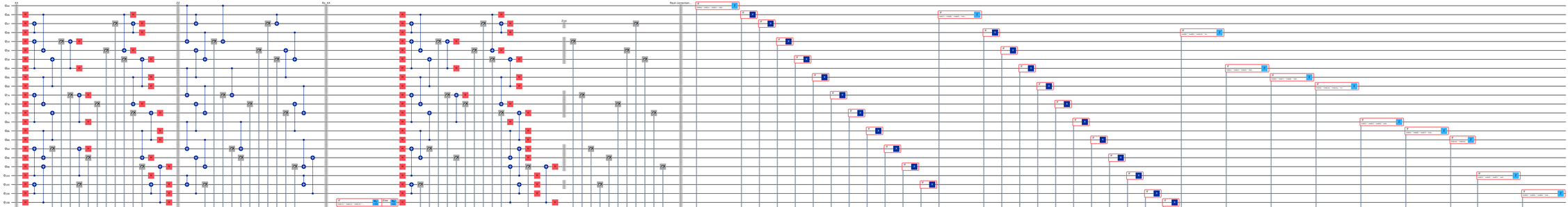
[Z10] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 18, 2q gates 18 (+0 vs. untranspiled from routing)



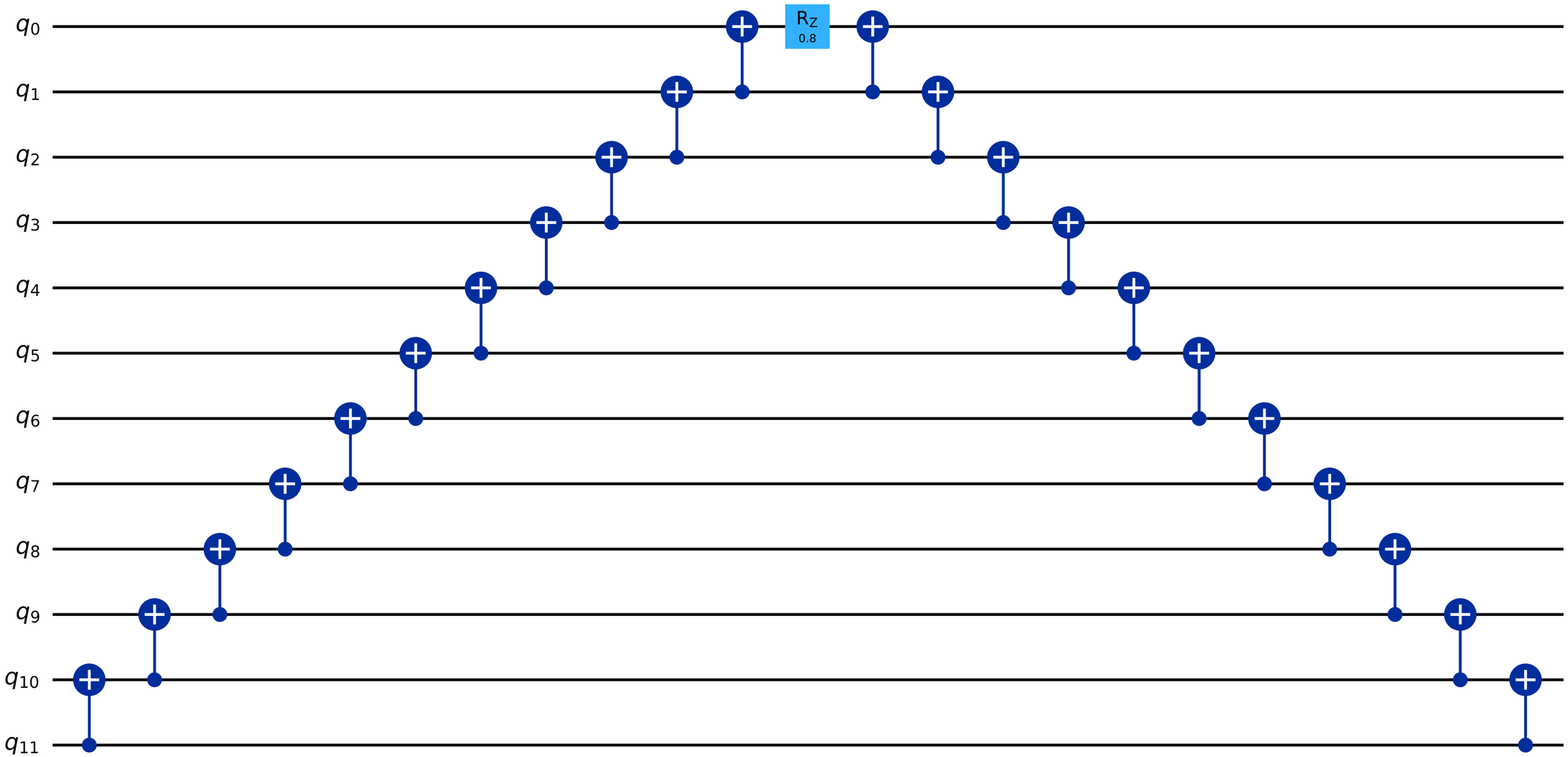
[Z10] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [26, 46, 66, 86, 106, 108, 88, 68, 48, 28]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 54



[Z12] DYNAMIC constant-depth circuit, NISQ IR as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
expr(9 Pi, 9+9.4 Pi + 20 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11) (11 logical qubits)
2q depth 6, 44 mid-circuit measurements, logical-physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]

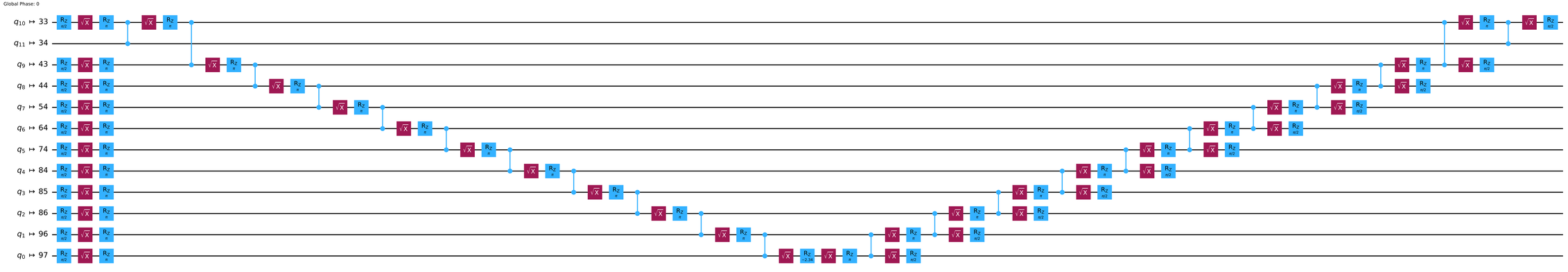


[Z12] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22



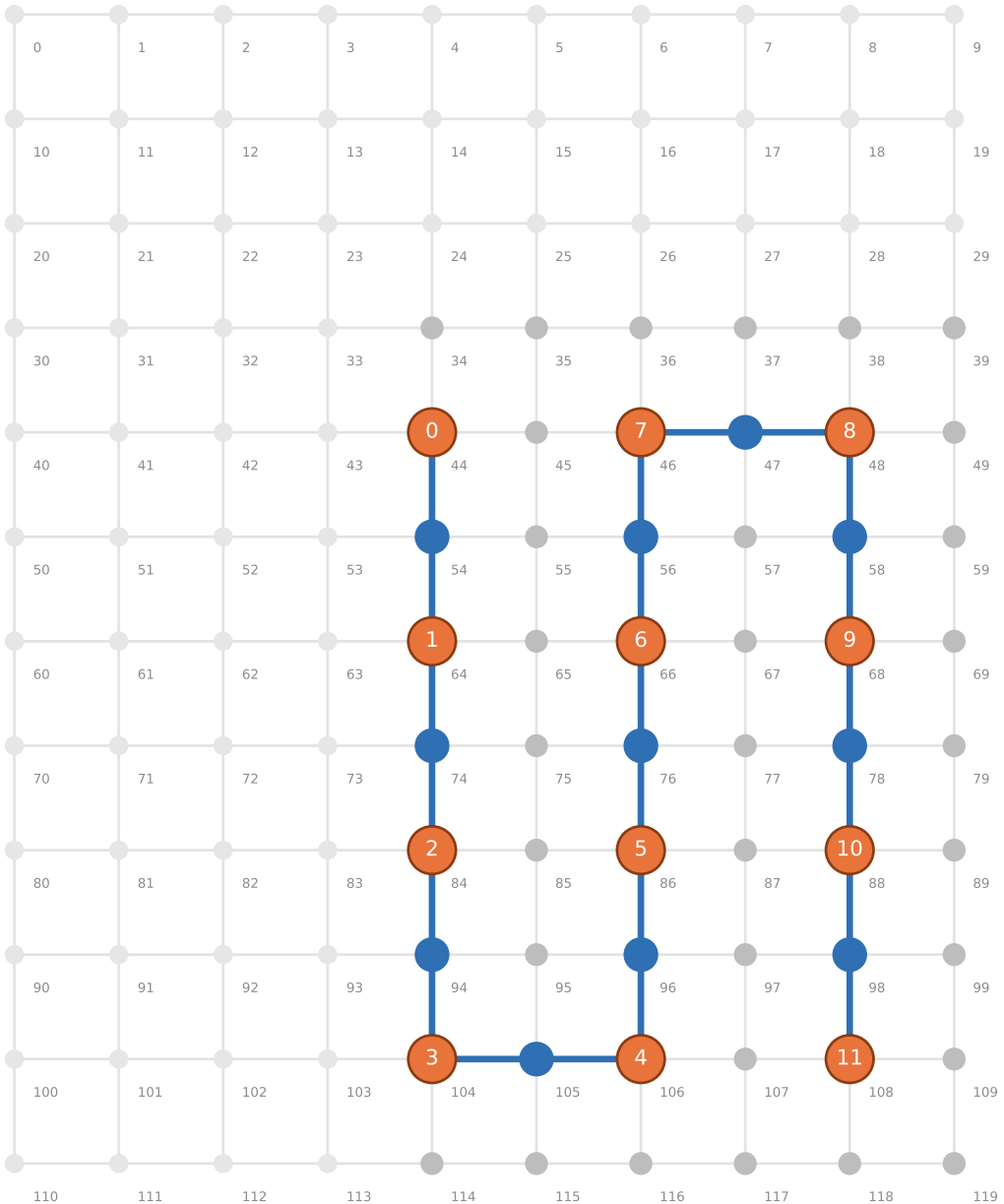
[2121] DYNAMIC circuit, W(Q) (# as a single qubit qubit), TRANSPILED for ibm_phoenix (ISA: n1 t68 sx 132 cz 66 measure 44 if_else 34 x 16)
expr: 4 fn, bits: 4 , # cx: 23 21 22 23 24 25 26 27 28 29 210 211 , (1) topical qubits)
31 depth 0, total depth 45

[Z12] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:55 sx:44 cz:22)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11   (12 logical qubits)
2q depth 22, total depth 71

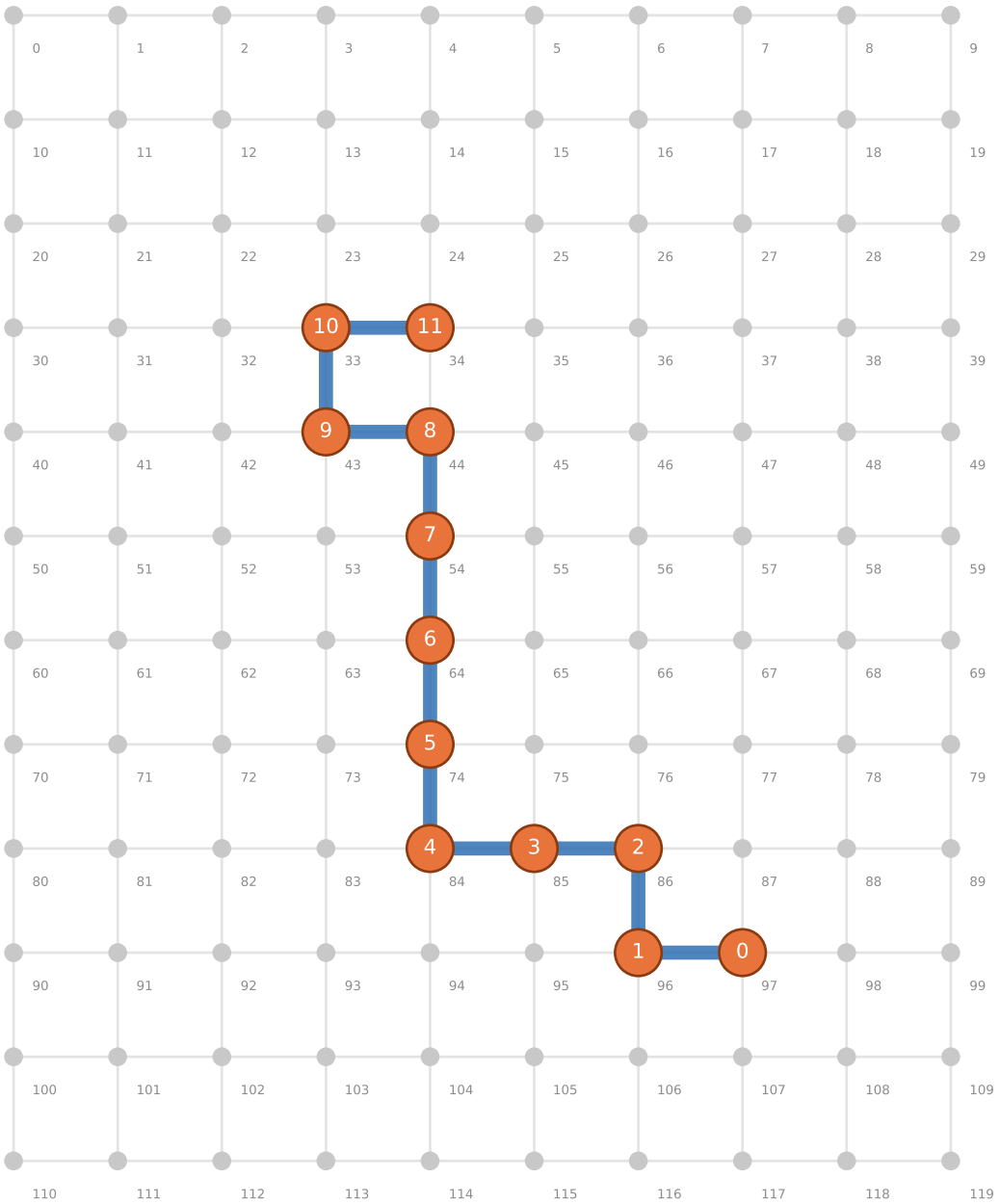


[Z12] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

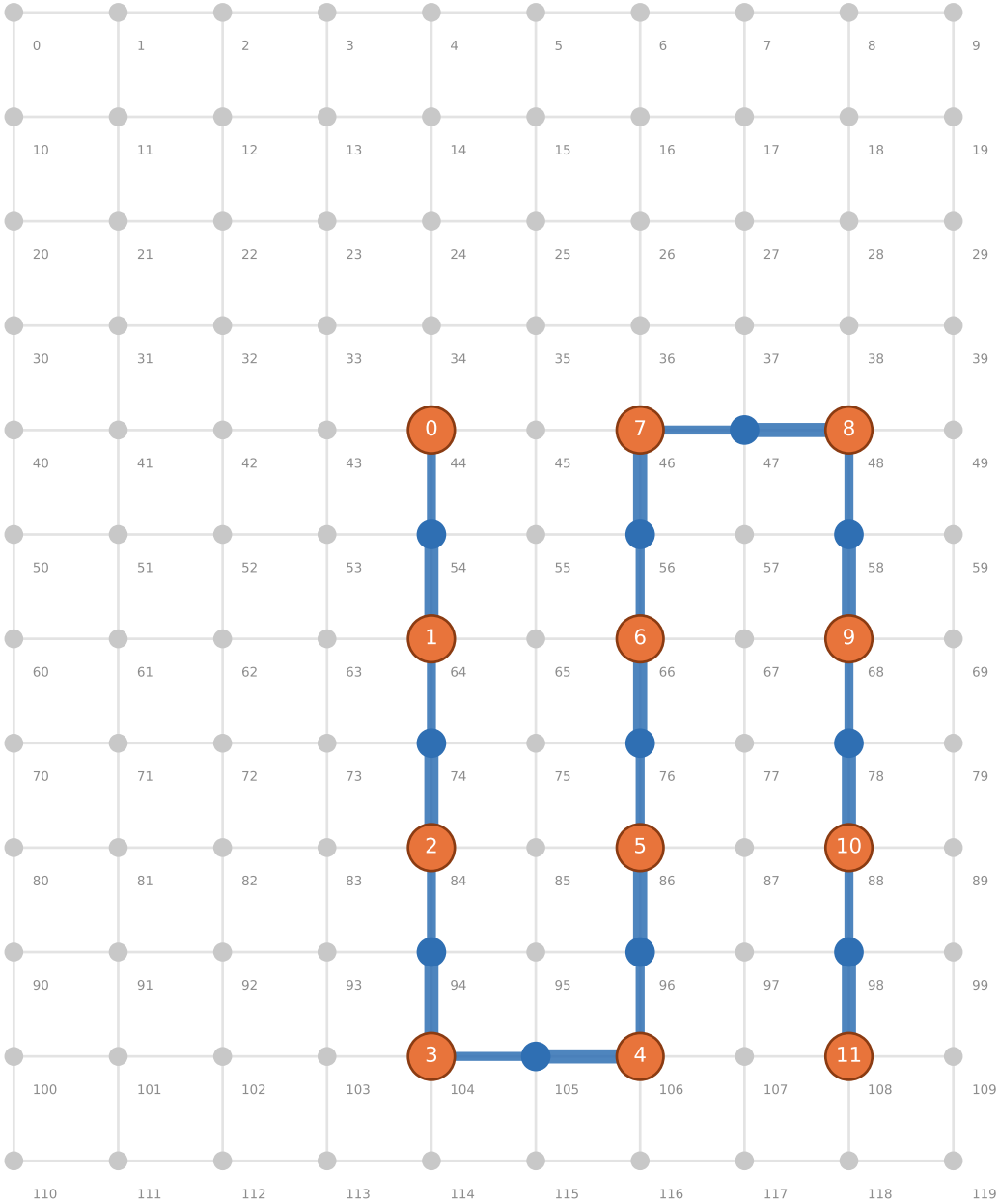
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11
chain length 23, 11 ancillas, 0 |0> fillers

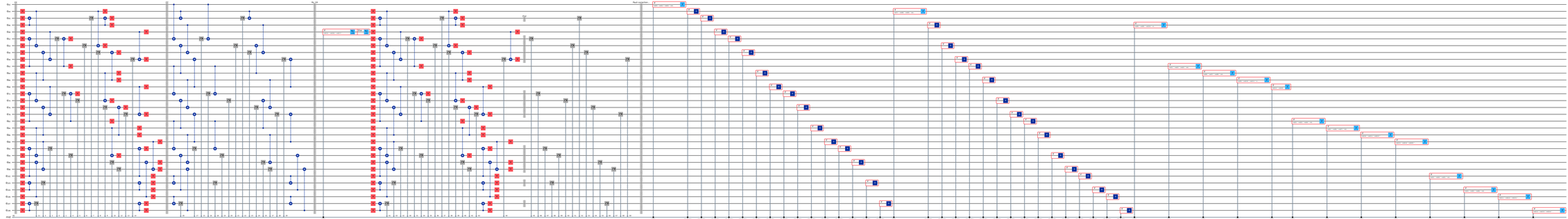


[Z12] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [97, 96, 86, 85, 84, 74, 64, 54, 44, 43, 33, 34] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 22, 2q gates 22 (+0 vs. untranspiled from routing)

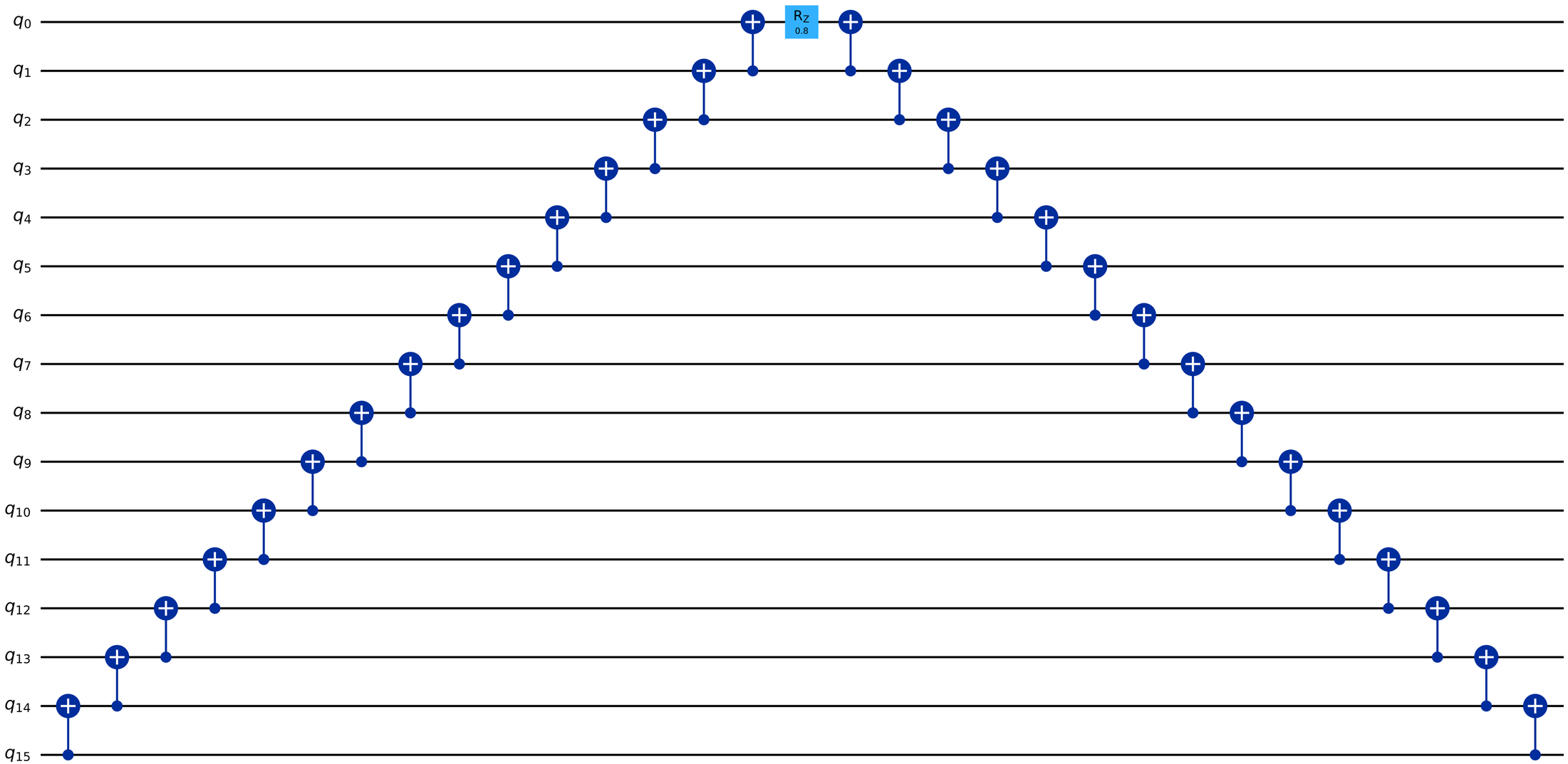


[Z12] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [44, 64, 84, 104, 106, 86, 66, 46, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 66

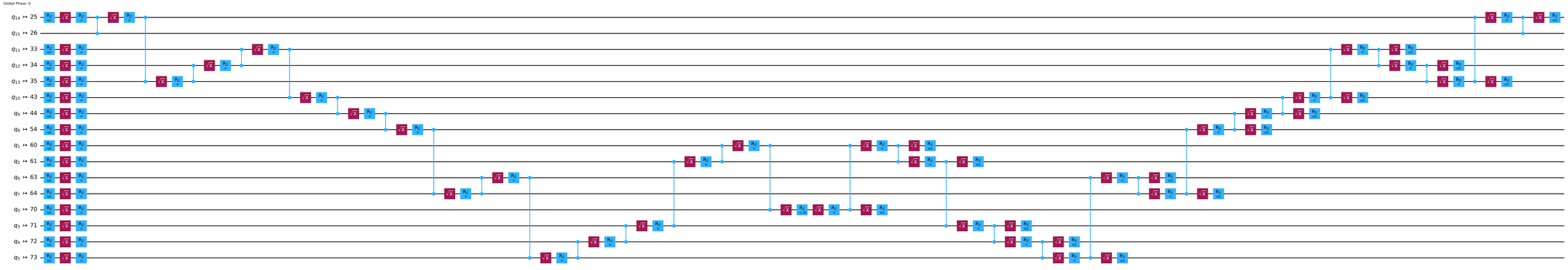



[2/16] DYNAMIC (unitary-depth circuit, MAQ 8) on a single qubit qdot (corresponds physical qubits of Ge4_physics)
[3] depth 8, 64 total gates, 1 qubits, 64 1q gates, 0 2q gates, 0 3q gates, 0 4q gates
[4] depth 8, 64 mid-circuit measurements, log2-of-vol-pol (n1, n2, n3, n4): (64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64, 64)
Rz_23
Pauli-Gaadmin

[Z16] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15   (16 logical qubits)
2q depth 30

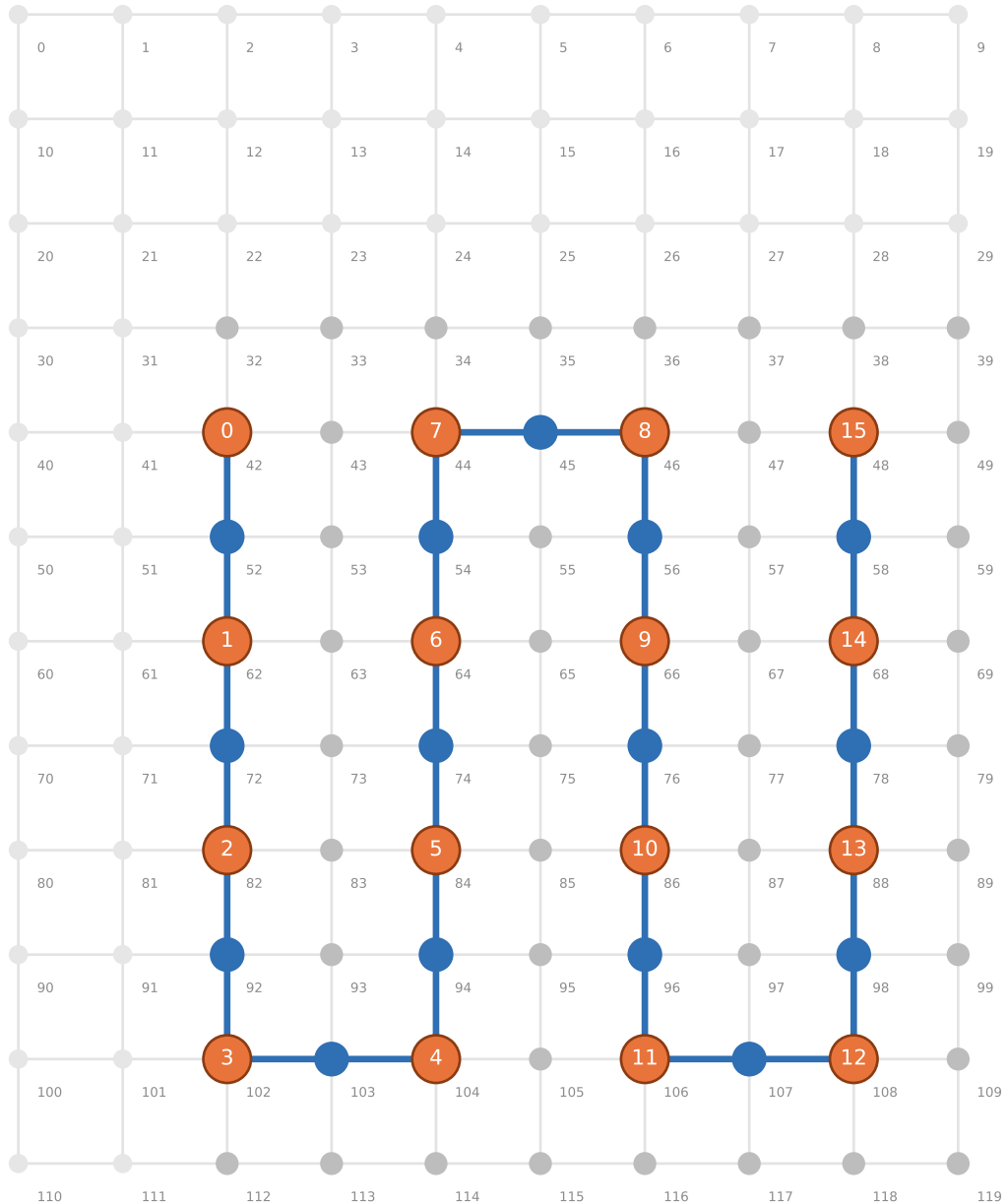


[Z16] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:75 sx:60 cz:30)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 (16 logical qubits)
2q depth 30, total depth 95

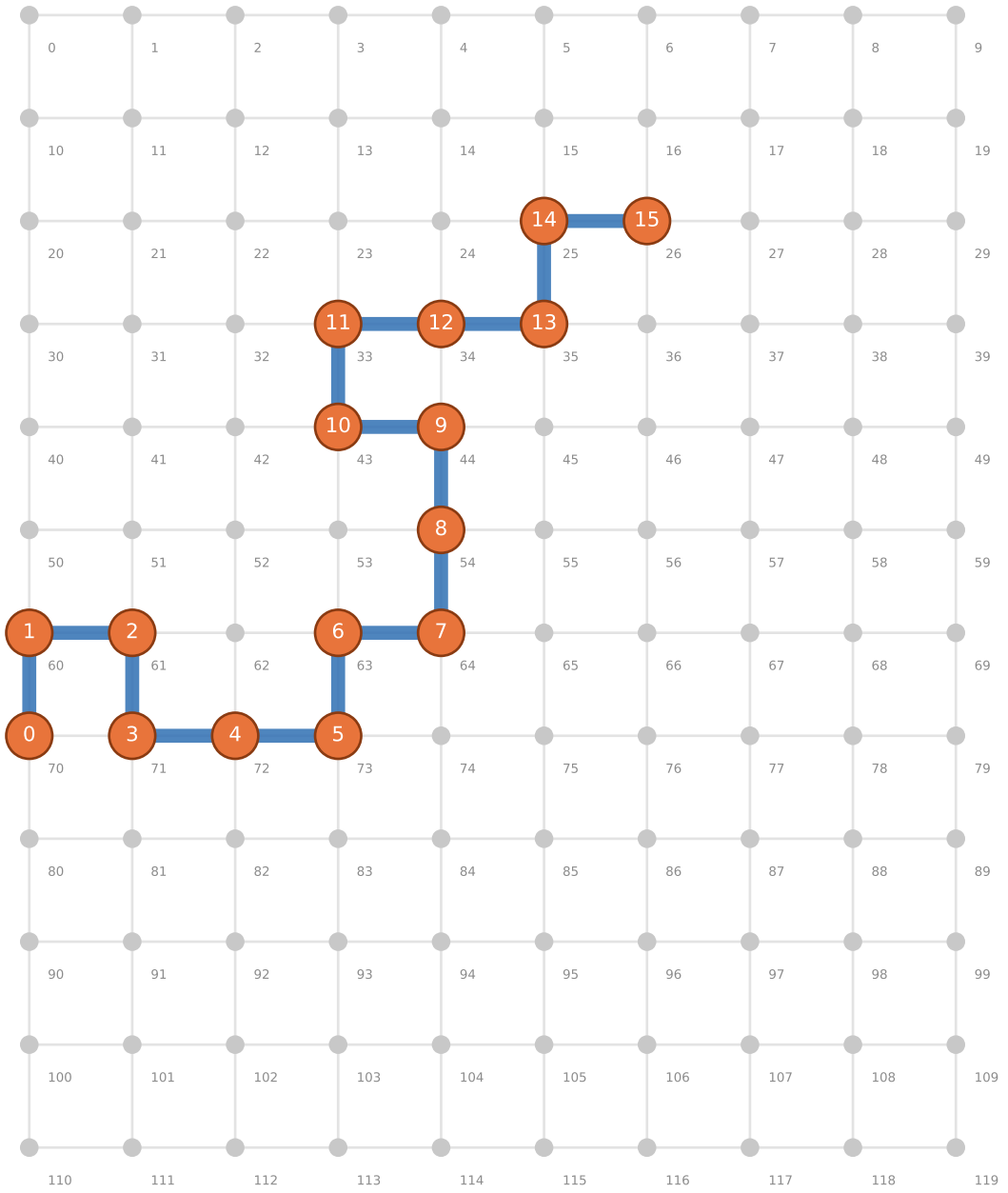


[Z16] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

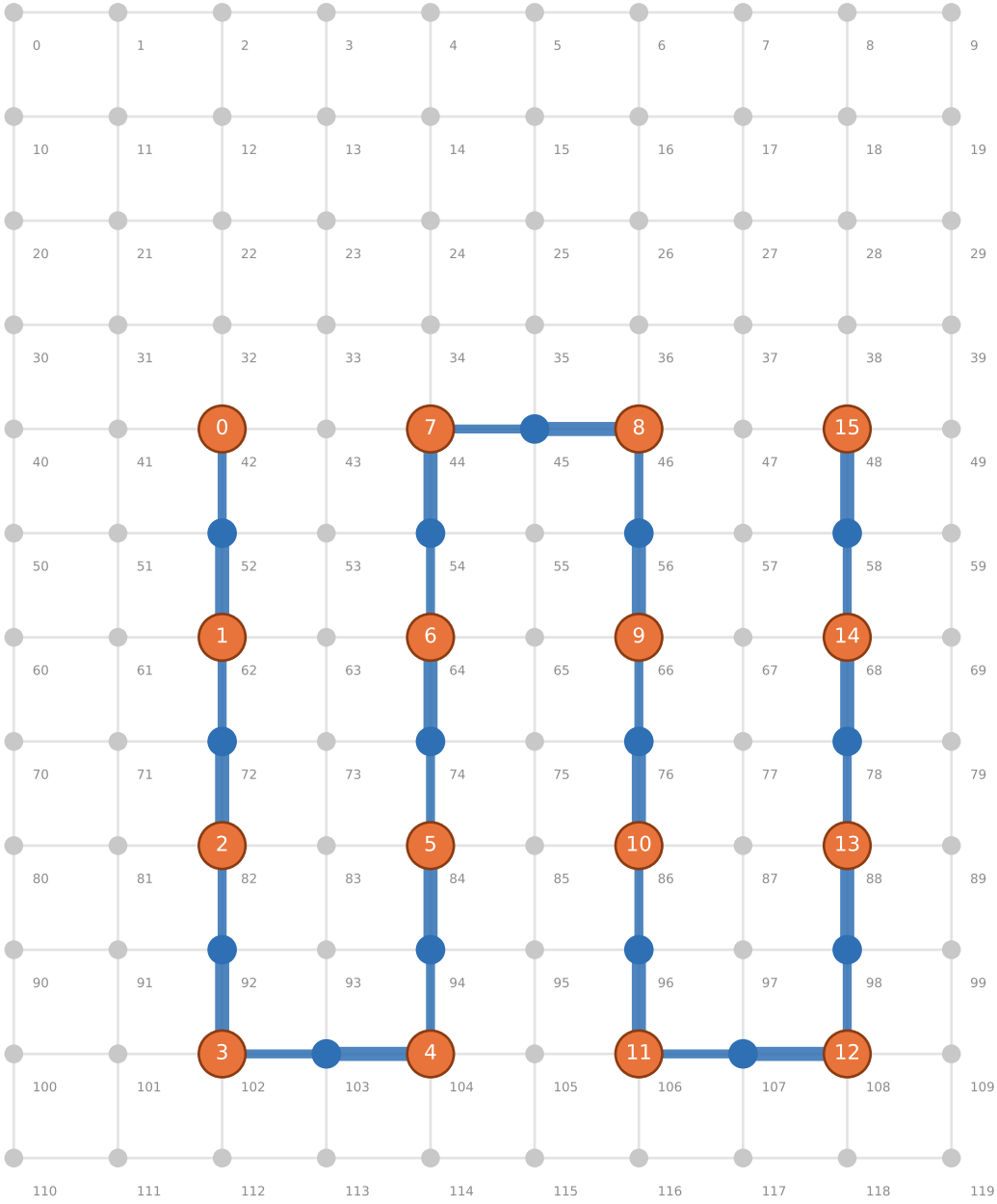
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15
chain length 31, 15 ancillas, 0 |0> fillers



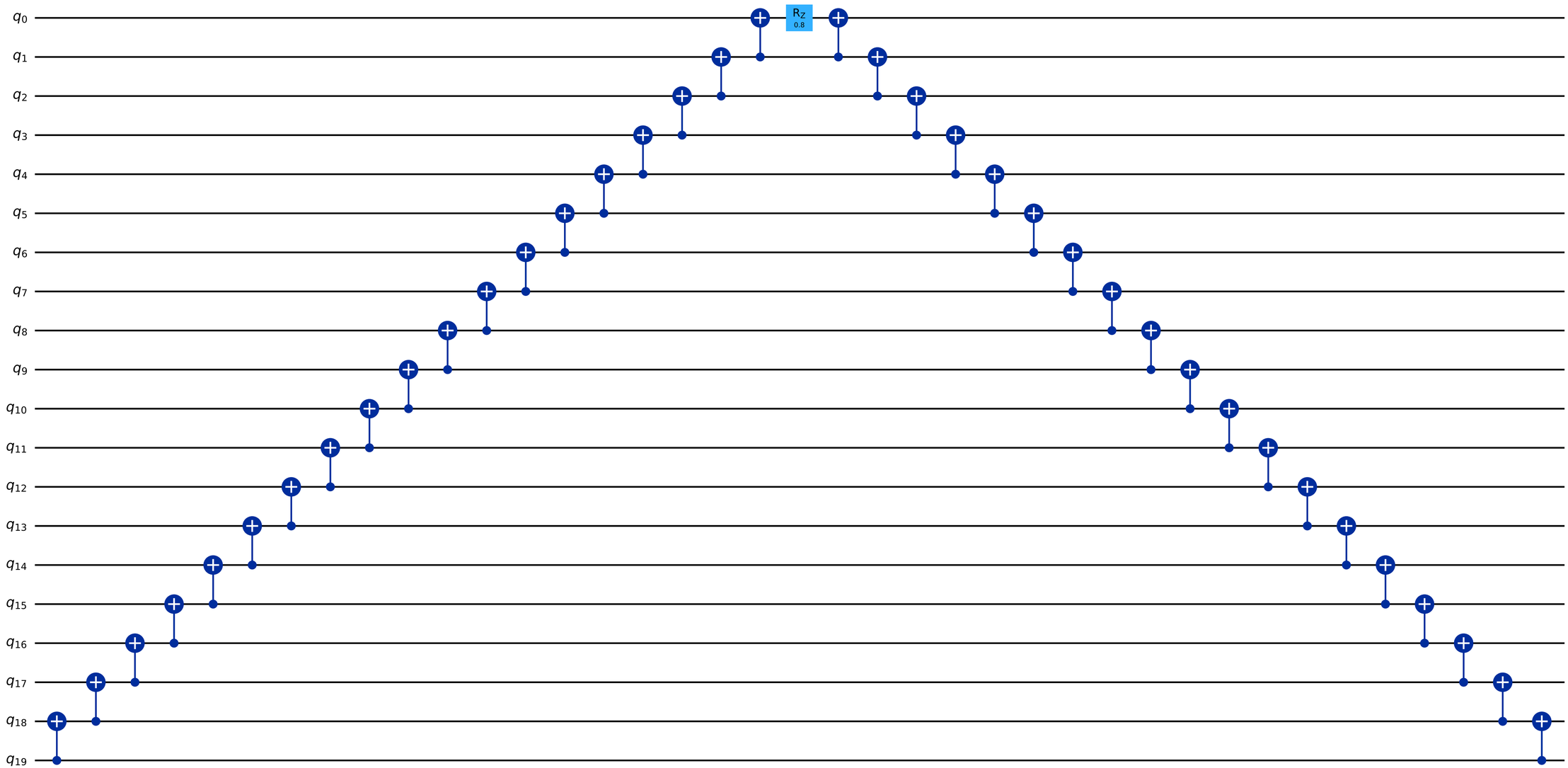
[Z16] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [70, 60, 61, 71, 72, 73, 63, 64, 54, 44, 43, 33, 34, 35, 25, 26] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 30, 2q gates 30 (+0 vs. untranspiled from routing)



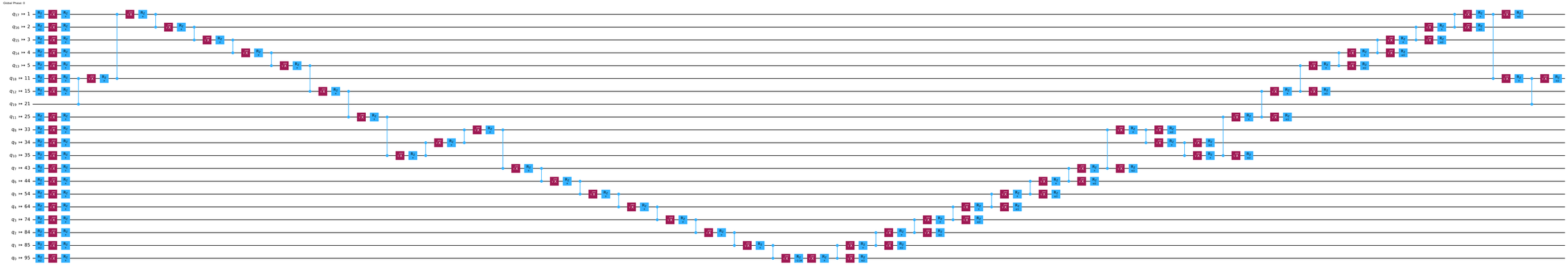
[Z16] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [42, 62, 82, 102, 104, 84, 64, 44, 46, 66, 86, 106, 108, 88, 68, 48]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 90



[Z20] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19   (20 logical qubits)
2q depth 38

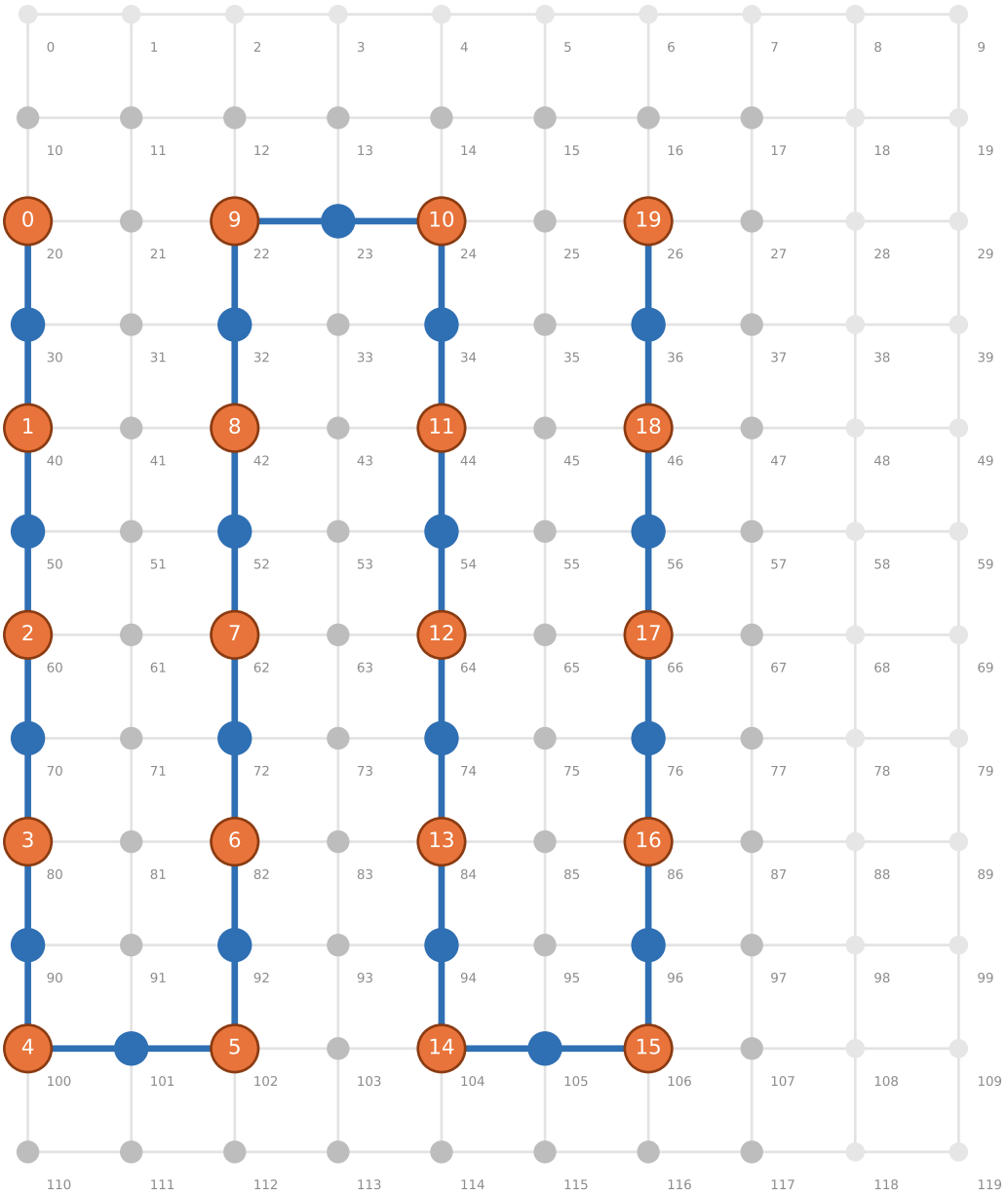


[Z20] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:95 sx:76 cz:38)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 (20 logical qubits)
2q depth 38, total depth 119

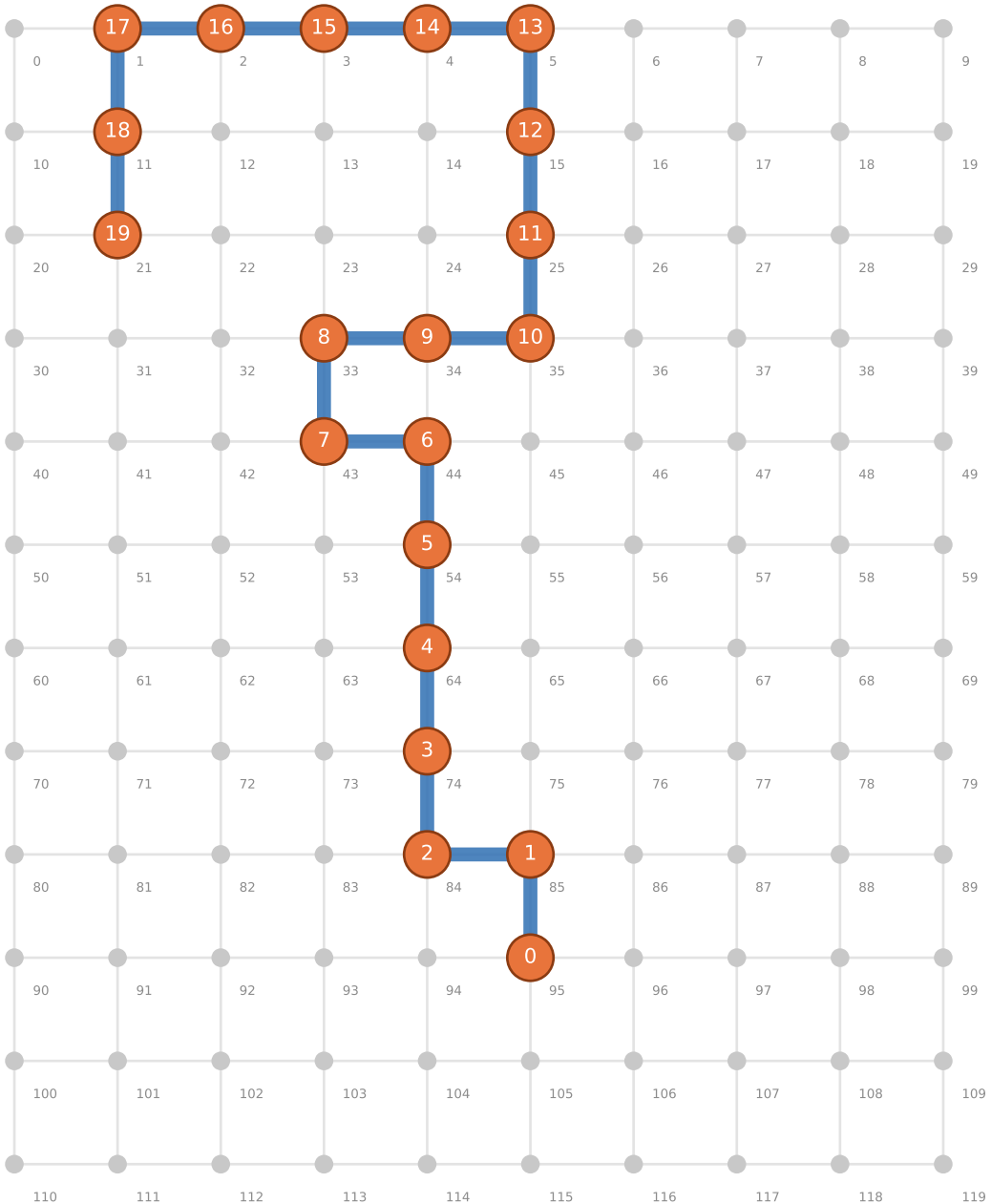


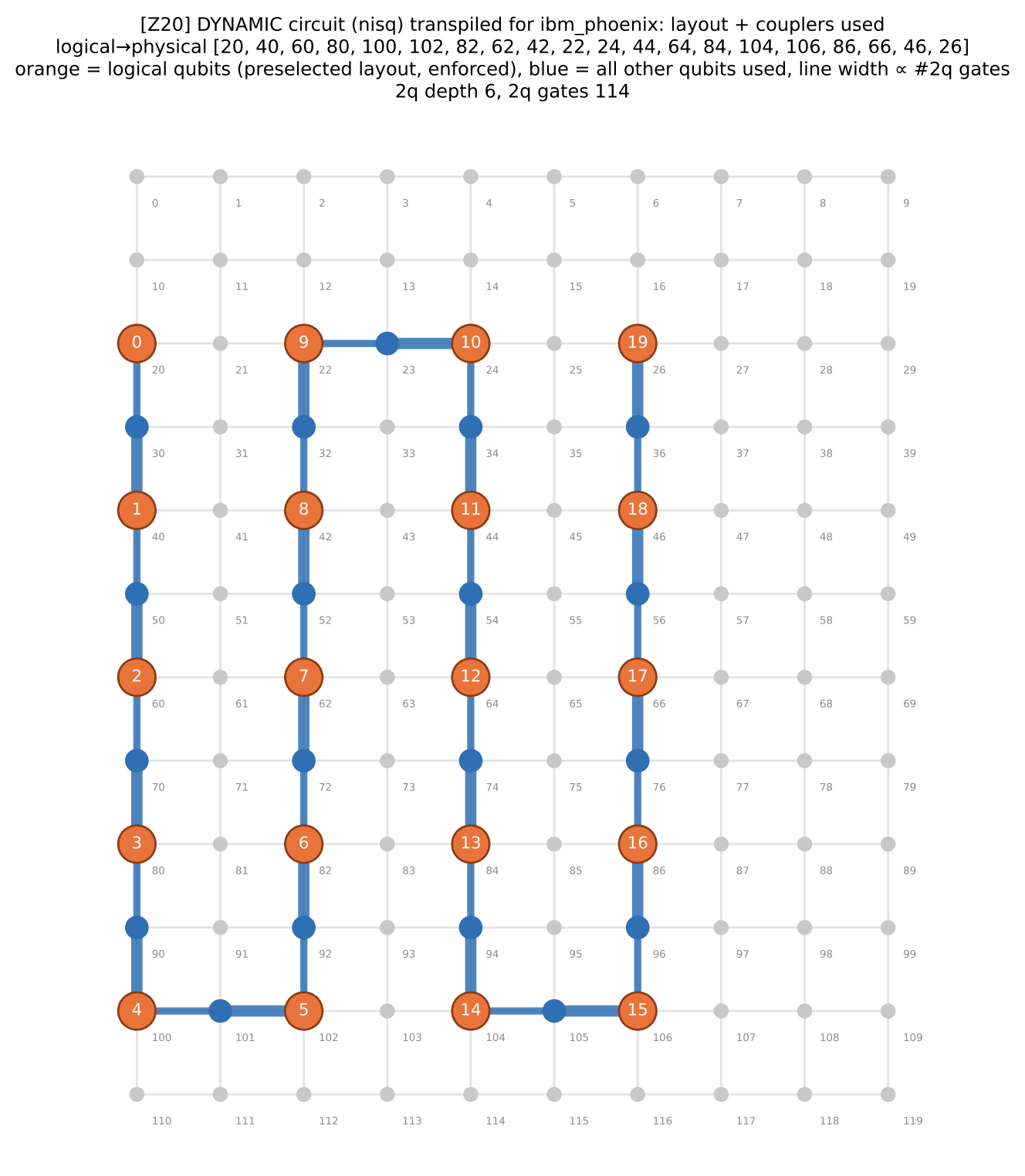
[Z20] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19
chain length 39, 19 ancillas, 0 |0> fillers

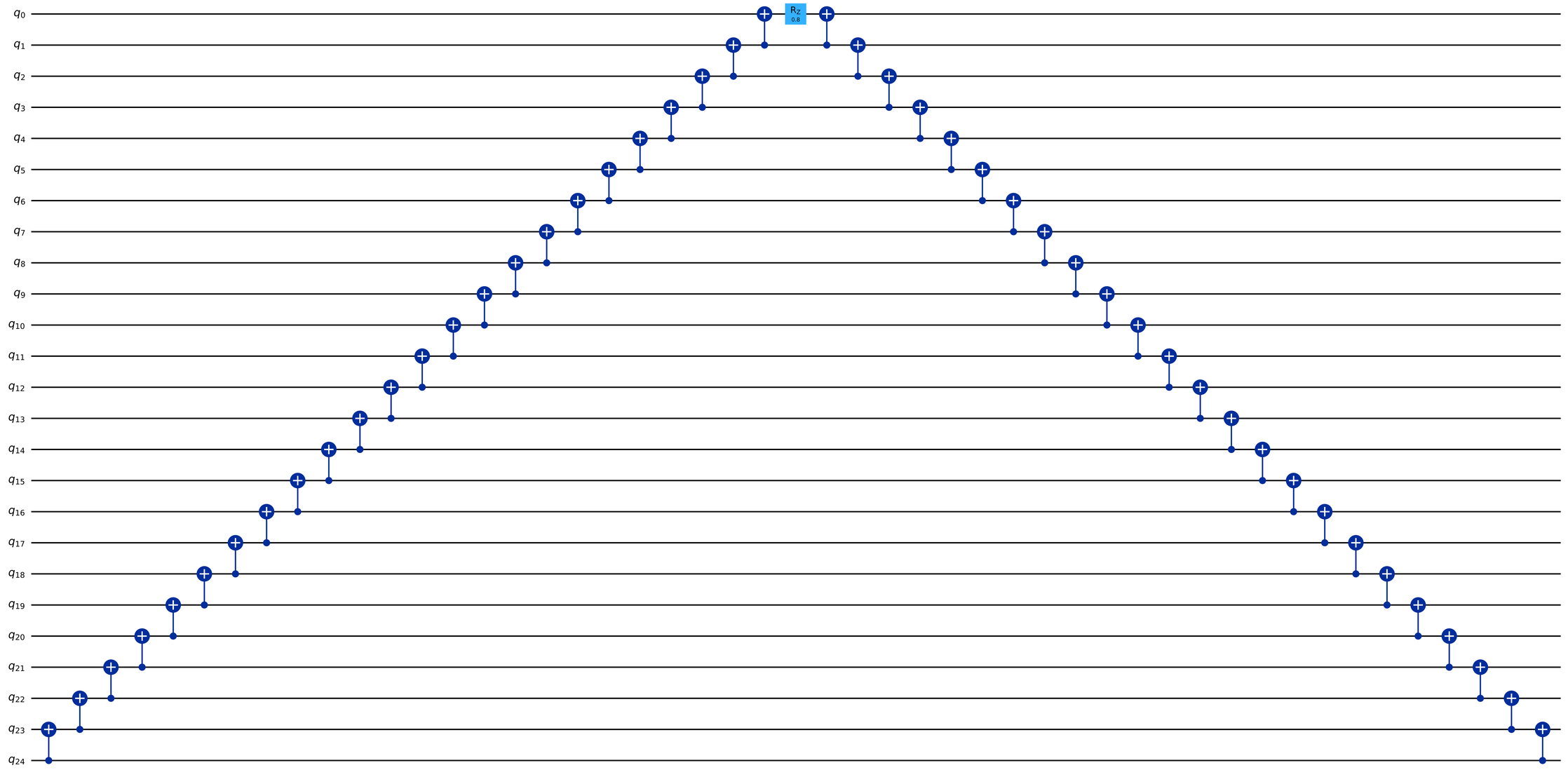


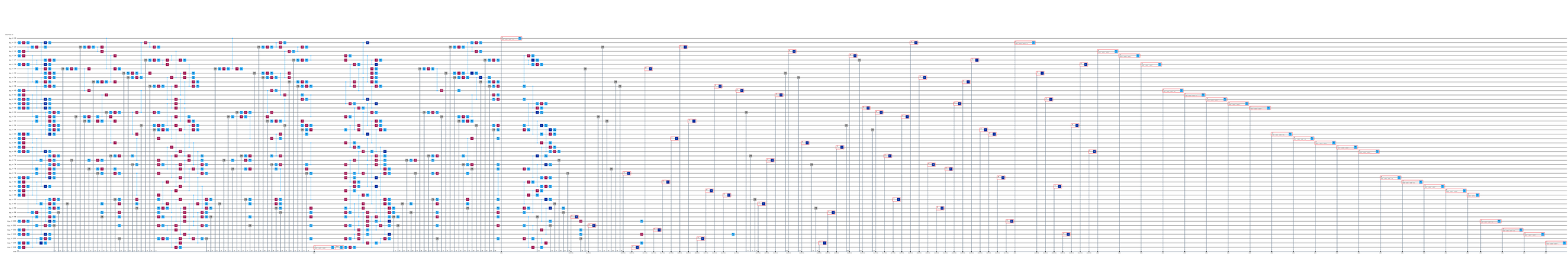
[Z20] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 84, 74, 64, 54, 44, 43, 33, 34, 35, 25, 15, 5, 4, 3, 2, 1, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 38, 2q gates 38 (+0 vs. untranspiled from routing)



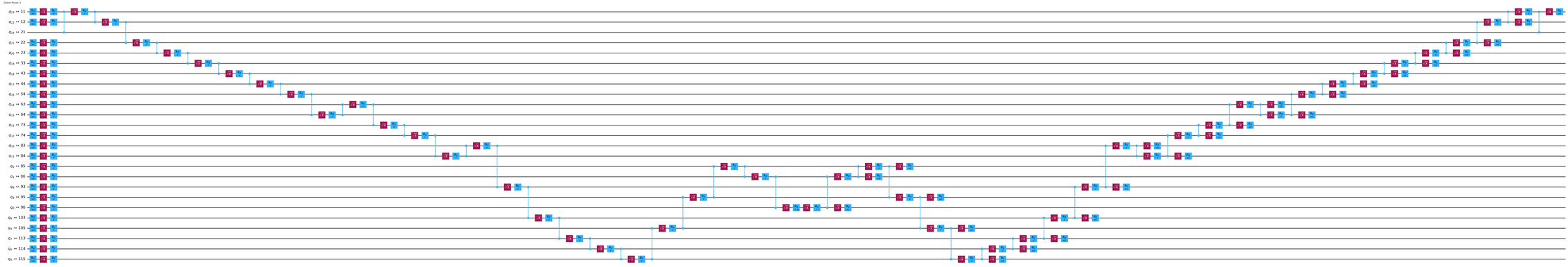

[Z20] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 114
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
0 9 10 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
1 8 11 18
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
2 7 12 17
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
3 6 13 16
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
4 5 14 15
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119

[Z25] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24   (25 logical qubits)
2q depth 48



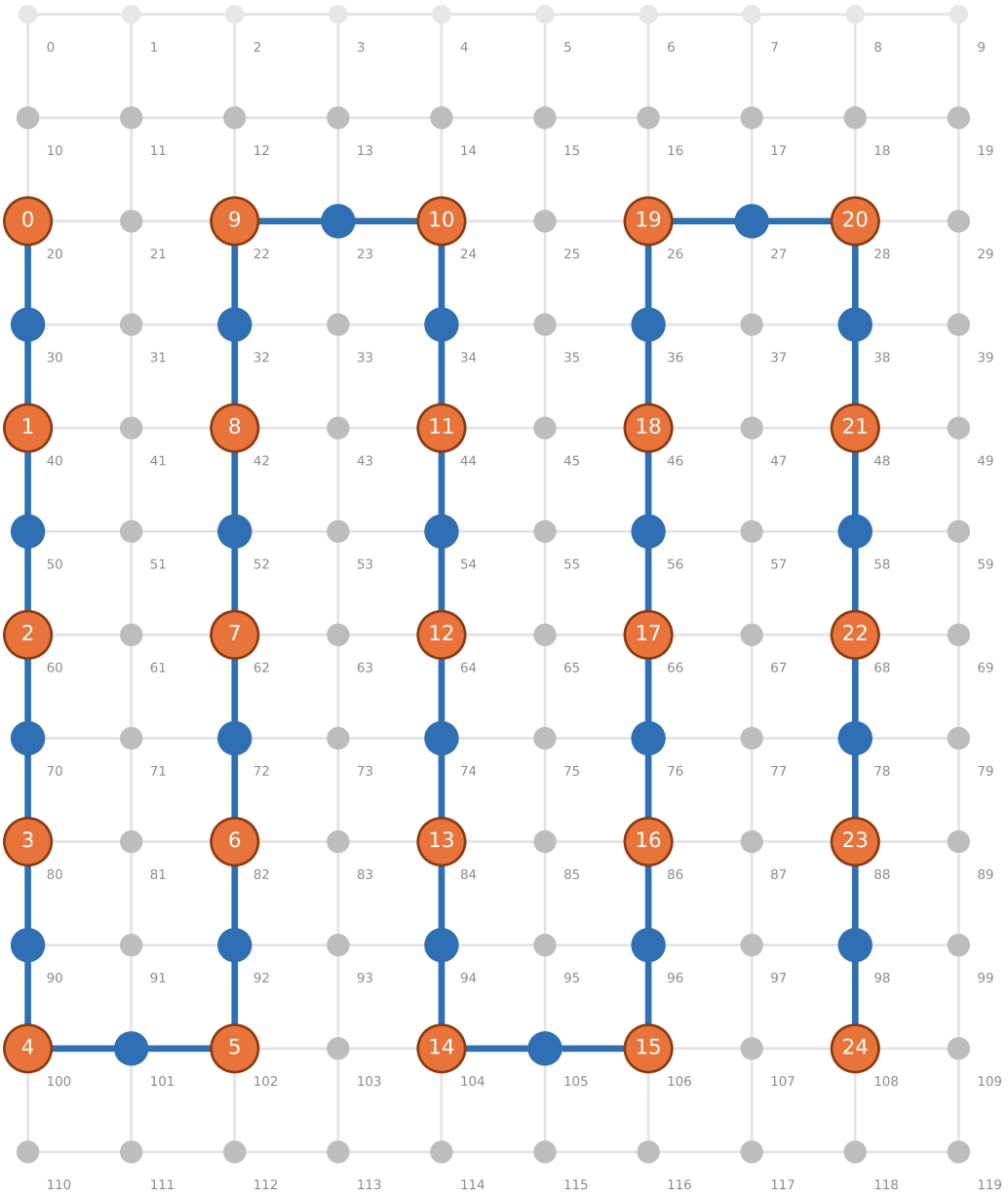


[225] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz,sx,x,cz,id) (q:48)
expi-t 8 PI, th=0.4: P = 20 21 22 23 24 25 26 27 28 29 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 (25 logical qubits)
2q depth 48, total depth 149

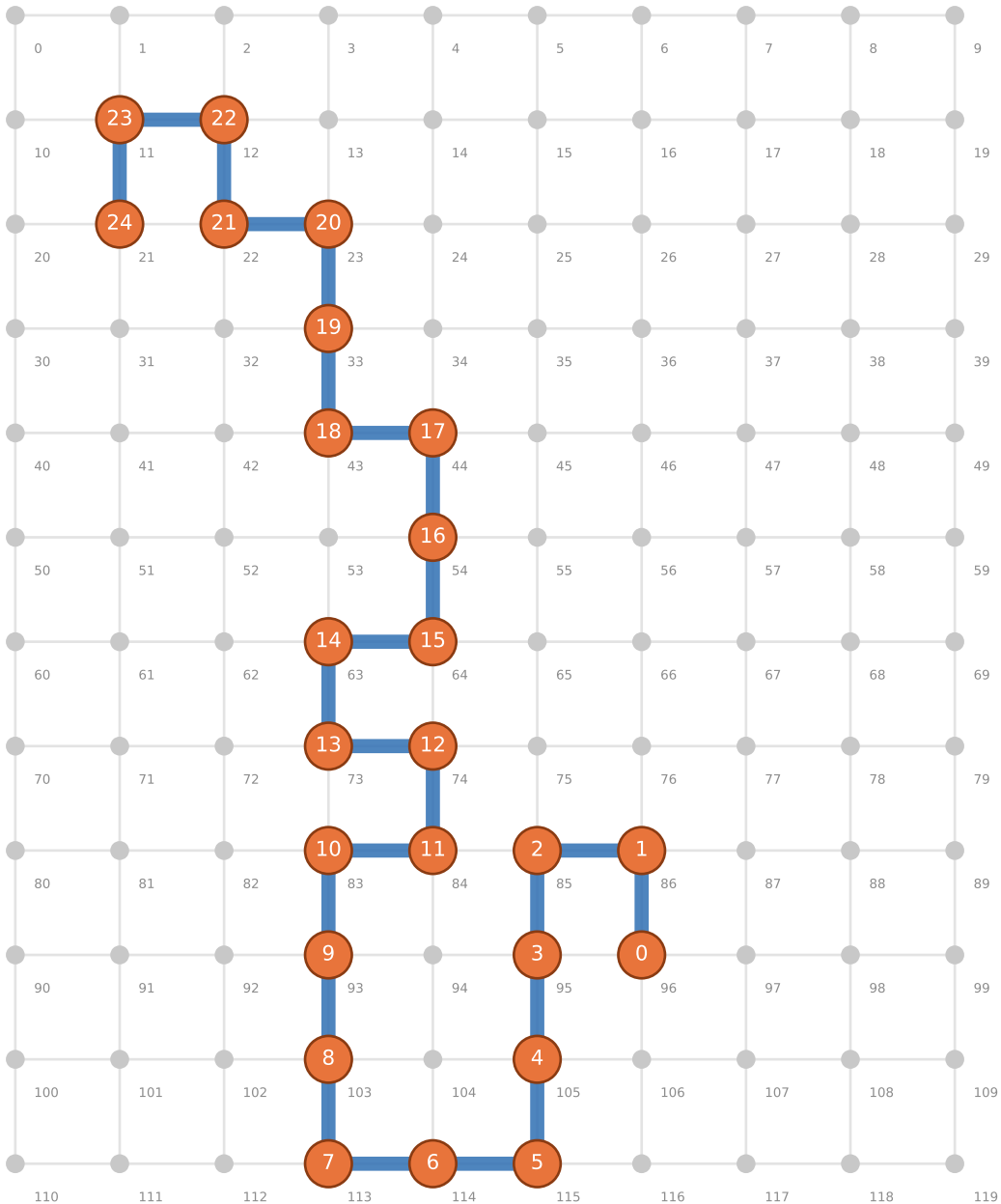


[Z25] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

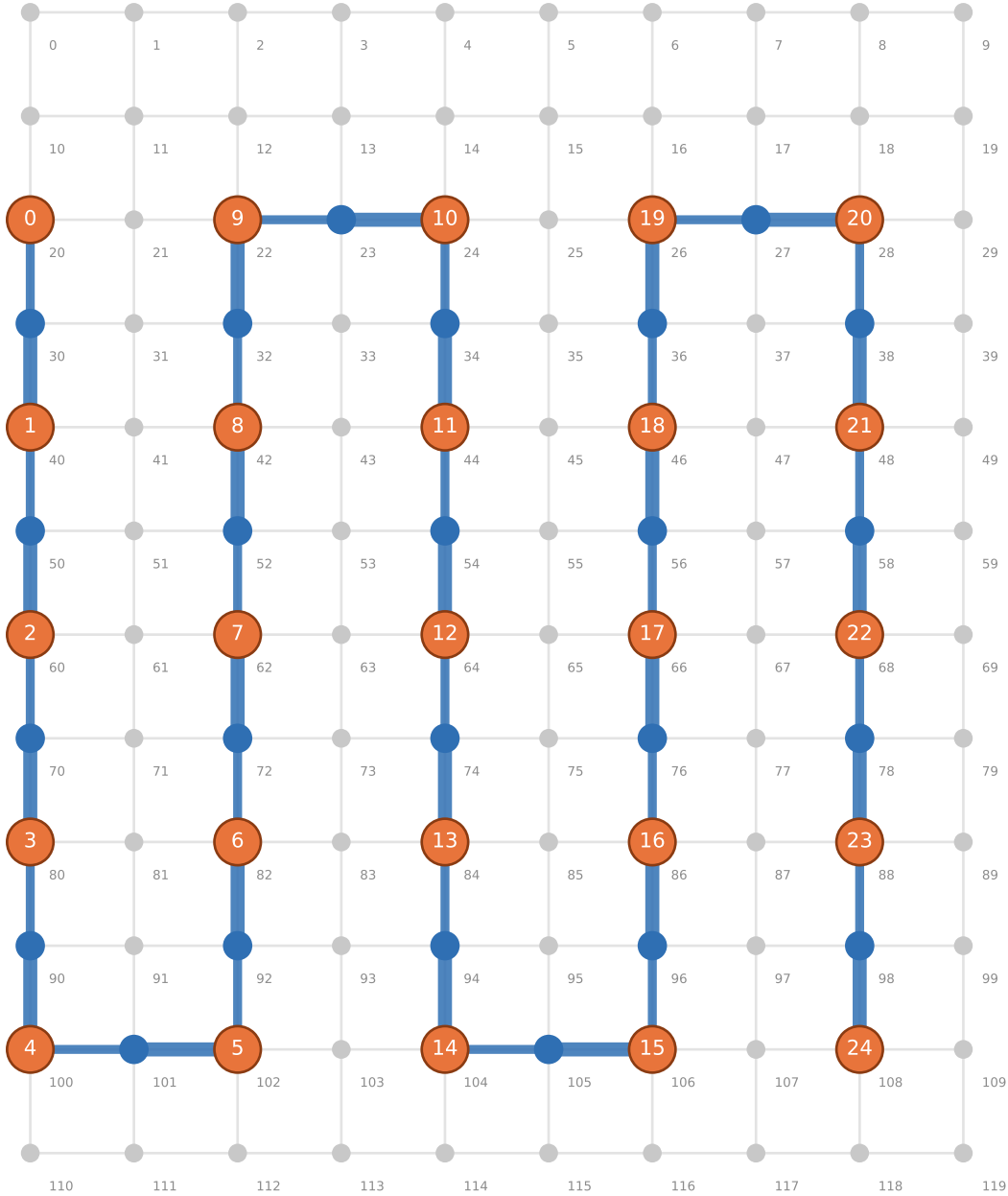
term 0: Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9 Z10 Z11 Z12 Z13 Z14 Z15 Z16 Z17 Z18 Z19 Z20 Z21 Z22 Z23 Z24
chain length 49, 24 ancillas, 0 |0> fillers



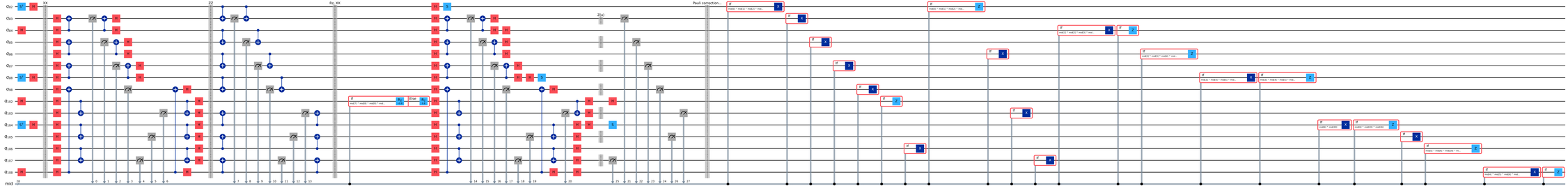
[Z25] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [96, 86, 85, 95, 105, 115, 114, 113, 103, 93, 83, 84, 74, 73, 63, 64, 54, 44, 43, 33, 23, 22, 12, 11, 21] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 48, 2q gates 48 (+0 vs. untranspiled from routing)



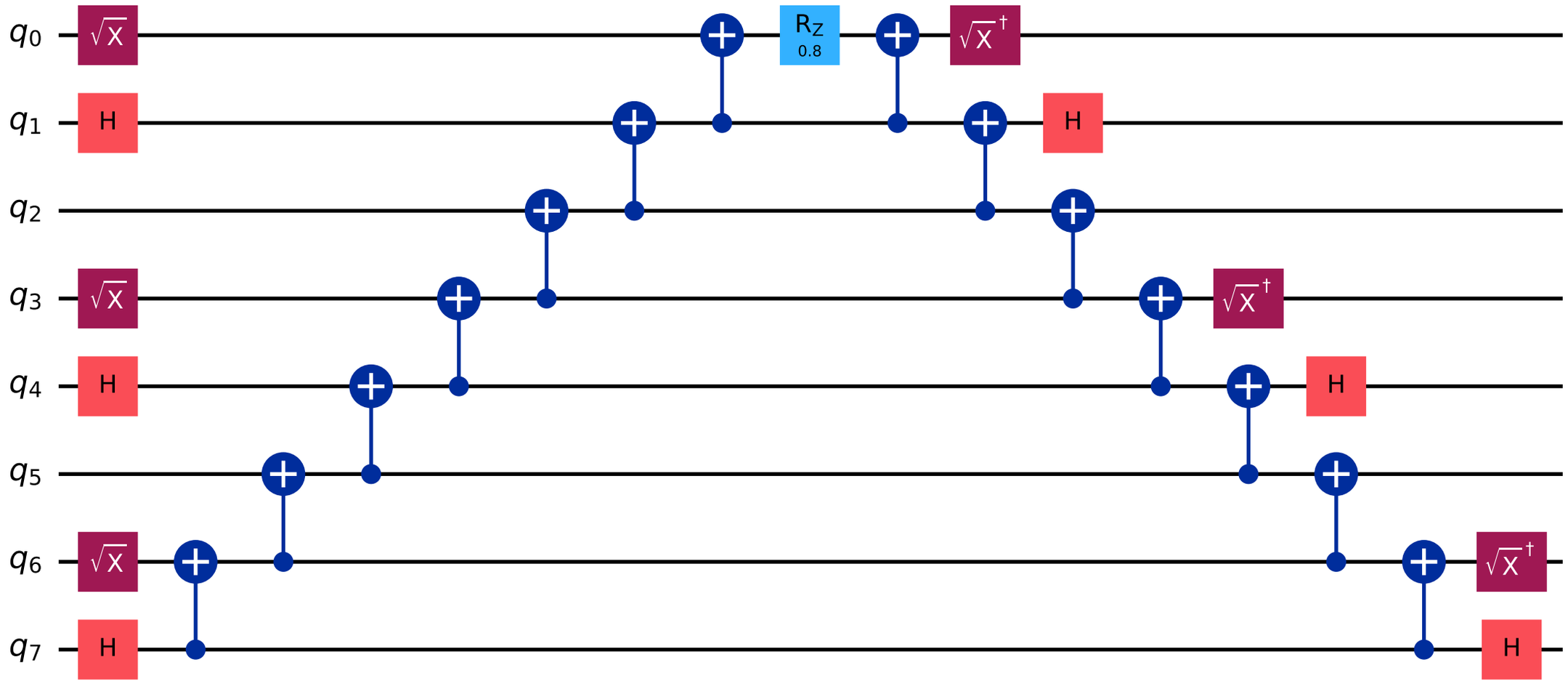
[Z25] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [20, 40, 60, 80, 100, 102, 82, 62, 42, 22, 24, 44, 64, 84, 104, 106, 86, 66, 46, 26, 28, 48, 68, 88, 108]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 144



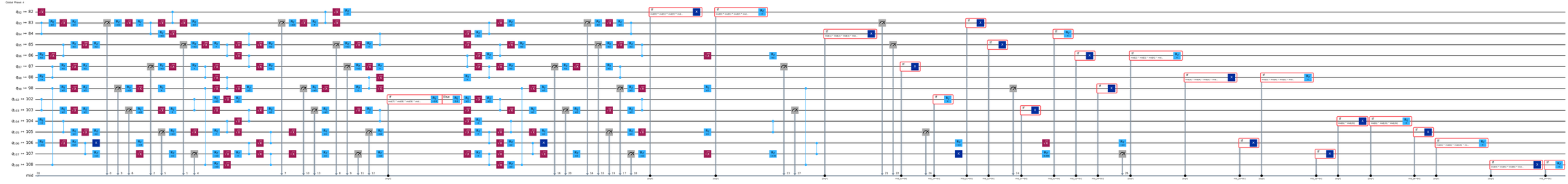
[XYZXYZXY] DYNAMIC constant-depth circuit, NISQ (8 as a single-qubit gate) (untranspiled, physical qubits of Ibm_phoenix)
exp(-i θ P_0 P_1 θ=0.4 $P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7$. (8 logical qubits)
2q depth 6, 28 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



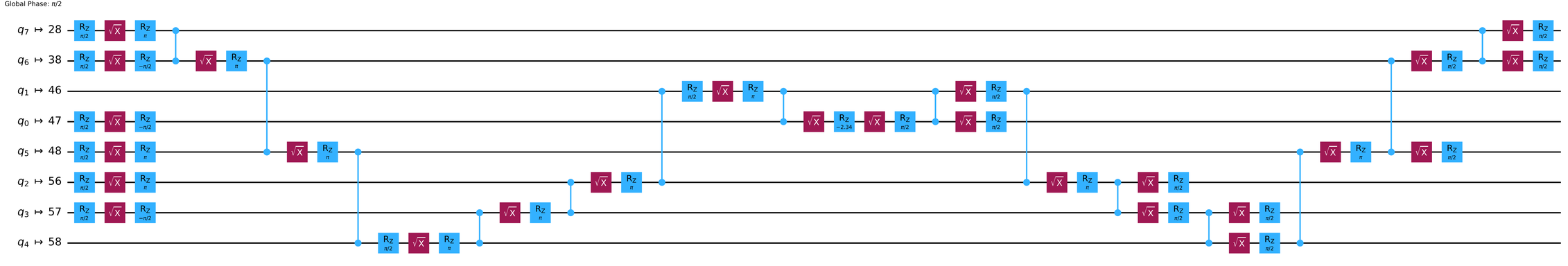
[XYZXYZXY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14

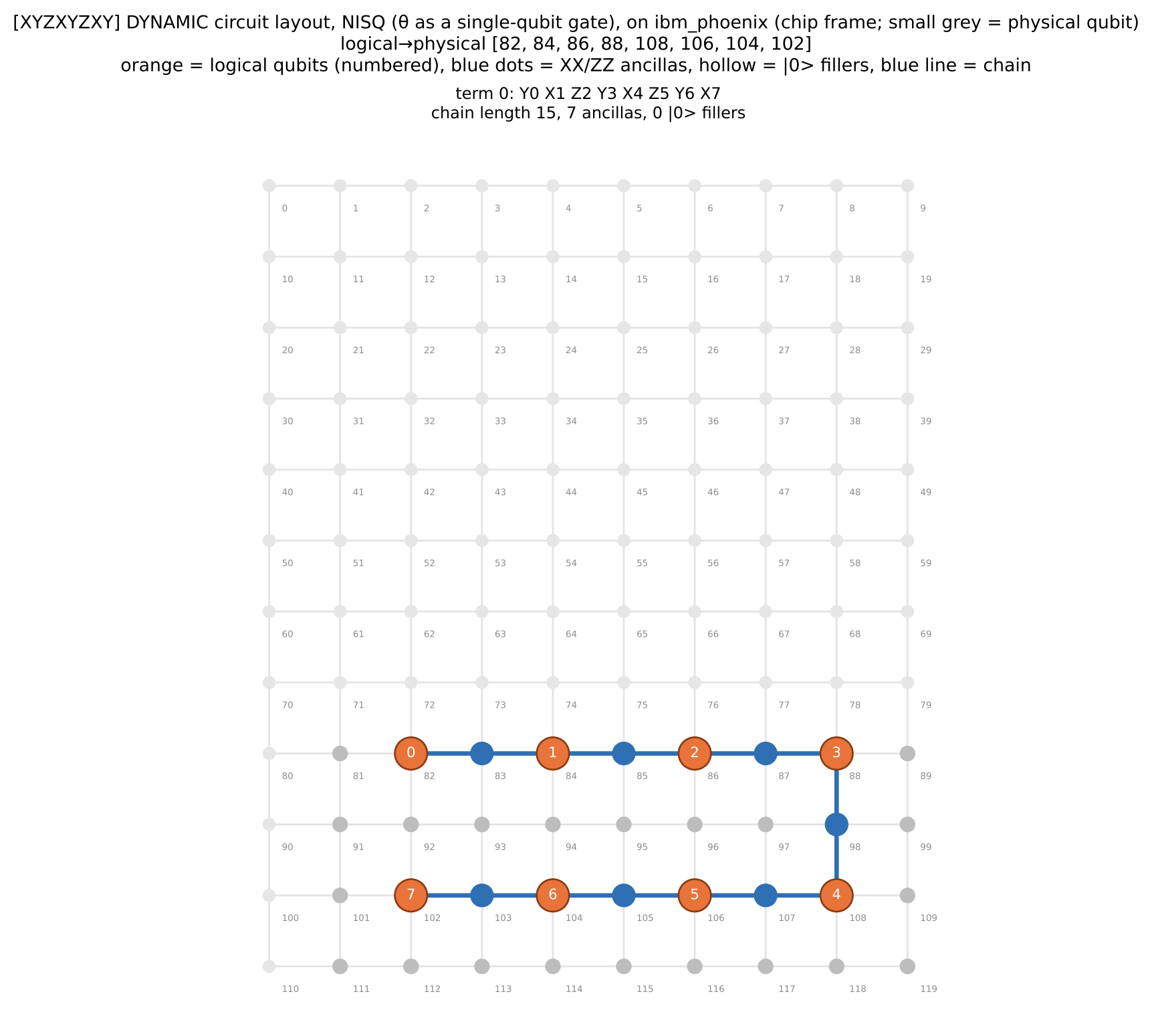


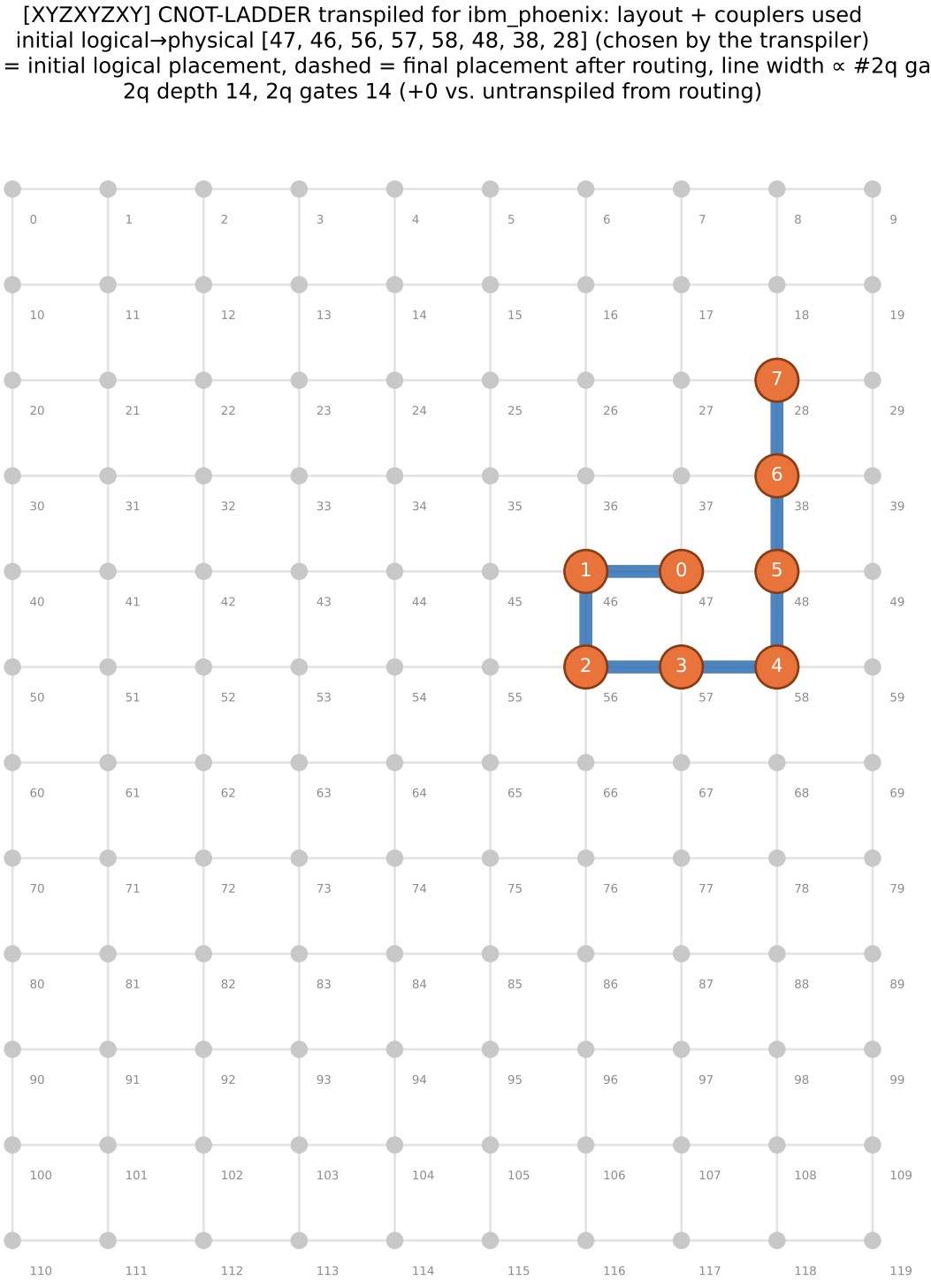
[XYZXYZXY] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:93 sx:76 cz:42 measure:28 if_else:23 x:3)
exp(i·θ·P), θ=π/4: P → YI XI ZZ YJ X4 Z5 Y6 X7 · (8 logical qubits)
2q depth 6, total depth 41



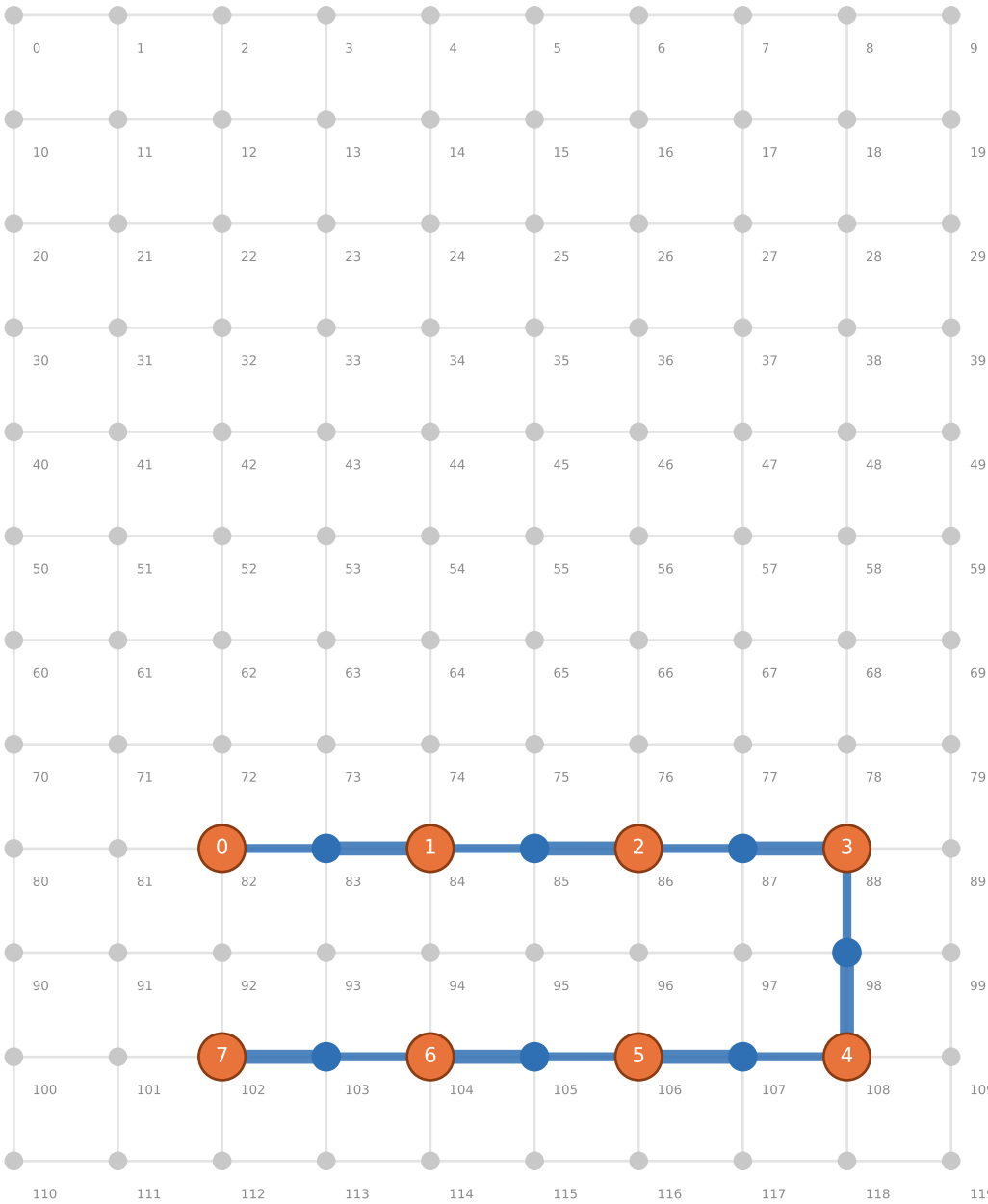
[XYZXYZXY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:34 sx:26 cz:14)
exp(-i θ P), θ=0.4: P = Y0 X1 Z2 Y3 X4 Z5 Y6 X7   (8 logical qubits)
2q depth 14, total depth 49



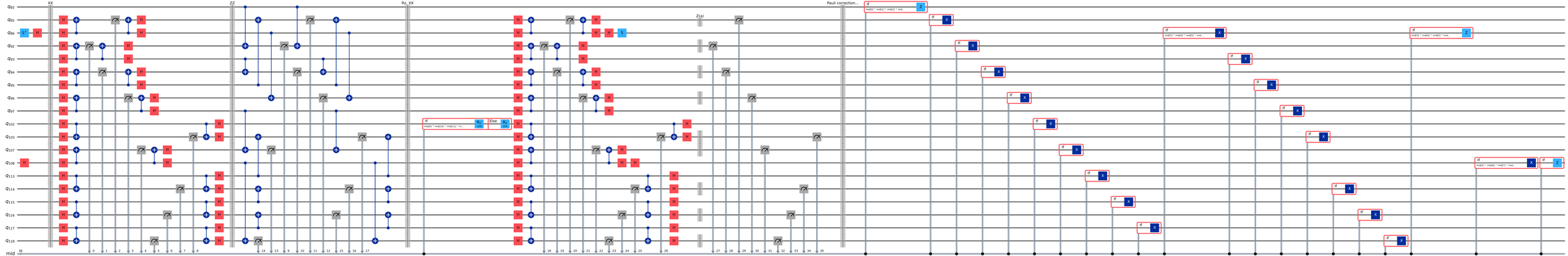

[XYZXYZXY] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain
term 0: Y0 X1 Z2 Y3 X4 Z5 Y6 X7
chain length 15, 7 ancillas, 0 |0> fillers
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
0
1
2
3
4
5
6
7


[XYZXYZXY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [47, 46, 56, 57, 58, 48, 38, 28] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 14, 2q gates 14 (+0 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
7
6
1
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5
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3
4

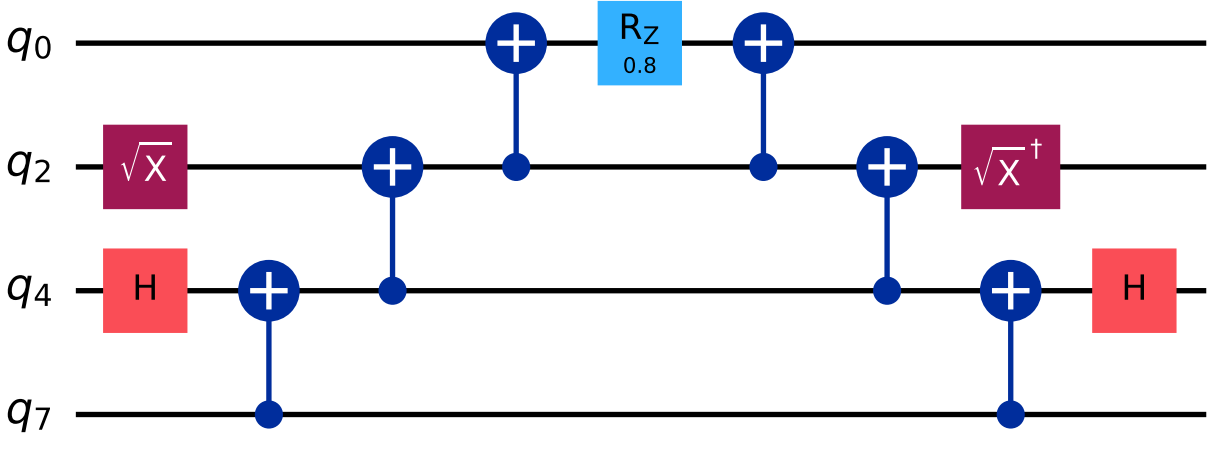
[XYZXYZXY] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 42



[ZIXYIZ_gapped] DYNAMIC constant-depth circuit, NISQ (θ as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = ZQ YZ X4 ZT (8 logical qubits)
2q depth 6, 36 mid-circuit measurements, logical→physical [82, 84, 86, 88, 108, 106, 104, 102]



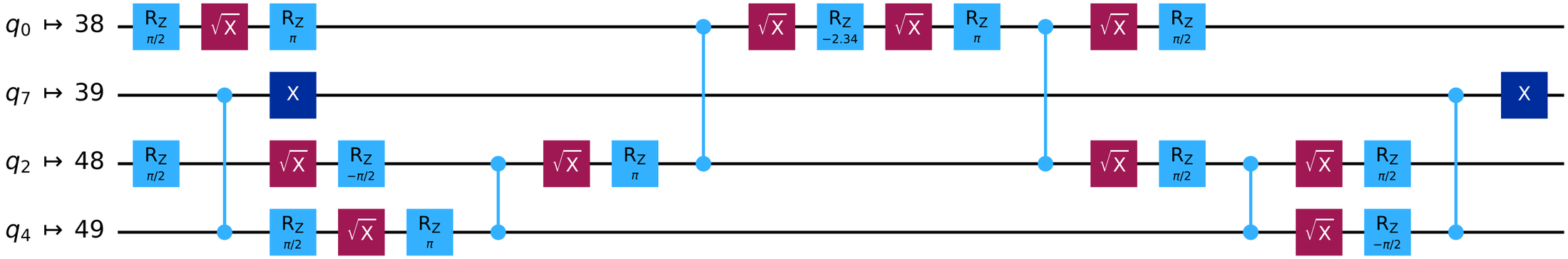
[ZIIXIYIZ_gapped] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7   (8 logical qubits)
2q depth 6



[ZIXIYIZ_gapped] DYNAMIC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:136 sx:104 cz:54 measure:36 if_else:22 x:15)
exp(-i 8 P1, 8=0.4) P = ZI YZ XX ZT (8 logical qubits)
2q depth 6, total depth 36

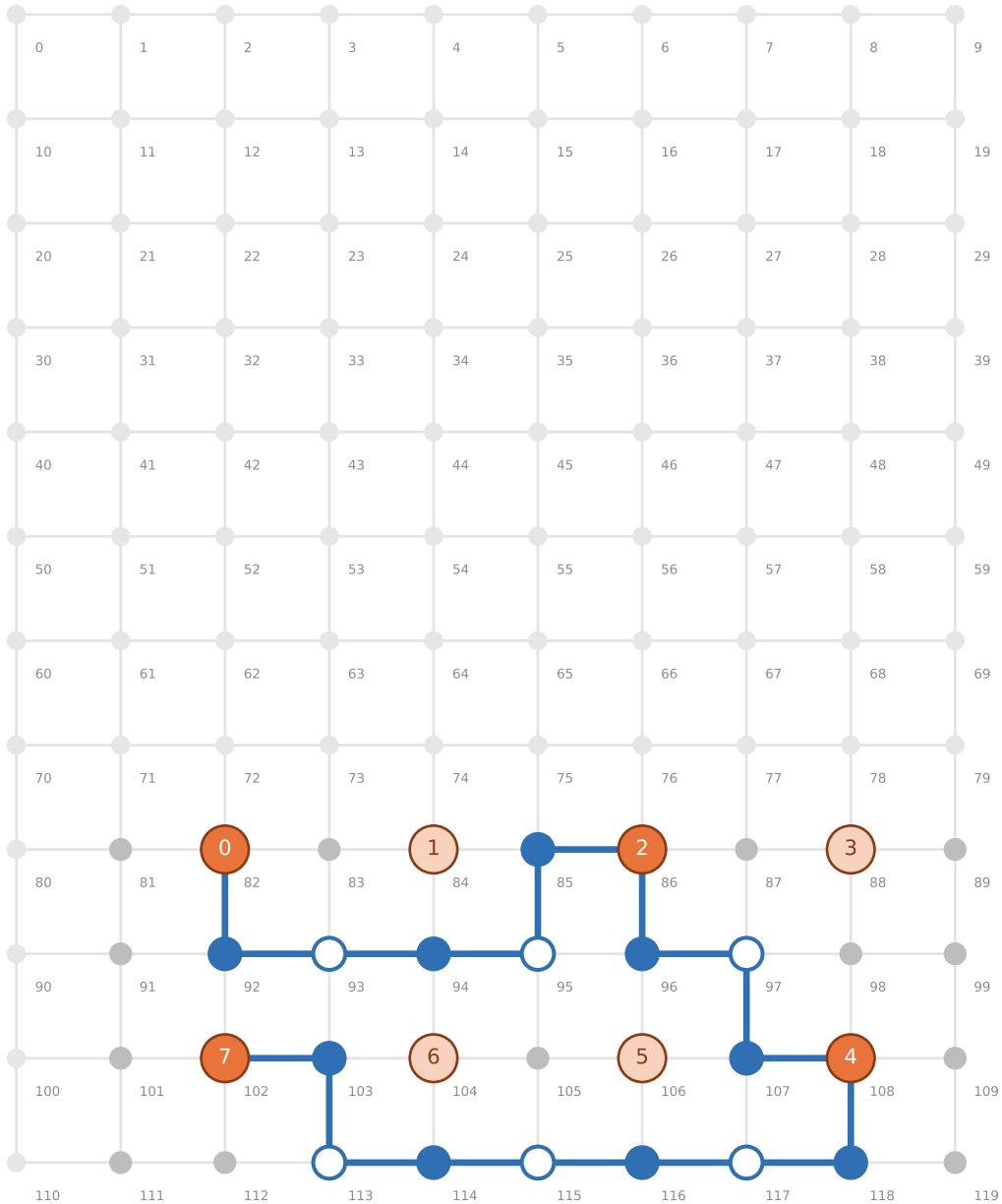


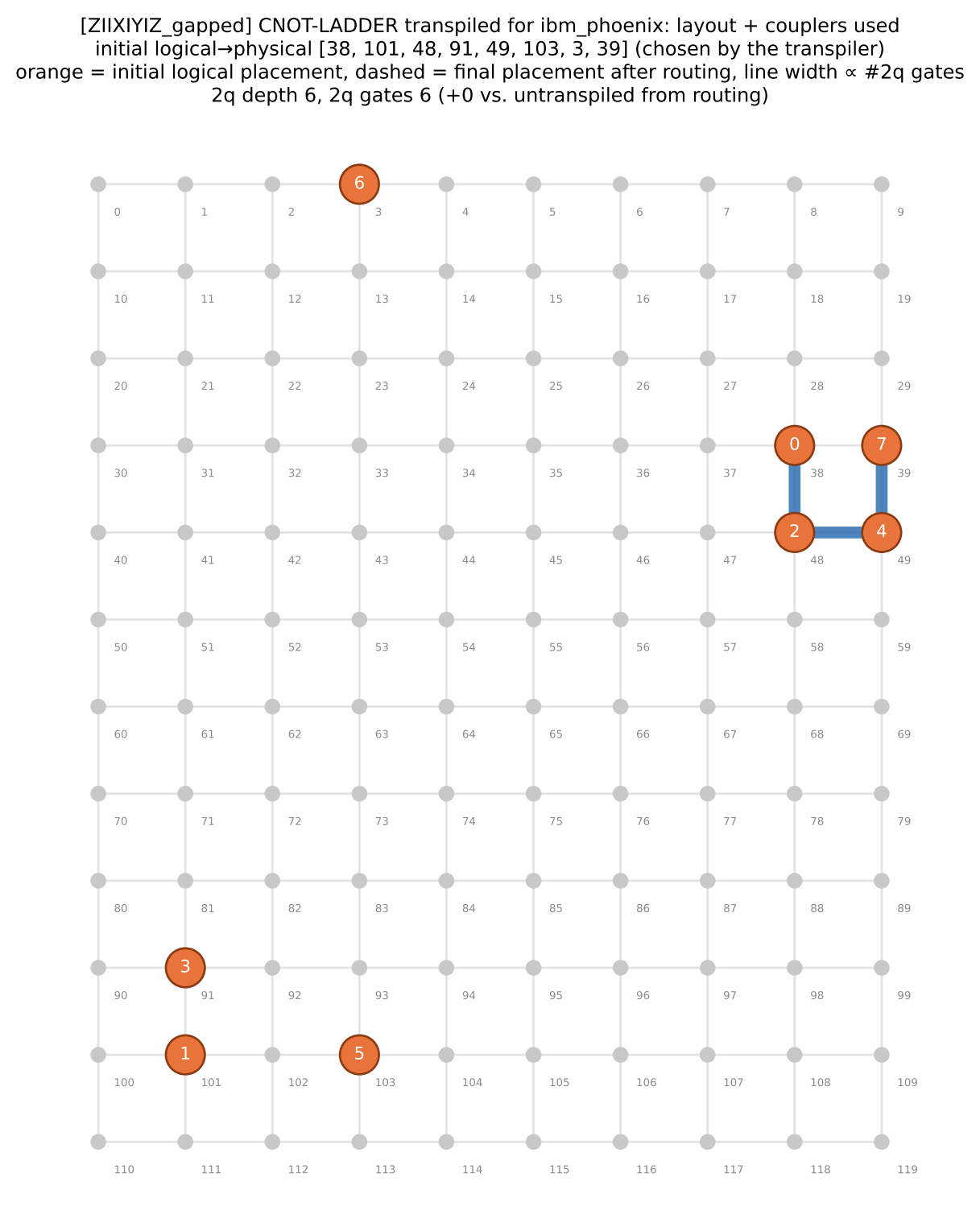
[ZIIXIYIZ_gapped] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:13 sx:10 cz:6 x:2)
exp(-i θ P), θ=0.4: P = Z0 Y2 X4 Z7 (8 logical qubits)
Global Phase: 0
2q depth 6, total depth 20



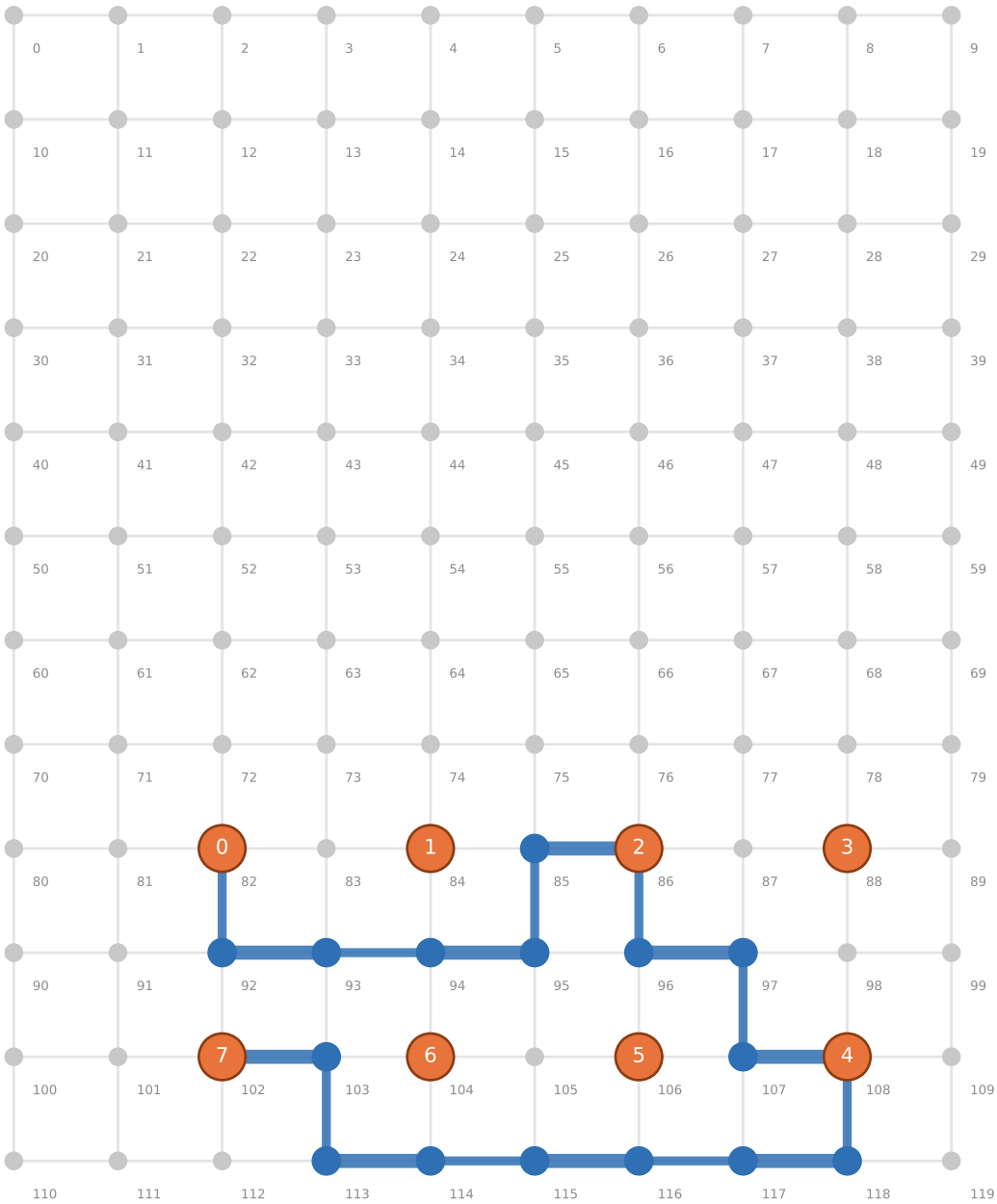
[ZIIXIYIZ_gapped] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: Z0 Y2 X4 Z7
chain length 19, 9 ancillas, 6 |0> fillers

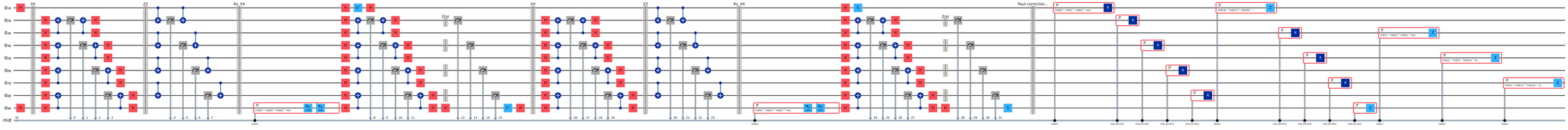



[ZIIXIYIZ_gapped] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [38, 101, 48, 91, 49, 103, 3, 39] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 6, 2q gates 6 (+0 vs. untranspiled from routing)
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
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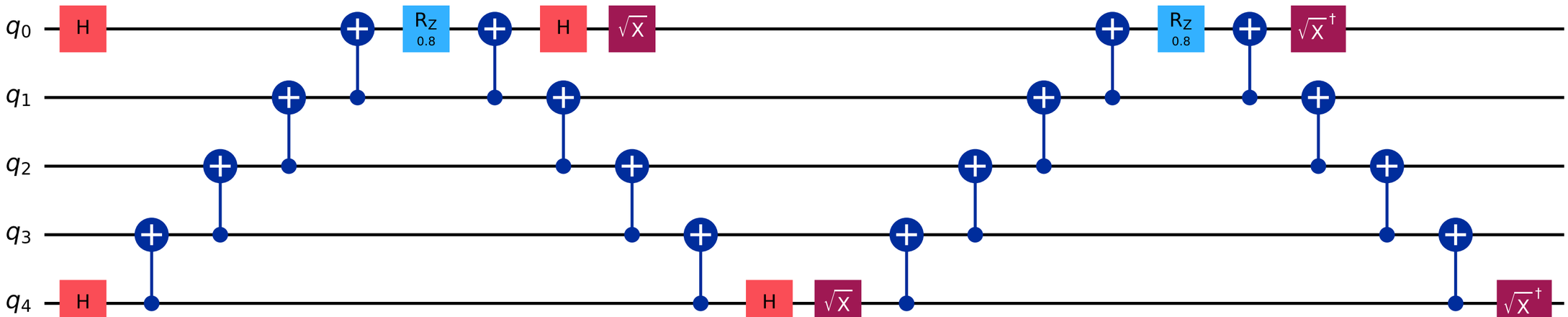
[ZIIXIYIZ_gapped] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [82, 84, 86, 88, 108, 106, 104, 102]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 6, 2q gates 54



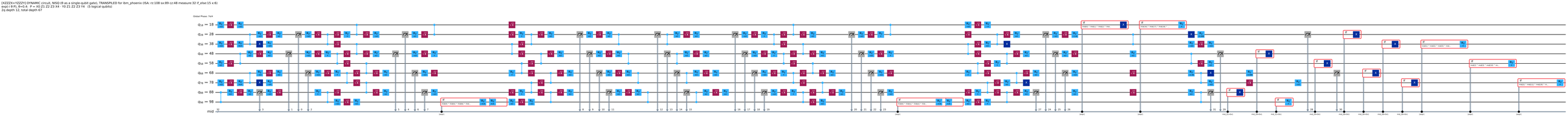
[XZZZX+YZZZY] DYNAMIC constant-depth-circuit, NISQ (8 as a single-qubit gate) (untranspiled, physical qubits of ibm_phoenix)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical-qubits)
2q depth 12, 32 mid-circuit measurements, logical→physical [18, 38, 58, 78, 98]



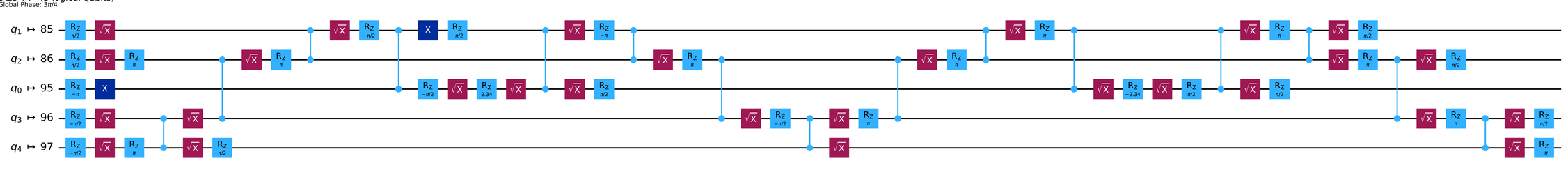
[XZZZX+YZZZY] CNOT-LADDER circuit (Qiskit PauliEvolutionGate, CNOT-chain synthesis; untranspiled, logical qubits, all-to-all)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4   (5 logical qubits)
2q depth 16



{XZZZX+YZZZY}:DYNAMC circuit, NISQ (8 as a single-qubit gate), TRANSPILED for ibm_phoenix (ISA: rz:108 sx:89 cz:48 measure:32 if_else:15 x:6)
exp(-i 9 P_8=3 x P -> X0 Z1 Z2 Z3 X4 -Y0 Z1 Z2 Z3 Y4 ; (5 logical qubits)
2q depth 12; total depth 67

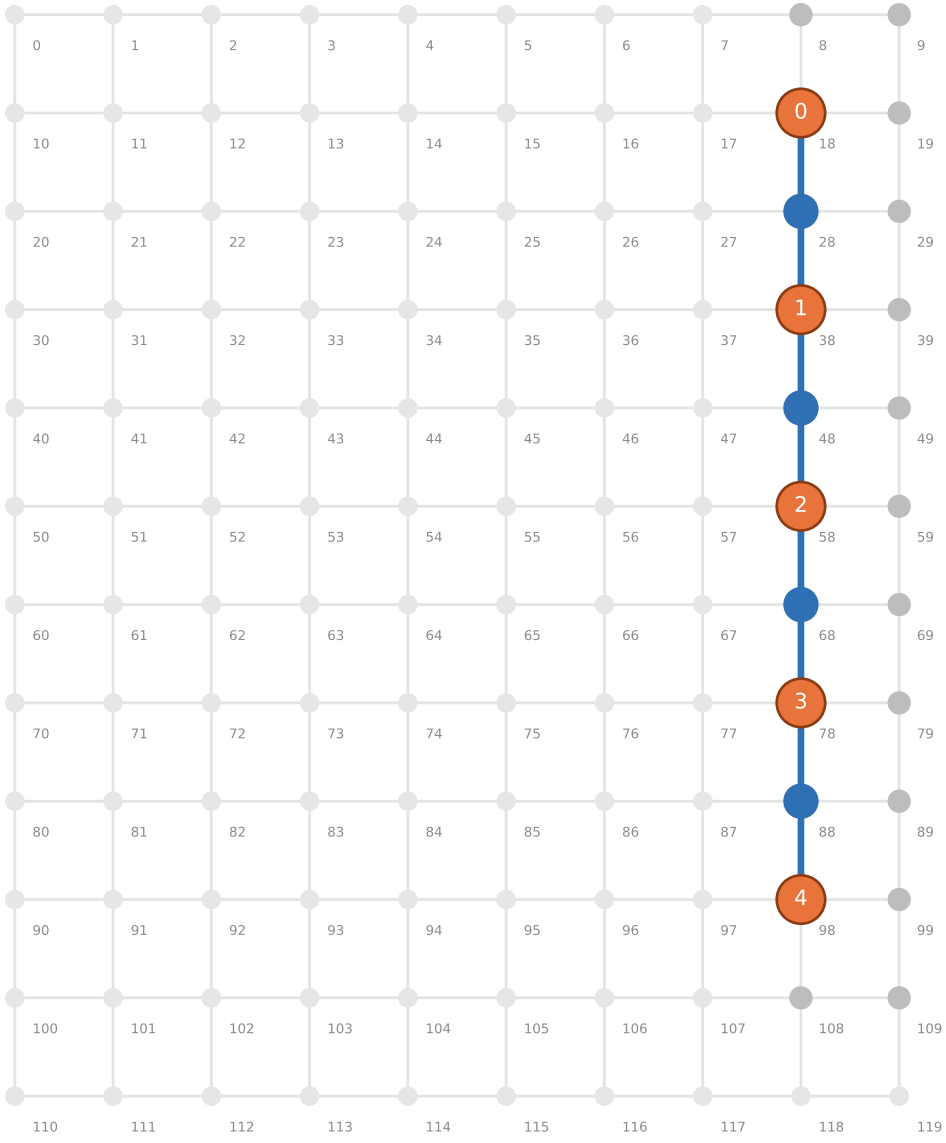


[XZZZX+YZZZY] CNOT-LADDER circuit TRANSPILED for ibm_phoenix (ISA: rz:30 sx:28 cz:15 x:2)
exp(-i θ P), θ=0.4: P = X0 Z1 Z2 Z3 X4 · Y0 Z1 Z2 Z3 Y4 (5 logical qubits)
2q depth 15, total depth 51

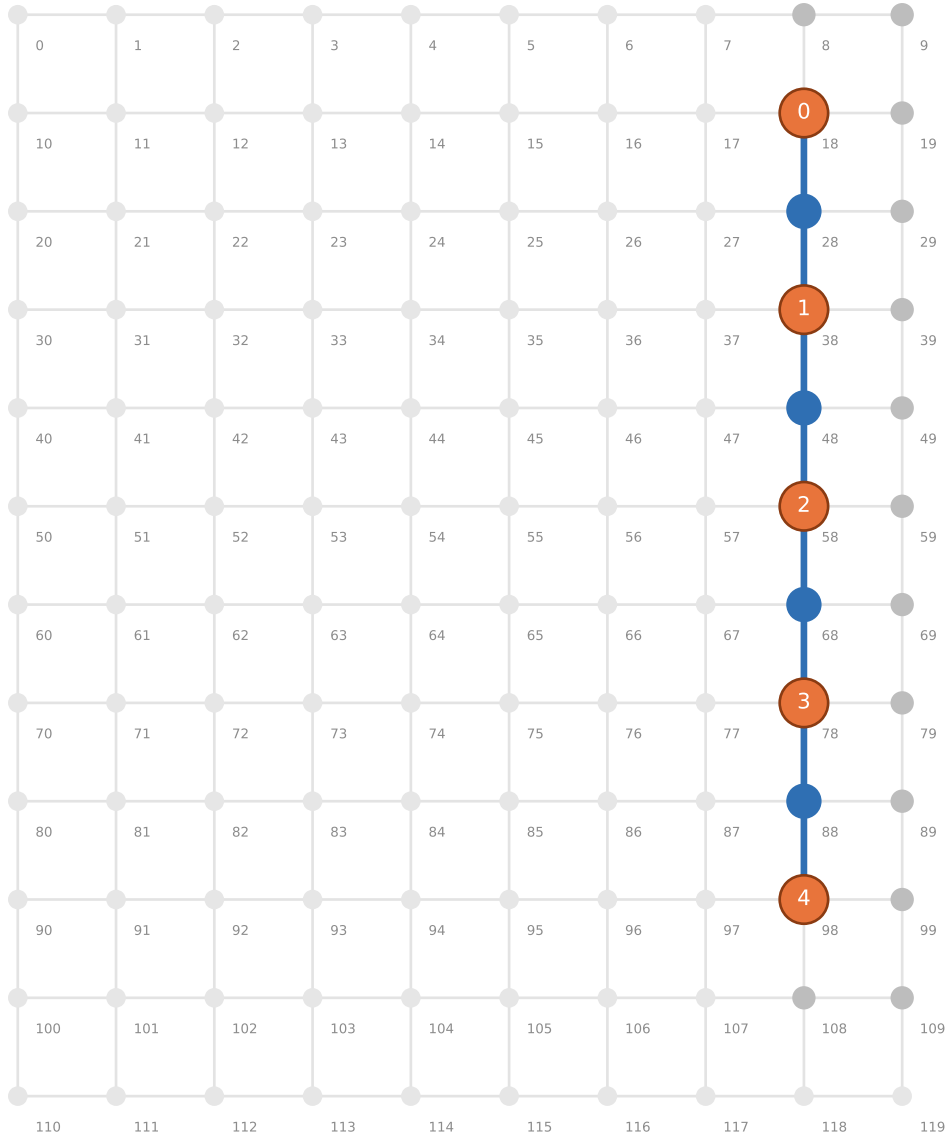


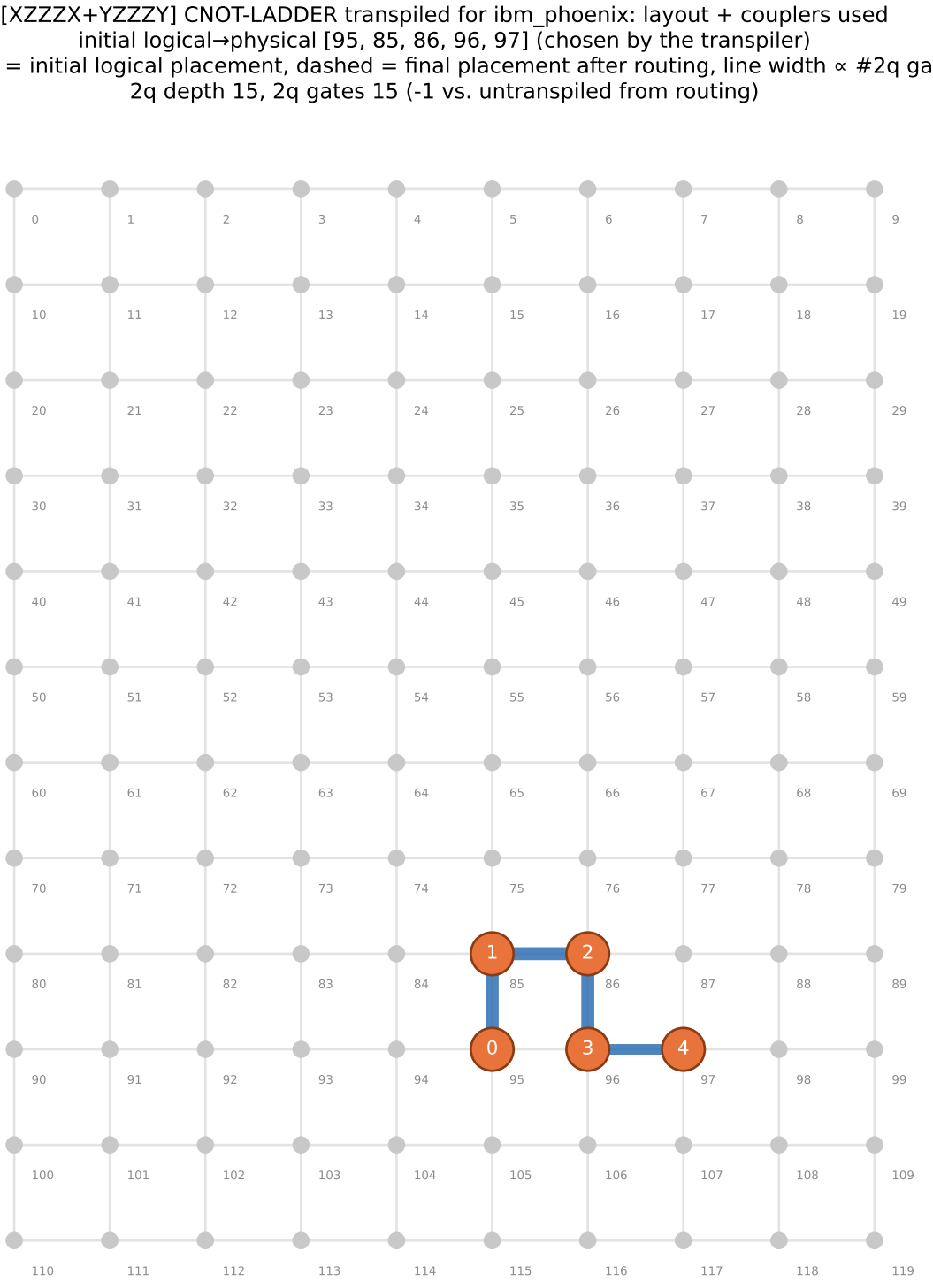
[XZZZX+YZZZY] DYNAMIC circuit layout, NISQ (θ as a single-qubit gate), on ibm_phoenix (chip frame; small grey = physical qubit)
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (numbered), blue dots = XX/ZZ ancillas, hollow = |0> fillers, blue line = chain

term 0: X0 Z1 Z2 Z3 X4
chain length 9, 4 ancillas, 0 |0> fillers



term 1: Y0 Z1 Z2 Z3 Y4
chain length 9, 4 ancillas, 0 |0> fillers




[XZZZX+YZZZY] CNOT-LADDER transpiled for ibm_phoenix: layout + couplers used
initial logical→physical [95, 85, 86, 96, 97] (chosen by the transpiler)
orange = initial logical placement, dashed = final placement after routing, line width ∝ #2q gates
2q depth 15, 2q gates 15 (-1 vs. untranspiled from routing)
0 1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29
30 31 32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49
50 51 52 53 54 55 56 57 58 59
60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87 88 89
90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109
110 111 112 113 114 115 116 117 118 119
1
2
0
3
4

[XZZZX+YZZZY] DYNAMIC circuit (nisq) transpiled for ibm_phoenix: layout + couplers used
logical→physical [18, 38, 58, 78, 98]
orange = logical qubits (preselected layout, enforced), blue = all other qubits used, line width ∝ #2q gates
2q depth 12, 2q gates 48

